MATHEMATICS

PARABOLA

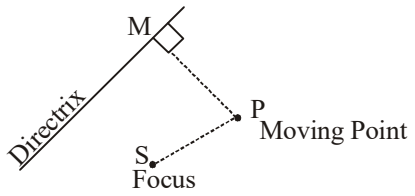
THEORY SHEET

ENGLISH

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1.1 CONIC SECTION : DEFINITION

A conic section, or conic is the locus of a point which moves in a plane so that its distance from a fixed point is in a constant ratio to its perpendicular distance from a fixed straight line.



$$\frac{PS}{PM} = e$$

- The fixed point is called the **Focus.**
- The fixed straight line is called the **Directrix.**
- The constant ratio is called the **Eccentricity** denoted by e.
- The line passing through the focus & perpendicular to the directrix is called the **Axis**.
- A point of intersection of a conic with its axis is called a **Vertex**.
- Point of intersection of axis & directrix is called foot of directrix.

1.2 GENERAL EQUATION OF A CONIC

If S is (α, β) & directrix is $\ell x + my + n = 0$

then $PS = \sqrt{(x-\alpha)^2 + (y-\beta)^2}$ & $PM = \dfrac{|\ell x + my + n|}{\sqrt{\ell^2 + m^2}}$

$\dfrac{PS}{PM} = e \quad \Rightarrow \quad (\ell^2 + m^2)[(x-\alpha)^2 + (y-\beta)^2] = e^2(\ell x + my + n)^2$

Which is of the form $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

The nature of the conic section depends upon the position of the focus S w.r.t. the directrix & also upon the value of the eccentricity e. Two different cases arise.

Case (i) When the focus lies on the directrix :

In this case $\Delta \equiv \mathbf{abc + 2fgh - af^2 - bg^2 - ch^2 = 0}$ & the general equation of a conic represents a pair of straight lines and if :

e > 1 the lines will be real & distinct intersecting at S.

e = 1 the lines will coincident.

e < 1 the lines will be imaginary.

Case (ii) When the focus does not lie on the directrix : In this case $\Delta \neq 0$. The conic represents:

a parabola	**an ellipse**	**a hyperbola**
$e = 1$; $\Delta \neq 0$	$0 < e < 1$; $\Delta \neq 0$	$\Delta \neq 0$; $e > 1$;
$h^2 = ab$	$h^2 < ab$	$h^2 > ab$

1. For conic $(x-3)^2+(y+2)^2=\frac{1}{5}(3x+4y-3)^2$

(i) find focus

Ans $(3,-2)$

(ii) find equation of directrix

Ans $3x+4y-3=0$

(iii) find eccentricity

Ans $e=\sqrt{5}$

2.1 PARABOLA : INTRODUCTION

General equation of conic $ax^2+2hxy+by^2+2gx+2ty+c=0$ will represent a parabola if

$\Delta \neq 0$, $h^2=ab$, $e=1$.

A parabola is the locus of a point which moves in such a way that its distance from a fixed point is equal to its perpendicular distance from a fixed straight line.

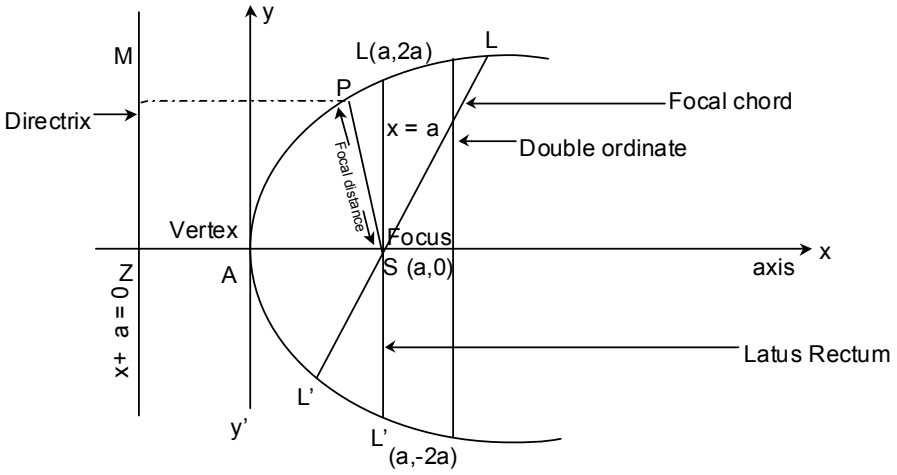
1. **Focus :** The fixed point is called the focus of the Parabola.

2. **Directrix :** The fixed line is called the directrix of the Parabola.

Standard equation of parabola is obtained by placing its focus at (a, 0) and taking its directrix as the line $x+a=0$ so that origin lies on the curve PS = PM

$(x-a)^2+y^2=(x+a)^2$

$y^2=4ax \qquad (a>0)$



Important Points

1. **Axis :** A straight line passes through the focus and perpendicular to the directrix is called the axis of parabola.
2. **Vertex :** The point of intersection of a parabola and its axis is called the vertex of the Parabola.
3. Z is foot of directrix.
4. **Focal Distance/ Focal radius :** The distance of any point P (x, y) on the parabola from the focus is called the focal length. i.e. The focal distance of P = the perpendicular distance of the point P from the directrix.

focal distance of P = PS = PM = $x_1 + a$ where P is (x_1, y_1).

5. **Focal chord :** Any chord of the parabola passing through the focus is called Focal chord.

6. **Double ordinate :** The chord which is perpendicular to the axis of Parabola or parallel to Directrix is called double ordinate of the Parabola.

7. **Latus Rectum :** If a double ordinate passes through the focus of parabola then it is called as latus rectum.

For ends of Latus rectum $y^2 = 4ax$

put $x = a \Rightarrow y^2 = 4a^2 \Rightarrow y = \pm 2a$

Ends of L.R. $\Rightarrow (a, 2a), (a, -2a)$

The length of the latus rectum = 2 (perpendicular distance of focus from the directrix)

The vertex is the middle point of the focus and the point of intersection of axis and directrix

Two parabolas are said to be equal if their latus rectum are of equal length.

Q.1 Parabola $(x-1)^2 + (y-1)^2 = \left(\frac{x+y-6}{\sqrt{2}}\right)^2$

(i) Find focus, equation of directirx
(ii) Find length of latus rectum
(iii) Find equation of axis of parabola
(iv) Find foot of directrix
(v) Find vertex
(vi) Find equation of tangent at vertex.

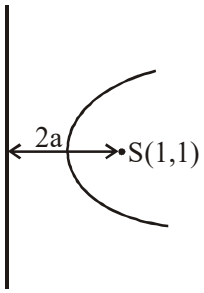
Ans (i) Focus(1, 1); Directrix: $x + y - 6 = 0$ (ii) $4\sqrt{2}$ (iii) $x - y = 0$
(iv) (3, 3) (v) (2, 2) (vi) $x + y = 4$

Sol. $(x-1)^2 + (y-1)^2 = \left(\frac{x+y-6}{\sqrt{2}}\right)^2$

$\Rightarrow \sqrt{(x-1)^2 + (y-1)^2} = \frac{|x+y-6|}{\sqrt{2}}$

$\Rightarrow \frac{\sqrt{(x-1)^2 + (y-1)^2}}{\left(\frac{|x+y-6|}{\sqrt{2}}\right)} = 1$ (Compare to $\frac{PS}{PM} = e$)

$\Rightarrow S \equiv (1,1)$ & Directrix : $x + y - 6 = 0$



$x + y - 6 = 0$

$\Rightarrow 2a = \frac{|1+1-6|}{\sqrt{2}} = \frac{4}{\sqrt{2}}$

$\Rightarrow a = \sqrt{2}$

$\therefore \ell_{L.R.} = 4\sqrt{2}$

Axis is $\perp$ to $x + y - 6 = 0$

$\therefore$ Slope of axis = 1

$\Rightarrow$ Axis : $y - 1 = 1(x - 1)$

$\Rightarrow x - y = 0$

Foot of Directrix will be P.O.I. of $x - y = 0$ & $x + y - 6 = 0$

$\Rightarrow (3,3)$

Vertex = mid point of foot of directrix & focus

$= \left(\frac{3+1}{2}, \frac{3+1}{2}\right) \equiv (2,2)$

Tangent at vertex : $y - 2 = -1(x - 2)$

$x + y = 4$

Parameters of the Parabola $y^2 = 4ax$

(i) Vertex A $\Rightarrow$ (0, 0)

(ii) Focus S $\Rightarrow$ (a, 0)

(iii) Directrix $\Rightarrow$ $x + a = 0$

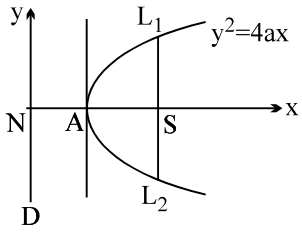
(iv) Axis $\Rightarrow$ $y = 0$ or x– axis

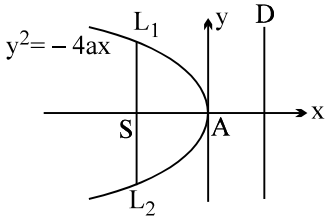
(v) Equation of Latus Rectum $\Rightarrow$ $x = a$

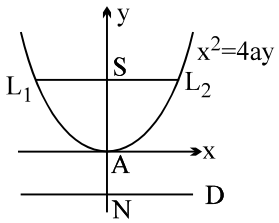
(vi) Length of L.R. $\Rightarrow$ 4a

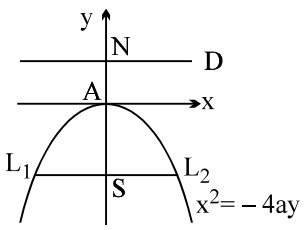
(vii) Ends of L.R. $\Rightarrow$ (a, 2a), (a, – 2a)

2.2 OTHER STANDARD PARABOLAS









Equation of Parabola	Vertex	Axis	Focus	Directrix	Equation of Latus rectum	Length of Latus rectum
$y^2 = 4ax$	(0, 0)	$y = 0$	(a, 0)	$x = -a$	$x = a$	4a
$y^2 = -4ax$	(0, 0)	$y = 0$	(–a, 0)	$x = a$	$x = -a$	4a

$x^2 = 4ay$	(0, 0)	x = 0	(0, a)	y = a	y = a	4a
$x^2 = -4ay$	(0, 0)	x = 0	(0, –a)	y = a	y = –a	4a

2. Find everything for the parabola

(a) $y^2 = 16x$

Ans Vertex (0, 0), Axis y = 0, Focus (4, 0), Z(–4, 0), $l_{LR} = 16$, Ends of LR (4, 8), (4, –8)
Equation of directrix x = –4

Sol. $y^2 = 16x$
$\Rightarrow y^2 = 4(4)x$
$\Rightarrow a = 4$
Vertex ≡ (0, 0)
Axis : y = 0
Focus ≡ (a, 0) ≡ (4, 0)
Equation of directrix : $x + a = 0 \Rightarrow x + 4 = 0$
Equation of L.R. : $x - a = 0 \Rightarrow x - 4 = 0$
$\ell_{L.R.} = 4a = 4(4) = 16$
Foot of directrix ≡ (–a, 0) ≡ (–4, 0)
ends of L.R. ≡ (a, 2a) & (a, –2a)
≡ (4, 8) & (4, –8)

(b) $x^2 = -24y$

Ans Vertex (0, 0), Axis x = 0, Focus (0, –6), Z(0, 6), $l_{LR} = 24$, Ends of LR (12, –6), (–12, –6)
Equation of directrix y = 6

Sol. $x^2 = -24y$
$x^2 = -4(6)y$
$\Rightarrow a = 6$
Vertex ≡ (0,0)
Axis : x = 0
Focus ≡ (0, –a) ≡ (0, –6)
Equation of directrix : $y - a = 0 \Rightarrow y - 6 = 0$
Equation of L.R. : $y + a = 0 \Rightarrow y + 6 = 0$
$\ell_{L.R.} = 4a = 4(6) = 24$
Foot of directrix ≡ (0, a) ≡ (0,6)
ends of L.R. ≡ (2a, a) & (–2a, –a)
≡ (12, –6) & (–12, –6)

2.3 SHIFTED PARABOLA

In case the vertex of the parabola is (h, k) then its equation can be taken as

(a) $(y - k)^2 = 4a(x - h)$ if its axis is parallel to x-axis.

(b) $(x - h)^2 = 4a(y - k)$ if its axis is parallel to y-axis.

2. Find everything for the parabola

(c) $x^2 - 8x + 24y + 88 = 0$

Ans Vertex (4, –3), Axis x = 4, Focus (4, –9), Z(4, 3), $l_{LR} = 24$, Ends of LR (16, –9), (–8, –9)
Equation of directrix y = 3

Sol. $x^2 - 8x + 24y + 88 = 0$
$\Rightarrow (x - 4)^2 - 16 + 24y + 88 = 0$

$\Rightarrow (x-4)^2 = -24y - 72$
$\Rightarrow (x-4)^2 = -24(y+3)$
$\Rightarrow (x-4)^2 = -4(6)(y+3)$
Let, $x - 4 = X$ & $y + 3 = Y$
$\Rightarrow X^2 = -4(6)Y$
For, X & Y
Vertex $\equiv (0, 0)$
Axis : $X = 0$
Focus $\equiv (0, -a) \equiv (0, -6)$
Equation of directrix : $Y - a = 0 \Rightarrow Y - 6 = 0$
Equation of L.R. : $Y + a = 0 \Rightarrow Y + 6 = 0$
$\ell_{L.R.} = 4a = 4(6) = 24$
Foot of directrix $\equiv (0, a) \equiv (0, 6)$
ends of L.R. $\equiv (2a, -a)$ & $(-2a, -a)$
$\equiv (12, -6)$ & $(-12, -6)$
For x & y
Vertex $\equiv (4, -3)$
Axis : $x = 4$
Focus $\equiv (4, -9)$
Equation of directrix : $y - 3 = 0$
Equation of L.R. : $y + 9 = 0$
$\ell_{L.R.} = 24$
Foot of directrix $\equiv (4, 3)$
ends of L.R. $\equiv (16, -9)$ & $(-8, -9)$

(d) $y^2 - 4y + 16x - 44 = 0$

Ans Vertex (3, 2), Axis $y = 2$, Focus (–1, 2), Z(7, 2), $l_{LR} = 16$, Ends of LR (–1, 10), (–1, –6)
Equation of directrix $x = 7$

Sol. $y^2 - 4y + 16x - 44 = 0$
$\Rightarrow (y-2)^2 = -4(4)(x-3)$
Let, $y - 2 = Y$ & $x - 3 = X$
$\Rightarrow Y^2 = -4(4)X$
For X and Y
Vertex $\equiv (0, 0)$
Axis : $Y = 0$
Focus $\equiv (-a,0) \equiv (-4, 0)$
Foot of directrix : $(a, 0) \equiv (4, 0)$
$\ell_{L.R.} \equiv 4a = 4(4) = 16$
Ends of L.R. $\equiv (-a, 2a)$ & $(-a, -2a) \Rightarrow (-4, 8)$ & $(-4, -8)$
Equation of directrix $\equiv X - 4 = 0$
For x & y
Vertex $\equiv (3, 2)$
Axis : $y - 2 = 0$
Focus $\equiv (-1, 2)$
Foot of directrix $\equiv (7, 2)$
$\ell_{L.R.} \equiv 16$
End of L.R. $\equiv (-1, 10)$ & $(-1, -6)$
Equation of directrix $\equiv x - 7 = 0$

3. Find equation of parabola

(i) Whose vertex is (4, –3), length of $l_{LR} = 4$, axis is parallel to x axis.

(ii) Vertex is (4, –3), directrix is parallel to x-axis and parabola passes through (–4, –7)

(iii) Vertex is (3,2), focus is (3,6)

(iv) focus (3,2) and foot of directrix is (7, 2)

(v) ends of LR are (5, -7) and (17, -7)

Ans (i) $(y + 3)^2 = 4(x – 4)$, $(y + 3)^2 = –4(x – 4)$

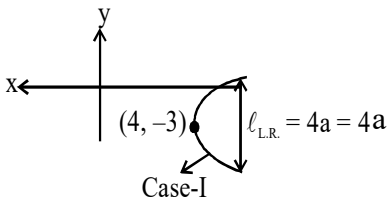
(ii) $(x – 4)^2 = –16(y + 3)$

(iii) $(x – 3)^2 = 16(y – 2)$

(iv) $(y – 2)^2 = –8(x – 5)$

(v) $(x – 11)^2 = 12(y + 10)$ $(x – 11)^2 = –12(y + 4)$

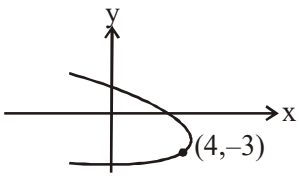
Sol. (i) Case I :



$\Rightarrow (y + 3)^2 = 4a(x – 4)$

$\Rightarrow (y + 3)^2 = 4(x – 4)$

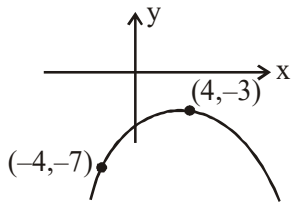
Case-II :



$(y + 3)^2 = –4a(x – 4)$

$(y + 3)^2 = –4 (x – 4)$

(ii)



$(x – 4)^2 = –4a(y + 3)$

$\because (–4, –7)$

$\Rightarrow (–8)^2 = –4a(–4)$

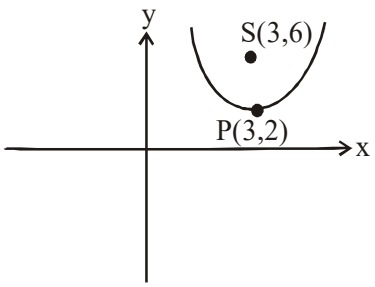
$\Rightarrow \frac{64}{16} = a$

$\Rightarrow a = 4$

$\Rightarrow (x – 4)^2 = –4(4) (y + 3)$

$\Rightarrow (x – 4)^2 = –16(y + 3)$

(iii)



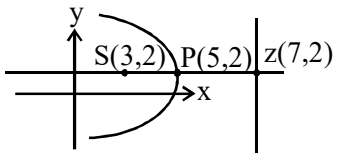
$\Rightarrow (x-3)^2 = 4a(y-2)$

$\because$ a = SP

a = 4

$\Rightarrow (x-3)^2 = 16(y-2)$

(iv)



$$P \equiv \left(\frac{3+7}{2}, \frac{2+2}{2}\right)$$

$P \equiv (5, 2)$

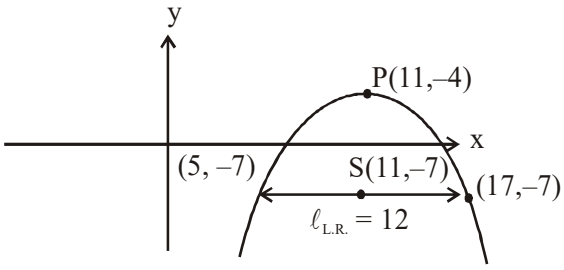
$\therefore (y-2)^2 = -4a(x-5)$

Now, $2a = SZ = 4$

$a = 2$

$\Rightarrow (y-2)^2 = -8(x-5)$

(v) Case-I :



$4a = 12$

$a = 3$

$$S \equiv \left(\frac{5+17}{2}, \frac{-7-7}{2}\right)$$

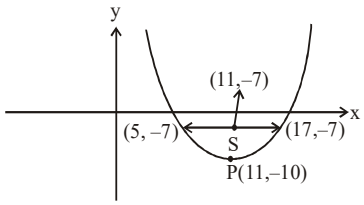
$S \equiv (11, -7)$

$\therefore P \equiv (11, -4)$

$\Rightarrow (x-11)^2 = -4a(y+4)$

$\Rightarrow (x-11)^2 = -12(y+4)$

Case-II :



$4a = 12$

$a = 3$

$P \equiv (11, -10)$

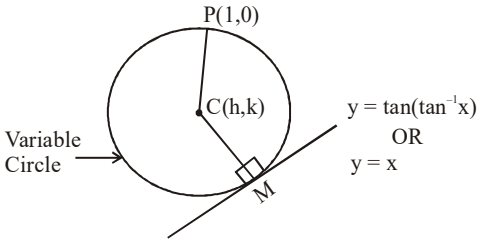
$\therefore (x - 11)^2 = 4a(y + 10)$

$(x - 11)^2 = 12(y + 10)$

4. A variable circle always passes through (1, 0) and touches the curve $y = \tan(\tan^{-1}x)$. Find the equation to the locus of its centre.

Ans $(x-1)^2 + y^2 = \left(\frac{x-y}{\sqrt{2}}\right)^2$

Sol.



Clearly, CP = CM

$$\Rightarrow \sqrt{(h-1)^2 + k^2} = \frac{|h-k|}{\sqrt{2}}$$

S.B.S.

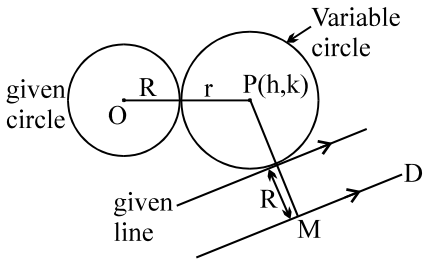
$$\Rightarrow (h-1)^2 + k^2 = \frac{(h-k)^2}{2}$$

$$\Rightarrow (x-1)^2 + y^2 = \frac{(x-y)^2}{2}$$

5. If a variable circle touches a fixed circle and a fixed line then prove that the locus of the centre of the variable circle is a parabola whose directrix is parallel to a given line at a distance equal to the radius of the given circle.

Ans.

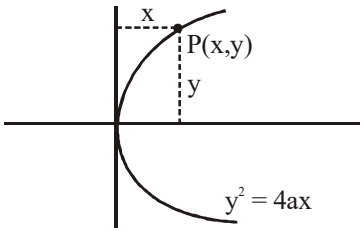
Sol.



Distance of P from 'O' is equal to PM

= distance of P from the directrix D = R + r
and PO $\Rightarrow$ locus of P is a parabola with 'O' as its focus and 'D' as its directrix]

2.4 ALTERNATE DEFINITION OF PARABOLA

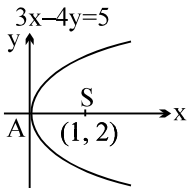


$$\left(\text{Distance of P from axis of parabola}\right)^2 = \text{length of latus rectum}\left|\text{Distance of P from tangent at vertex}\right|$$

6. For some parabola, tangent at the vertex is $3x - 4y = 5$ and its focus is (1, 2). Find its equation.

Ans $\left(\frac{4x+3y-10}{5}\right)^2 = -8\left(\frac{3x-4y-5}{5}\right)$

Sol

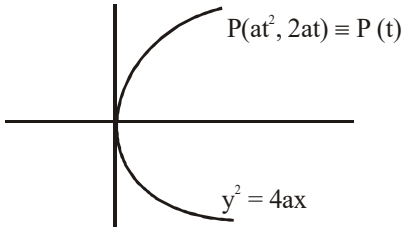


Equation of Parabola : $\left(\frac{4x+3y-10}{5}\right)^2 = -8\left(\frac{3x-4y-5}{5}\right)$

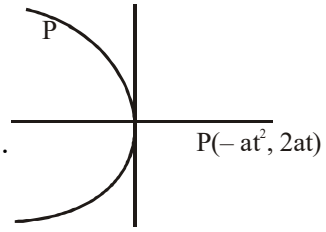
($\because$ (1, 2) & (x, y) lies on same side)

3 PARAMETRIC COORDINATES

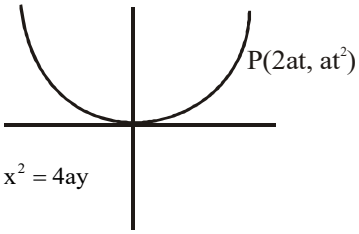
1.



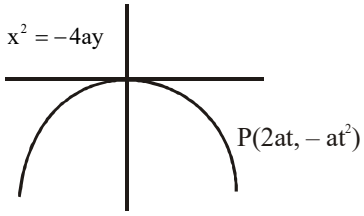
2.



3.



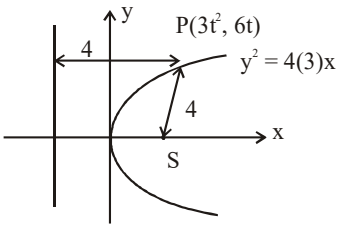
4.



1. Find point P on parabola $y^2 = 12x$ whose focal distance is (i) 4 (ii) 3 (iii) 2

Ans (i) $\left(1, \pm 2\sqrt{3}\right)$ (ii) (0, 0) (iii) ϕ

Sol.



$\Rightarrow 3t^2 + 3 = 4$

$\Rightarrow t^2 = \frac{1}{3}$

$\Rightarrow t = \pm\frac{1}{\sqrt{3}} = \pm\frac{\sqrt{3}}{3}$

$\therefore P \equiv \left(1, 2\sqrt{3}\right)$ & $\left(1, -2\sqrt{3}\right)$

(ii) $3t^2 + 3 = 3$

$3t^2 = 0$

$t = 0$

$\therefore P \equiv (0, 0)$

(iii) $3t^2 + 3 = 2$

$t^2 = -\frac{1}{3}$

$t \in \phi$

Not possible

2. Prove that the area of the triangle whose vertices are (x_i, y_i), i = 1, 2, 3 and inscribed in the parabola $y^2 = 4ax$ is $\frac{1}{8a}|(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)|$

Sol. Let points be $A\left(\frac{y_1^2}{4a}, y_1\right), B\left(\frac{y_2^2}{4a}, y_2\right), C\left(\frac{y_3^2}{4a}, y_3\right)$

$$\Delta = \frac{1}{2}\begin{vmatrix} \frac{y_1^2}{4a} & y_1 & 1 \\ \frac{y_2^2}{4a} & y_2 & 1 \\ \frac{y_3^2}{4a} & y_3 & 1 \end{vmatrix}$$

$$= \frac{1}{8a}\begin{vmatrix} y_1^2 & y_1 & 1 \\ y_2^2 & y_2 & 1 \\ y_3^2 & y_3 & 1 \end{vmatrix}$$

$$\Delta = \frac{1}{8a}\left|(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)\right|$$

3. Point P is a variable point on parabola $y^2 = 4ax$. Find locus of mid point of OP.

Ans. $y^2 = 2ax$

Sol. O(0,0) $P(at^2, 2at)$

Mid point $M\left(\frac{at^2}{2}, \frac{2at}{2}\right)$

$M\left(\frac{at^2}{2}, at\right)$

M(h, k)

$h = \frac{at^2}{2}$ $k = at$

$h = \frac{a\left(\frac{k}{a}\right)^2}{2}$

$h = \frac{ak^2}{2a^2}$

$k^2 = 2ah$

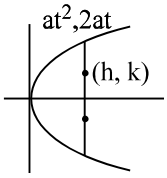
$y^2 = 2ax$

4. Obtain the locus of the point of trisection of double ordinates of the parabola $y^2 = 4ax$.

Ans. $y^2 = \frac{4ax}{9}$

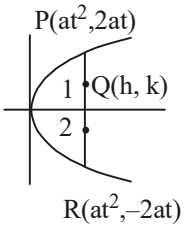
[Sol. $h = at^2$

$6|k| = 4at \Rightarrow 3|k| = 2at$



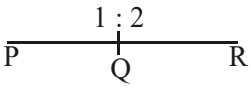
now eliminate t

Alternate :



$y^2 = 4ax$

$h = at^2$



$$h = \frac{at^2 + 2at^2}{3} \Rightarrow at^2$$

$$k = \frac{-2at + 4at}{3} = \frac{2at}{3}$$

$$t = \frac{3k}{2a}$$

$$h = a\left(\frac{3k}{2a}\right)^2$$

$$h = a\frac{9k^2}{4a^2}$$

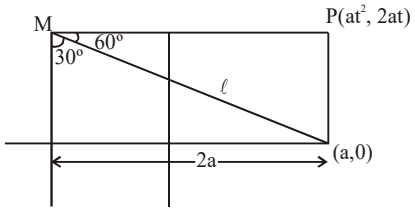
$$4ah = 9k^2$$

$$y^2 = \frac{4}{9}ax$$

5. P is a point on parabola $y^2 = 4ax$ whose focus is S and foot of perpendicular from P on directrix is M. If traingle formed by P, S and M is equilateral then find its area.

Ans $4\sqrt{3}\,a^2$

Sol.



$$\sin 30^\circ = \frac{2a}{\ell}$$

$$\frac{1}{2} = \frac{2a}{\ell}$$

$$\ell = 4a$$

$$\text{are of } \Delta = \frac{\sqrt{3}}{4}(\text{side})^2$$

$$= \frac{\sqrt{3}}{4} \times 16a^2$$

$$\Delta = 4\sqrt{3}a^2$$

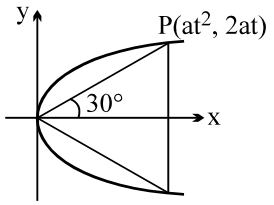
6. Find the side of an equilateral triangle inscribed in $y^2 = 4ax$ if one of its vertex coincides with the vertex of the parabola.

Ans. $8\sqrt{3}\,a$

Sol. $\tan 30° = \frac{1}{\sqrt{3}} = \frac{2}{t}$

$\therefore \quad t = 2\sqrt{3}$

Hence P is $(12a, 4\sqrt{3}\,a)$ $\Rightarrow$ side $= 8\sqrt{3}\,a$



7. Prove that the parabolas $y^2 = 4b(x - 2a + b)$ and $x^2 + 4a(y - 2b - a) = 0$ intersect orthogonally at the common end of their latus rectum.

Sol. $y^2 = 4b(x - (2a - b))$ or $y^2 = 4bX$ where $x - (2a - b) = X$

$x^2 + 4a(y - (a + 2b))$ or $x^2 = -4aY$ where $y - (a + 2b) = Y$

for $y^2 = 4bX$, extremities of latus rectum (b, 2b) and (b, – 2b) w.r.t. X Y axis

i.e. (2a, 2b) and (2a, – 2b) w.r.t. xy axis

for $x^2 = -4aY$, extremities of latus rectum (2a, – a) and (–2a, –a) w.r.t. XY axis

i.e. (2a, 2b) and (–2a, 2b)

Hence the common end of latus rectum (2a, 2b)

now for 1st parabola $2y\frac{dy}{dx} = 4b \Rightarrow \frac{dy}{dx} = \frac{2b}{y_1} = 1$ at (2a, 2b)

also for 2nd parabola $2x = -4a\frac{dy}{dx}$ or $\frac{dy}{dx} = -\frac{x}{2a} = -1$ at (2a, 2b)

8. A variable parabola is drawn to pass through A & B, the ends of a diameter of a given circle with centre at the origin and radius c & to have as directrix a tangent to a concentric circle of radius 'a' (a >c) ; the axes being AB & a perpendicular diameter, prove that the locus of the focus of the parabola is the standard ellipse

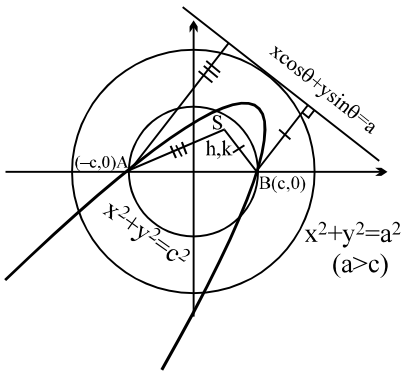
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $b^2 = a^2 - c^2$. **[O]**

Sol. $(h - c)^2 + k^2 = (c\cos\theta - a)^2$ ....(1)

sub. $(h + c)^2 + k^2 = (c\cos\theta + a)^2$ ....(2)

$4ch = 4ca\cos\theta \Rightarrow h = a\cos\theta$ .....(3)

add $2(c^2 + h^2 + k^2) = 2(c^2\cos^2\theta + a^2)$ .....(4)



put $\cos\theta = \frac{h}{a}$ in equation (4)

we get $c^2 + h^2 + k^2 = c^2 \cdot \frac{h^2}{a^2} + a^2$

$c^2a^2 + h^2a^2 + k^2a^2 = c^2h^2 + a^4$

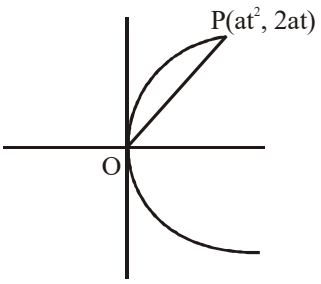
$(a^2 - c^2)h^2 + k^2a^2 = a^2(a^2 - c^2)$

$\frac{h^2}{a^2} + \frac{k^2}{a^2 - c^2} = 1$

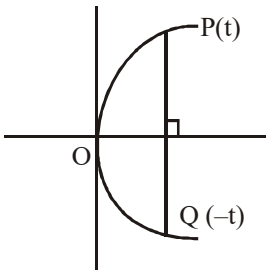
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Important Points

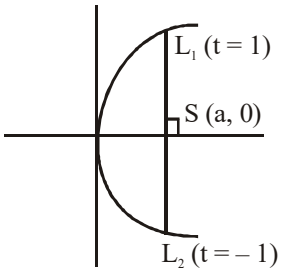
1. Slope of radius vector $m_{OP} = \frac{2at - 0}{at^2} = \frac{2}{t}$



2. For double ordinate

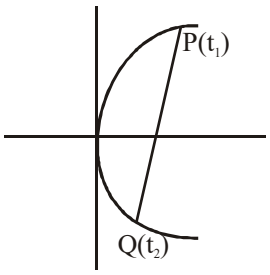


3. For latus rectum



4. Slope of line joining two points

$$m_{PQ} = \frac{2at_1 - 2at_2}{at_1^2 - at_2^2}, \quad m_{PQ} = \frac{2}{t_1 + t_2}$$

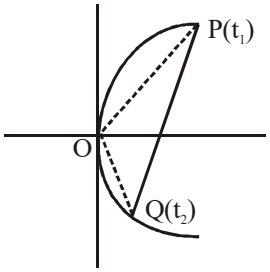


5. If chord PQ substends 90° at vertex

$(m_{OP})\ (m_{OQ}) = -1$

$$\left(\frac{2}{t_1}\right)\left(\frac{2}{t_2}\right) = -1$$

$t_1 t_2 = -4$

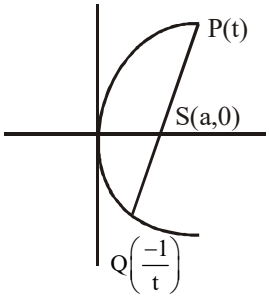


6. Length of focal chord

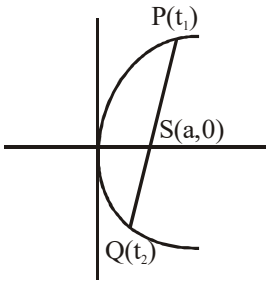
PQ = PS + QS

$= (at^2 + a) + \left(\frac{a}{t^2} + a\right)$

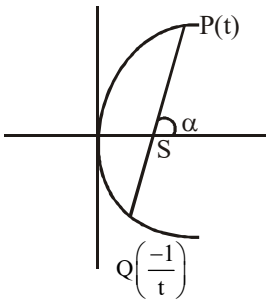
$$= a\left(t^2 + \frac{1}{t^2} + 2\right) = a\left(t + \frac{1}{t}\right)^2$$



7. Focal chord



$t_1t_2 = -1$



8. $l_{PQ} = 4a\ \text{cosec}^2\alpha$

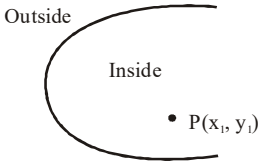
$$\frac{2}{t-\frac{1}{t}} = \tan\alpha$$

$$t-\frac{1}{t} = 2\cot\alpha$$

$$\left(t+\frac{1}{t}\right)^2 = \left(t-\frac{1}{t}\right)^2 + 4 = 4\cot^2\alpha + 4 = \text{cosec}^2\alpha$$

4 POSITION OF A POINT WRT A PARABOLA

The point (x_1, y_1) lies outside, on or inside the parabola $y^2 = 4ax$ according as the expression $y_1^2 - 4ax_1$ is positive, zero or negative.



$S_1 : y_1^2 - 4ax_1$

$S_1 < 0 \rightarrow$ Inside

$S_1 > 0 \rightarrow$ Outside

Q.1 For the parabola $y^2 = 8x$ point (2, 5) is

(A) Inside the parabola (B) Focus (C) Outside the parabola (D) On the parabola

Ans. C

Sol. $(y^2 - 8x)_{x=2,\ y=5} = (5)^2 - 8 \times 2 = 9 > 0$

$\Rightarrow$ Point (2, 5) is outside parabola $y^2 = 8x$

Q.2 Find the value of α for which the point $(\alpha - 1, \alpha)$ lies inside the parabola $y^2 = 4x$.

Ans $\alpha \in \phi$

Sol. $\because$ Point $(\alpha - 1, \alpha)$ lies inside the parabola $y^2 = 4x$

$\therefore$ $y_1^2 - 4ax_1 < 0$

$\Rightarrow$ $\alpha^2 - 4(\alpha - 1) < 0$

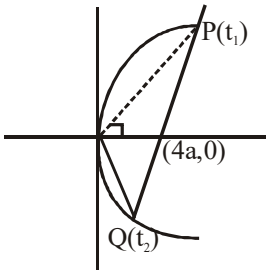
$\Rightarrow$ $\alpha^2 - 4\alpha + 4 < 0$

$(\alpha - 2)^2 < 0 \Rightarrow \alpha \in \phi$

5 CHORD JOINING TWO POINTS

Let the points are $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ then equation of chord is-

$$(y - 2at_1) = \frac{2at_2 - 2at_1}{at_2^2 - at_1^2}(x - at_1^2)$$



$$\Rightarrow y - 2at_1 = \frac{2}{t_1 + t_2}(x - at_1^2)$$

$$\Rightarrow (t_1 + t_2)\ y = 2x + 2at_1\ t_2$$

It passes through (c, 0)

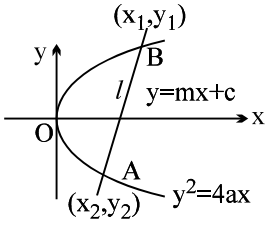
$$0 = 2c + 2at_1t_2 \quad \Rightarrow \quad t_1t_2 = \frac{-c}{a}$$

If chord PQ subtend 90º at vertex, it will always pass through (4a, 0).

$t_1t_2 = -4$

$c = 4a$

Length of chord of the parabola intercepted on $y = mx + c$



$l^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

$= (x_2 - x_1)^2 [1 + m^2]$

$= [(x_2 + x_1)^2 - 4x_1x_2][1 + m^2] \qquad \left(\therefore \frac{y_2 - y_1}{x_2 - x_1} = m\right)$

now $x_1 + x_2 = -\frac{2(cm - 2a)}{m^2}$; $x_1 x_2 = \frac{c^2}{m^2}$ from (1)

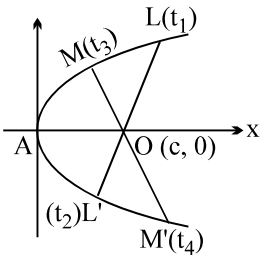
hence $l = \frac{4}{m^2}\sqrt{a(1 + m^2)(a - cm)}$

Questions on Chord joining two points

1 LOL' and MOM' are two chords of parabola $y^2 = 4ax$ with vertex A passing through a point O on its axis. Prove that the radical axis of the circles described on LL' and MM' as diameters passes through the vertex of the parabola. **[O]**

Sol. Equation of LL' $2x - (t_1 + t_2)y + 2at_1t_2 = 0$

passes through (c, 0)



$\therefore \quad t_1 t_2 = -\frac{c}{a}$; ||ly $t_3 t_4 = -\frac{c}{a}$

now circle with LL' as diameter

$$(x - at_1^2)(x - at_2^2) + (y - 2at_1)(y - 2at_2) = 0$$

$x^2 + y^2 - a(t_1^2 + t_2^2)x - 2a(t_1 + t_2)y + c^2 - 4ac = 0$ ....(1)

(as $a^2t_1^2t_2^2 = c^2$ & $4a^2t_1t_2 = 4ac$)

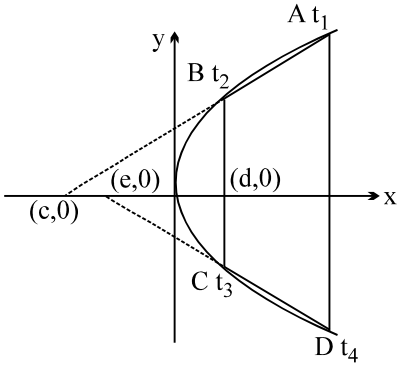
|||ly circle on MM' as diameter

$x^2 + y^2 - a(t_3^2 + t_4^2)x - 2a(t_3 + t_4)y + c^2 - 4ac = 0$ ....(2)

radical axis i.e. (1) – (2) obviously passes through the origin.

2. A quadrilateral is inscribed in a parabola $y^2 = 4ax$ and three of its sides pass through fixed points on the axis. Show that the fourth side also passes through fixed point on the axis of the parabola.

Sol. Given $t_1t_2 = -\frac{c}{a}$ ....(1) ; $t_2t_3 = -\frac{d}{a}$ ....(2)



$t_3t_4 = -\frac{e}{a}$ ....(3)

(1) × (3)

$t_1t_2t_3t_4 = \frac{ce}{a^2}$

using (2)

$t_1t_4\left(-\frac{d}{a}\right) = \frac{ce}{a^2} = \frac{-ce}{ad}$ (Constant)

$\therefore$ AD also passes through fixed point

3. A circle and a parabola $y^2 = 4ax$ intersect in four points ; show that the algebraic sum of the ordinates of the four points is zero.

Sol. Let the equation of the circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$

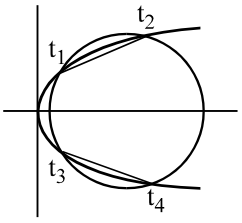
solving it with $x = at^2$, $y = 2at$

$a^4t^4 + 4a^2t^2 + 2gat^2 + 4aft + c = 0$

$a^4t^4 + 2a(2a + g)t^2 + 4aft + c = 0$

Hence, $t_1 + t_2 + t_3 + t_4 = 0$ ....(1)

or $2a(t_1 + t_2 + t_3 + t_4) = 0$ Hence proved

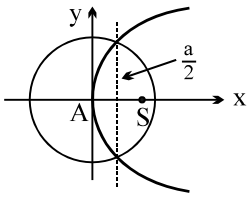


4. A circle is described whose centre is the vertex and whose diameter is three quarters of the latus rectum of a parabola, prove that the common chord of the circle and parabola bisects the distance between the vertex and the focus. **[O]**

Sol. Diameter of the circle $= \frac{3}{4}\cdot 4a = 3a$

$\therefore$ equation of the circle is $x^2 + y^2 = \frac{9a^2}{4}$

solving it with $y^2 = 4ax$



$$x^2 + 4ax - \frac{9a^2}{4} = 0$$

$$4x^2 + 16ax - 9a^2 = 0$$

$$4x^2 + 18ax - 2ax - 9a^2 = 0$$

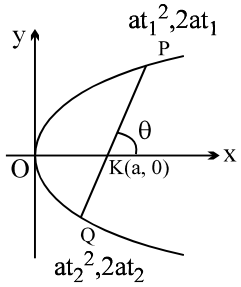
$$2x(2x + 9a) - a(2x + 9a) = 0$$

$$(2x - a)(2x + 9a) = 0$$

$\therefore \quad x = \frac{a}{2}$ or $x = -\frac{9a}{2}$ (rejected)

5. Prove that on the axis of any standard parabola there is a certain point K (2a, 0) which has the property that, if a chord PQ of the parabola be drawn through it, then $\frac{1}{PK^2} + \frac{1}{QK^2}$ is the same for all positions of the chord. **[O]**

Sol. $\frac{x-a}{\cos\theta} = \frac{y-0}{\sin\theta} = r$



$x = r\cos\theta + c$; $y = r\sin\theta$; substitute in $y^2 = 4ax$

$r^2\sin^2\theta = 4a(r\cos\theta + c)$

$r^2\sin^2\theta - (4a\cos\theta)\, r - 4ac = 0$

$r_1 + r_2 = \frac{4a\cos\theta}{\sin^2\theta}$; $r_1 r_2 = -\frac{4ac}{\sin^2\theta}$

now, $\frac{1}{PK^2} + \frac{1}{QK^2} = \frac{r_1^2 + r_2^2}{(r_1 r_2)^2} = \frac{(r_1 + r_2)^2 - 2r_1 r_2}{(r_1 r_2)^2} = \frac{\left(\frac{16a^2\cos^2\theta}{\sin^4\theta} + \frac{8ac}{\sin^2\theta}\right)\sin^4\theta}{16a^2c^2}$

$= \left(\frac{16a^2\cos^2\theta + 8ac\sin^2\theta}{16a^2c^2}\right)$ which is a constant if $c = 2a$ and constant $= \frac{1}{4a^2}$

6. If the curve $x^2 + 2y^2 = 2$ intersects the line $x + y = 1$ at two points P and Q, then the angle subtended by the line segment PQ at the origin is : **[JEE(Main)Feb.2021]**

यदि वक्र, $x^2 + 2y^2 = 2$, रेखा $x + y = 1$ को दो बिंदुओं P तथा Q पर काटता है, तो रेखा खंड PQ द्वारा मूल बिन्दु पर अंतरित कोण होगा : **[JEE(Main)Feb.2021]**

Ans. $\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{4}\right)$

Sol. H.F. $=\frac{x+y}{1}$

$x^2+2y^2=2(x+y)^2$

$x^2+4xy=0$

$L_1: x=0$ $\qquad$ $L_2: x+4y=0$

$m\to\infty$ $\qquad$ $m_2=\frac{-1}{4}$

$$\tan\theta=\left|\frac{1}{\frac{-1}{4}}\right|=4$$

$\theta=\tan^{-1}(4)$

obtuse angle $=\pi-\tan^{-1}(4)$

$$=\pi-\left(\cot^{-1}\left(\frac{1}{4}\right)\right)$$

$$=\pi-\left(\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{4}\right)\right)$$

$$=\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{4}\right)$$

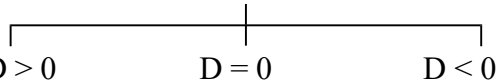
6.1 LINE AND PARABOLA

$y=mx+c$ & $y^2=4ax$

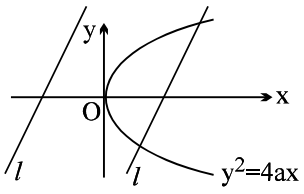
solving them together,

$(mx+c)^2-4ax=0$

$m^2x^2+2(cm-2a)x+c^2=0$ ....(1)



$D>0$	$D=0$	$D<0$
line is a secant	Tangent	$\overline{T}\cap\overline{S}$



$\Downarrow$

now, $c=\frac{a}{m}$ gives the condition of tangency

hence $y=mx+\frac{a}{m}$ is always a tangent to $y^2=4ax$ for all $m\neq 0$.

6.2 DIRECTOR CIRCLE

Locus of the point of intersection of the perpendicular tangents to the parabola $y^2 = 4ax$ is called the **director circle.** It's equation is **$x + a = 0$** which is parabola's own directrix.

Find equation of director circle for parabola $y^2 = 4ax$

It is the locus of the point P from which tangents are drawn to the parabola $y^2 = 4ax$ having slopes m_1 and m_2 such that $m_1m_2 = -1$

Equation of tangent to $y^2 = 4ax$ is $y = mx + a/m$

Let it passes through P(h, k)

$\therefore$ $m^2h - mk + a = 0$ Its roots are m_1 and m_2

$$m_1m_2 = -1 \Rightarrow \frac{a}{h} = -1 \Rightarrow \quad h = -a$$

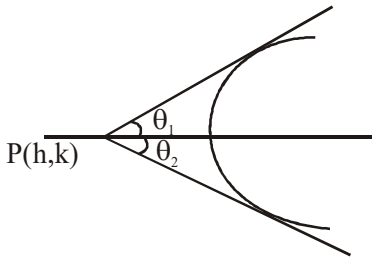
$\therefore$ locus of P(h, k) is **$x + a = 0$**

1. From a variable point P tangents are drawn to the parabola $y^2 = 4ax$ such they are making angle of θ_1 and θ_2 with x-axis. Find locus of P if

(i) sum of their slopes is 5.

Ans. $5x - y = 0$

Sol.



Sum of their slopes is 5.

Let equation of tangent be

$$y = mx + \frac{a}{m}$$

$$k = mh + \frac{a}{m}$$

$$mk = m^2h + a$$

$$m^2h - mk + a = 0 \left\langle \begin{matrix} m_1 \\ m_2 \end{matrix} \right.$$

$$m_1 + m_2 = \frac{k}{h}$$

$$5 = \frac{k}{h}$$

$$k = 5h$$

$$5x - y = 0$$

(ii) sum of reciprocals of their slopes is 3.

Ans. $y = 3a$

Sol. $m_1 + m_2 = \frac{k}{h}$

$m_1 m_2 = \frac{a}{h}$

$\frac{1}{m_1} + \frac{1}{m_2} = 3$

$\frac{m_2 + m_1}{m_1 m_2} = 3$

$\frac{\frac{k}{h}}{+\frac{a}{h}} = 3$

$k = 3a$

$y = 3a$

(iii) $\theta_1 + \theta_2 = \frac{\pi}{4}$

Ans. $y = x - a$

Sol. $\theta_1 + \theta_2 = \frac{\pi}{4}$

$\frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1 \tan\theta_2} = 1$

$\frac{m_1 + m_2}{1 - m_1 m_2} = 1$

$\frac{k}{h} = 1 - \frac{a}{h}$

$k = h - a$

$y = x - a$

2. Find condition if line is tangent to the parabola.

(i) $y^2 = 12x$, $y = 5x + c$

Ans. $c = \frac{3}{5}$

Sol. $y^2 = 12x$, $y = 5x + c$

$c = \frac{a}{m}$ $\qquad 4a = 12$

$c = \frac{3}{5}$ $\qquad a = 3$

(ii) $y^2 = 12x, y = ax + 4.$

Ans. $a = \frac{3}{4}$

Sol. $y^2 = 12x, y = ax + 4$

$c = \frac{a}{m}$

$4 = \frac{3}{a}$

$a = \frac{3}{4}$

(iii) $y^2 = 16x, 3x + ay = 5$

Ans. $\phi\left(a^2 = \frac{-15}{4}\right)$

Sol. $y^2 = 16x, 3x + ay = 5$

$4a = 16$ $\qquad y = -\frac{3x}{a} + \frac{5}{a}$

$a = 4$

$\frac{5}{a} = \frac{4 \times a}{-3}$

$a^2 = -\frac{15}{4}$

7.1 TANGENT

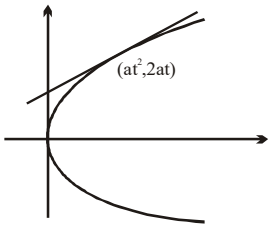
1. Point Form :

The equation of tangent to the parabola $y^2 = 4ax$ at the point $P(x_1, y_1)$ is $yy_1 = 2a(x + x_1)$ or $T = 0$

2. Parametric Form : The equation of the tangent to the parabola at t i.e. $P(at^2, 2at)$ is

$ty = x + at^2$

3. Slope Form : The equation of the tangent of the parabola $y^2 = 4ax$ is $\quad y = mx + \frac{a}{m}$



$\text{COT } c = \frac{a}{m}$ $\qquad m = \frac{1}{t}$

7.2 NORMAL

1. Point Form : The equation to the normal at the point (x_1, y_1) of the parabola $y^2 = 4ax$ is given by

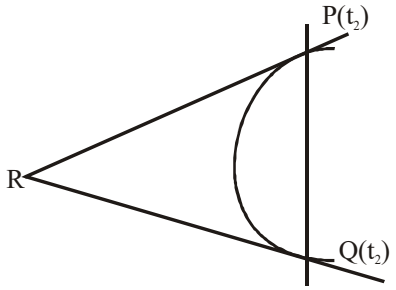
$y - y_1 = \frac{-y_1}{2a}(x - x_1)$

2. **Parametric Form :** The equation to the normal at the point $(at^2, 2at)$ is $y + tx = 2at + at^3$

3. **Slope Form :** Equation of normal in terms of slope m is $y = mx - 2am - am^3$.
Condition of a line $y = mx + c$ to be a normal is $c = -2am - am^3$; $m = -t$

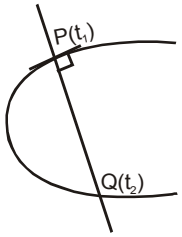
Important Notes

1. Point of intersection of tangents at points $P(t_1)$and $Q(t_2)$ of parabola is $R(at_1t_2, a(t_1+t_2))$



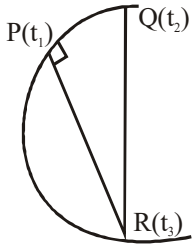
2. Point of intersection of normals at t_1 & t_2 is $(a(t_1^2 + t_2^2 + t_1t_2 + 2), -a\,t_1\,t_2\,(t_1 + t_2))$.

3. If the normals to the parabola $y^2 = 4ax$ at the point t_1, meets the parabola again at the point t_2, then $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$.



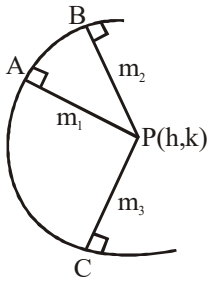
4. Tangents at the end of focal chord are perpendicular to each other & meet at directrix.

5. Normal at $P(t_1)$ & $Q(t_2)$ meet at parabola at $R(t_3)$ then $t_1t_2 = 2$; $t_1 + t_2 = -t_3$



7.3 NORMAL TO PARABOLA FROM A POINT

Normal to parabola from a point.



$A\left(am_1^2, -2am_1\right)$

$B\left(am_2^2, -2am_2\right)$

$C\left(am_3^2, -2am3\right)$

Equation of normal $y = mx - 2am - am^3$

It passes throght P (h, k)

$k = mh - 2am - am^3$

$am^3 - m(h-2a) + k = 0$ $\rightarrow m_1, m_2, m_3$

Note :

1. Normal to a parabola from a point P, atmost 3 normals can be drawn & atleast 1 normal can be drawn.
2. Algebraic sum of slopes of three concurrent normals is zero.
3. A, B & C are called co-normal points.
4. Sum of ordinates of A, B & C is zero.
5. Centroid of triangle formed by three co-normal points. will lie on axis of parabola.
6. If three are 3 distinct normals from a point. P (h, k) to the parabola $y^2 = 4ax$ then required condition is $27ak^2 < 4(h-2a)^3$.

Questions on Tangent & Normal

1. Find equation of tangent to the parabola $y^2 = 16x$ whose slope is 2 and then find point of tangency also.

Ans. $y = 2x + 2$; (1,4)

Sol. $y^2 = 16x$

$4a = 16$

$a = 4$

$$y = 2x + \frac{4}{2}$$

$y = 2x + 2$

$$t = \frac{1}{m} = \frac{1}{2}$$

POT $(at^2, 2at)$

$$\left(4\times\frac{1}{4}, 2\times4\times\frac{1}{2}\right)$$

(1, 4)

2. Find equation of tangent to the parabola $y^2 = 12x$

(i) if point of tangency is (3,6)

Ans. $y = x + 3$

Sol. $$6y = 12\left(\frac{x+3}{2}\right)$$

$6y = 6x + 3$

$y = x + 3$

(ii) If tangent is perpendicular to the line $3x + 4y = 5$

Ans. $4x - 3y + \frac{27}{4} = 0$

Sol. Slope $= -\frac{3}{4}$

Slope of tangent $= \frac{4}{3}$

Equation $y = \frac{4}{3}x + \frac{3 \times 3}{4}$

$y = \frac{4}{3}x + \frac{9}{4}$

$12y = 4 \times 4x + 27$

$4x - 3y + \frac{27}{4} = 0$

(iii) If tangent passes through (2, 5)

Ans. $y - 5 = m(x - 2)$

$m = 1$, or $\frac{3}{2}$

Sol. **Method I :**

$y - 5 = m(x - 2)$

$mx - y = 5 - 2m$

$5 - 2m = \frac{3}{m}$

$2m^2 - 5m + 3 = 0$

$(m - 1)(2m - 3) = 0$

$m = 1, \frac{3}{2}$

Method II :

$y = mx + \frac{3}{m}$

$(2, 5)$

$5 = 2m + \frac{3}{m}$

$m = 1, \frac{3}{2}$

Method III :

$ty = x + at^2$

$(2, 5)$

$5t = 2 + 3t^2$

$t = 1, \frac{2}{3}$

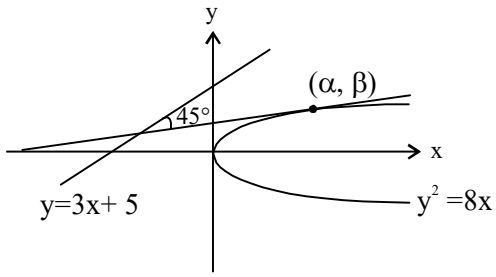
3. A tangent to the parabola $y^2 = 8x$ makes an angle of 45° with the line $y = 3x + 5$. Find its equation and also its point of contact.

Ans. $2x + y + 1 = 0$ at $(1/2, -2)$; $x - 2y + 8 = 0$ at $(8, 8)$]

Sol. $y^2 = 8x$

$4a = 8$

$a = 2$



Let equation of tangent be $y = mx + \frac{2}{m}$

$\tan 45^\circ = \left|\frac{m-3}{1+3m}\right|$

$\pm(1 + 3m) = (m - 3)$

Taking positive sine

$1 + 3m = m - 3$

$2m = -4m = -2$

$\boxed{m = -2} \Rightarrow$ eqn. of tangent is $2x + y + 1 = 0$

$\Rightarrow t = -1/2 \quad \Rightarrow$ P.O.C. $\equiv (2t^2, 4t) \equiv (1/2, -2)$

taking nagitive sine

$-1 - 3m = m - 3$

$4m = 2$

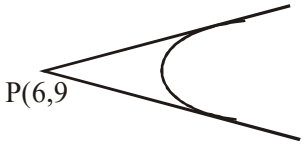
$\boxed{m = \frac{1}{2}} \Rightarrow$ eqn. of tangent is $x - 2y + 8 = 0$

$\Rightarrow t = 2 \quad \Rightarrow$ P.O.C. $\equiv (2t^2, 4t) \equiv (8, 8)$

4. From point (6,9) tangents are drawn to the parabola $y^2 = 12x$. Find point of tangency.

Ans. P(3,6), Q (12, 12)

Sol.



$y = mx + \frac{3}{m}$

$\because (6,9)$

$9 = 6m + \frac{3}{m}$

$2m^2 - 3m + 1 = 0$

$m = 1, \frac{1}{2} \qquad \Rightarrow t = 1, 2$

P. O.C. $\equiv (3t^2, 6t)$

If $t = 1 \Rightarrow (3, 6)$

If $t = 2 \Rightarrow (12, 12)$

5. If line $y = 2x + \lambda$ is normal to the parabola $y^2 = 4x$ then find λ and foot of normal.

Ans. $\lambda = -12$; foot of normal P(4, –4)

Sol. $y = 2x + \lambda$ $\qquad y^2 = 4x$

Slope = 2 $\qquad a = 1$

$m = -t$

$2 = -t$

$t = -2$

Point $(at^2, 2at)$

$(4, -4)$

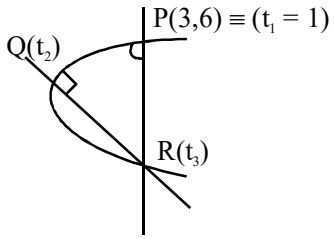
$-4 = 8 + \lambda$

$\lambda = -12$

6. Normals to the parabola $y^2 = 12x$ at P(3,6) and Q meet the parabola again at R. Find Q and R.

Ans. Q(12,12), R(27,–18)

Sol.



$y^2 = 12x$

$a = 3$

We know that $t_1 t_2 = 2$

$\Rightarrow t_2 = 2$

$t_1 + t_2 = -t_3$

$1 + 2 = -t_3$

$t_3 = -3$

$P(3, 6) = P(at_1^2, 2at_1)$

$2at_1 = 6$

$6t_1 = 6$

$t_1 = 1$

$a(at_2^2, 2at_2) = (3 \times 4, 2 \times 3 \times 2)$

$= (12, 12)$

$R(at_3^2, 2at_3) = (3 \times 9, 2 \times 3 \times (-3))$

$= (27, -18)$

7. If a chord which is normal to the parabola $y^2 = 4ax$ at one end subtend a right angle at the vertex, prove that it is inclined at an angle $\tan^{-1}\sqrt{2}$ to the axis and the normal chord passes through (4a, 0).

Sol. $y = mx - 2am - am^3$; homogenise

$P(at_1^2, 2at_1)$ $Q(at_2^2, 2at_2)$

$$y - 2at_1 = \frac{2at_2 - 2at_1}{at_2^2 - at_1^2}\left(x - at_1^2\right)$$

$(t_1 + t_2)y = 2x + 2at_1t_2$

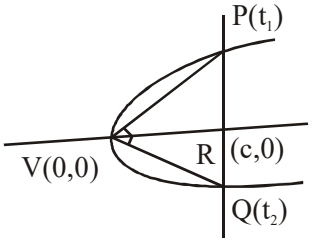
$(c, 0)$: $0 = 2c + 2at_1t_2$

$$t_1t_2 = -\frac{c}{a} \qquad \{\because\ t_1t_2 = -4\}$$

$-4a = -c$

$c = 4a$

$R(4a, 0)$



Equation of normal

$y = mx - 2am - am^3$

$(4a, 0)$ $0 = 4am - 2am - am^3$

$am^3 - 2am = 0$

$am(m^2 - 2) = 0$

$m = 0$ (Not possible) $m = \pm\sqrt{2}$

$\tan\theta = \sqrt{2}$

$\theta = \tan^{-1}\left(\sqrt{2}\right)$

8. From a variable point P three normals are drawn to the parabola $y^2 = 4ax$. Find locus of P if

(i) Sum of slopes of two normals is zero.

Ans. $k = 0$, $y = 0$

Sol. Sum of slopes of two normals is zero

$$am^3 - m(h - 2a) + k \left\langle \begin{matrix} m_1 \\ m_2 \\ m_3 \end{matrix} \right.$$

$m_1 + m_2 + m_3 = 0$

$m_3 = 0$

$k = 0$

$y = 0$

(ii) Two normals are perpendicular to each other.

Ans. $y^3 - axy + 3a^2y = 0$

Sol. two normals are $\perp$ to each other.

$$m_1m_2m_3 = -\frac{k}{a}$$

$$(-1)m_3 = -\frac{k}{a}$$

$$m_3 = \frac{k}{a}$$

$$a\left(\frac{k^3}{a^3}\right) - \frac{k}{a}(h-2a) + k = 0$$

$$\frac{k^3}{a^2} - \frac{kh}{a} + 3k = 0$$

$\Rightarrow y^3 - axy + 3a^2y = 0$

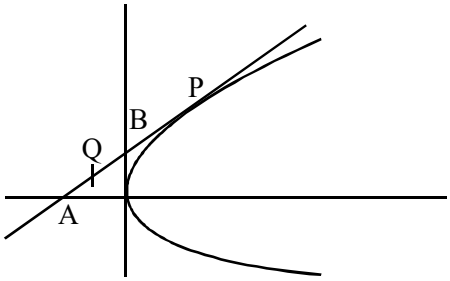
9. Find locus of mid-point of portion of the tangent, to the parabola $y^2 = 4ax$, intercepted between co-ordinate axis.

Ans. $ax + 2y^2 = 0$

Sol. $ty = x + at^2$

$A(-at^2, 0)$

$B(0,at)$



$$Q \equiv \left(-\frac{at^2}{2}, at\right)$$

$$h = -\frac{at^2}{2} \qquad k = \frac{at}{2}$$

$$h = -\frac{a}{2}\left(\frac{2k}{a}\right)^2 \qquad t = \frac{2k}{a}$$

$$h = -\frac{a}{2} \times \frac{4k^2}{a^2}$$

$ah = -2k2$

$ah + 2k^2 = 0$

$ax + 2y^2 = 0$

10. Find locus of foot of perpendicular from origin on a variable tangent to the standard parabola $y^2 = 4ax$.

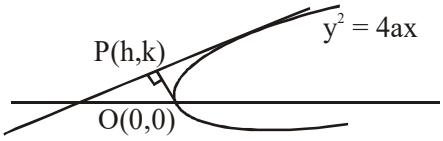
Ans. $x(x^2 + y^2) = -ay^2$

Sol. Let tangent be $y = mx + \frac{a}{m}$

Passes through (h, k)

$$k = mh + \frac{a}{m}$$

$$k = \frac{-h}{k}h + \frac{ak}{-h}$$



$$k = \frac{-h^3 - ak^2}{kh}$$

$k^2h = -h^3 - ak^2$

$h^3 + k^2h = -ak^2$

$x^3 + xy^2 = -ay^2$

$x(x^2 + y^2) = -ay^2$

7.4 OTHER QUESTIONS ON TANGENT & NORMAL

1. Let the tangent to the parabola $y^2 = 4ax$ meets the axis in T and the tangent at the vertex A is y. If the rectangle TAYG is completed find the locus of G.

Ans. $y^2 + ax = 0$

Sol

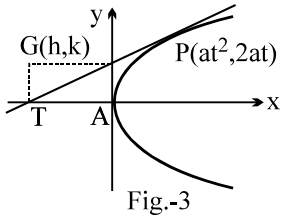


Fig.-3

tangent at P

$ty = x + at^2$

T : (y, 0) $\quad x = -at^2$

$T(-at^2, 0)$

Y : (x = 0) : $\quad ty = at^2$

Y(0, at)

$C_1 : (h, k) : C_1 (-at^2, at)$

$h = -at^2, k = at$

$$h = -a\left(\frac{k}{a}\right)^2$$

$$h = -a \times \frac{k^2}{a^2}$$

$ah = -k^2$

$ax + y^2 = 0$

2. The normal at any point P for parabola $y^2 = 4ax$ meets the axis in G and the tangent at the vertex in Y, if A be the vertex and the rectangle GAYQ be completed, prove that the equation to the locus of Q is $\mathbf{x^3 = 2ax^2 + ay^2}$.

Sol.

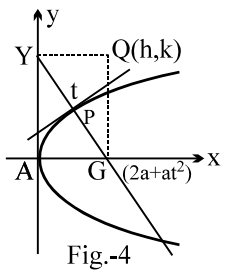


Fig.-4

$y + tx = 2at + at^3$
$G \equiv (2a + at^2, 0)$
$Q \equiv (h, k)$

$\Rightarrow \boxed{h = 2a + at^2}$

$\frac{h}{a} = 2 + t^2$

$t^2 = \frac{h}{a} - 2$ .....(1)

$Y \equiv (0, 2at + at^3)$
$Q \equiv (h, k)$

$\boxed{k = 2at + at^3}$

$k = t(2a + at^2)$

$k = at(2 + t^2)$

$k = at\left(\frac{h}{a}\right)$

$k = th$
$k^2 = t^2h^2$
Using (1)

$k^2 = \left(\frac{h}{a} - 2\right)h^2$

$y^2 = \left(\frac{h - 2a}{a}\right)x^2$

$ay^2 = (h - 2a)x^2$
$ay^2 = (x - 2a)x^2$
$x^3 = 2ax^2 + ay^2$

3. Prove that the equation to the circle, which passes through the focus and touches the parabola $y^2 = 4ax$ at the point $(at^2, 2at)$ is $x^2 + y^2 - ax(3t^2 + 1) - ay(3t - t^3) + 3a^2t^2 = 0$. If t varies, prove also that the locus of its centre is the curve $27ay^2 = (2x - a)(x - 5a)^2$.

Sol. Tangent at $P(at^2, 2at)$ is
$ty = x + at^2$
family of circle type IV
$(x - at^2)^2 + (y - 2at)^2 + \lambda(x - ty + at^2) = 0$
it passes through (a, 0)
$a^2 + a^2t^4 - 2a^2t^2 + 4a^2t^2 - a\lambda - at^2\lambda = 0$
$a^2 + a^2t^4 + 2a^2t^2 - a\lambda - at^2\lambda = 0$
$\lambda(1 + t^2) = a + at^4 + 2at^2$

$\lambda = \frac{a\left(1 + 2t^2 + t^4\right)}{1 + t^2} = a\left(1 + t^2\right)$

$\Rightarrow (x - at^2)^2 + (y - 2at)^2 + a(1 + t^2)(ty - x - at^2) = 0$
$x^2 + y^2 - ax(3t^2 + 1) - ay(3t - t^3) + 3a^2t^2 = 0$

Center (h, k) : $2h = a(3t^2 + 1)$ ...(1)

$2k = a(3t - t^3)$

$\Rightarrow t^2 = \frac{\frac{2h}{a} - 1}{3} = \frac{2h - a}{3a}$

$2k = at(3 - t^2)$

$$2k = at\left(3 - \frac{2h - a}{3a}\right)$$

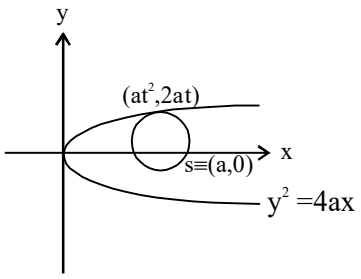
$$2k = t\left(\frac{10a - 2h}{3}\right)$$

$$t = \frac{6k}{10a - 2h} = \frac{3k}{5a - h}$$

from (i)

$$2h = a\left(3\left(\frac{3k}{5a - h}\right)^2 + 1\right)$$

$\Rightarrow 27ay^2 = (2x - a)(x - 5a)^2$



4. Through a point P are drawn tangents PQ and PR to a parabola and circles are drawn through the focus to touch the parabola in Q and R respectively. Prove that the common chord of these circles passes through the centroid of the triangle PQR.

Sol. Equation of circle touching the parabola $y^2 = 4ax$ at P and passing through its focus is

$$x^2 + y^2 - a(3t_1^2 + 1)x - a(3t_2 - t_1^3)y + 3a^2 t_1^2 = 0 \quad ....(1)$$

|||ly circle touching at Q is

$$x^2 + y^2 - a(3t_2^2 + 1)x - a(3t_2 - t_2^3)y + 3a^2 t_2^2 = 0 \quad ....(2)$$

common chord between (1) and (2) is

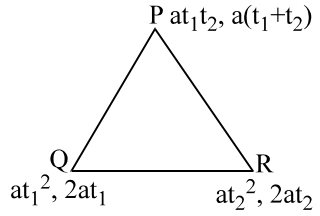
$$3(t_2 + t_1)x + [3 - (t_1^2 + t_2^2 + t_1t_2)]y - 3a(t_2 + t_1) = 0 \quad ....(3)$$

Now centroid of the circle PQR is

$$x_g = \frac{a}{3}(t_1^2 + t_2^2 + t_1t_2)$$

$$y_g = a(t_1 + t_2)$$

these two satisfies equation (3)



5. A pair of tangents are drawn which are equally inclined to a straight line $y = mx + c$ whose inclination to the axis is α, prove that the locus of their point of intersection is the straight line $y = (x - a)\tan 2\alpha$. **[O]**

Sol. We have $\theta = \alpha_1 - \alpha = \alpha - \alpha_2$

$\therefore \quad 2\alpha = \alpha_1 + \alpha_2$

$$\tan 2\alpha = \tan(\alpha_1 + \alpha_2) = \frac{m_1 + m_2}{1 - m_1m_2}$$

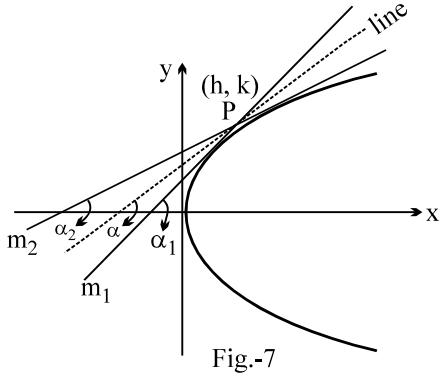


Fig.-7

but (h, k) lies on $y = mx + \frac{a}{m} \Rightarrow m^2h - km + a = 0$

hence $m_1 + m_2 = \frac{k}{h}$ and $m_1 m_2 = \frac{a}{h}$; $\tan 2\alpha = \frac{k/h}{1 - a/h} = \frac{k}{h - a}$

$\Rightarrow \quad y = (x - a) \tan 2\alpha$

6. If tangents are drawn to $y^2 = 4ax$ from any point P on the parabola $y^2 = a(x + b)$ then show that the normals drawn at their point of contact meet on a fixed line. **[O]**

Sol. Let P $\left(at_1t_2, a(t_1+t_2)\right)$; P must satisfy $y^2 = a(x + b)$

Hence $a^2 (t_1 + t_2)^2 = a ((at_1t_2) + b))$

$a (t_1^2 + t_2^2 + t_1t_2) = b$ is true.

Now point of intersection of normals at t_1 and t_2 are

$h = a(t_1^2 + t_2^2 + t_1t_2 + 2)$ ....(1)

and $k = - at_1t_2 (t_1 + t_2)$ ....(2)

from (1) $h = b + 2a \Rightarrow x = b + 2a$

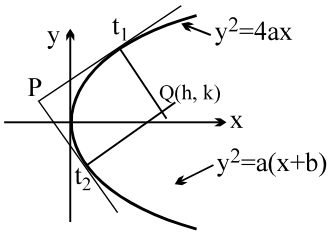


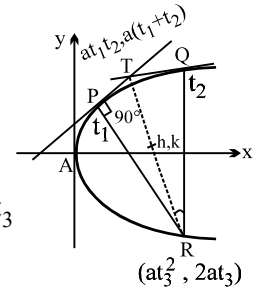
Fig.-6 (a)

7. Prove that the equation to the circle passing through the points $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ and the intersection of the tangents to the parabola at these points is

$x^2 + y^2 - a [(t_1 + t_2)^2 + 2] x - a (t_1 + t_2)(1 - t_1t_2) y + a^2t_1t_2(2 - t_1t_2) = 0$. **[O]**

8. TP and TQ are tangents to the parabola $y^2 = 4ax$ and the normals at P and Q meet at a point R on the curve, prove that the centre of the circle circumscribing the triangle TPQ lies on the parabola **$2y^2 = a (x - a)$**.

Sol. We have $t_1t_2 = 2$ ....(1)



now $2k = a(t_1 + t_2) + 2at_3 = a(t_1 + t_2 + t_3) + at_3$

$\therefore \quad t_3 = \dfrac{2k}{a}$

also $\quad 2h - 2a + at_3^2$

$$2h = 2a + a \cdot \frac{4k^2}{a^2} \quad \Rightarrow \quad 2h = 2a + \frac{4k^2}{a} \quad \Rightarrow \quad ha = a^2 + 2k^2$$

$\Rightarrow \quad 2y^2 = a(x - a)$ Henced proved.

9. Find the locus of a point which is such that **[O]**

(a) two of the normals drawn from it to the parabola are at right angles,

(b) the three normals through it cut the axis in points whose distances from the vertex are in arithmetical progression.

Ans : (a) $y^2 = a(h - 3a)$; (b) $27ay^2 = 2(x - 2a)^3$]

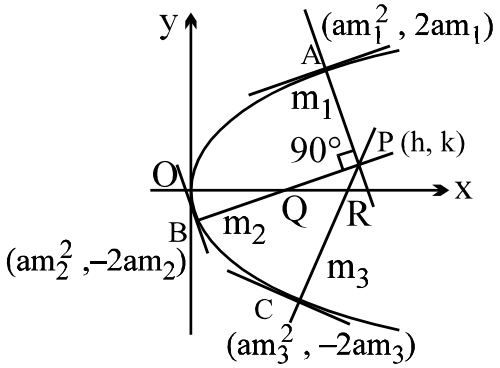
Sol. (a) we have $m_1 m_2 = -1$

also $\quad m_1 m_2 m_3 = -\dfrac{k}{a}$

$\therefore \quad m_3 = \dfrac{k}{a}$

put $m_3 = -\dfrac{k}{a}$ is a root of

$am^3 + (2a - h)m + k = 0$



(b) $\quad y = mx - 2am - am^3$

hence $\quad 2a + am_1^2, 2a + am_2^2, 2a + am_3^2$

$\therefore \quad 2m_2^2 = m_1^2 + m_3^2$

$$3m_2^2 = m_1^2 + m_2^2 + m_3^2 = (m_1 + m_2 + m_3)^2 - 2\left(\sum m_1 m_2\right) = \frac{2(h - 2a)}{a}$$

$$m_2^2 = \frac{2(h - 2a)}{3a}$$ which is root of $am^3 + (2a - h)m + k = 0$

10. A circle circumscribing the triangle formed by three co–normal points(w.r.t. (h,k)) passes through the vertex of the parabola and its equation is, $2(x^2 + y^2) - 2(h + 2a)x - ky = 0$.

8. RULES FOR TRANSFORMATION

$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
x	x	y	y
y	y	x	x
a	−a	a	−a
m	m	$\frac{1}{m}$	$\frac{1}{m}$
$T: y = mx + \frac{a}{m}$	$y = mx - \frac{a}{m}$	$x = \frac{y}{m} + am$	$x = \frac{y}{m} - am$
$N: y = mx - 2am - am^3$	$y = mx + 2am + am^3$	$x = \frac{y}{m} - \frac{2a}{m} - \frac{a}{m^3}$	$x = \frac{y}{m} + \frac{2a}{m} + \frac{a}{m^3}$

$(y-\beta)^2 = 4a(x-\alpha)$	$(x-\alpha)^2 = 4a(y-\beta)$
$x-\alpha$	$y-\beta$
$y-\beta$	$x-\alpha$
a	a
m	$\frac{1}{m}$
$y-\beta = m(x-\alpha) + \frac{a}{m}$	$y-\beta = m(x-\alpha) - am^3$
$y-\beta = m(x-\alpha) - 2am - am^3$	$y-\beta = m(x-\alpha) + 2a + \frac{a}{m^2}$

9. COMMON TANGENTS

For common tangent, first assume its equation as $y = mx + c$ & then apply **COT** w.r.t both curves.

1. Find equation of common tangent

(i) $y^2 = 4x$, $x^2 = 4y$

Ans. $x + y + 1 = 0$

(ii) $y^2 = 4x$, $x^2 + y^2 = \frac{1}{2}$

Ans. $x - y + 1 = 0$, $x + y + 1 = 0$

(iii) $y^2 = 4x$, $x^2 = -32y$

Ans. $x - 2y + 4 = 0$

Sol. (i) Let equation of tangent for $y^2 = 4x$ be $y = mx + \frac{1}{m}$

Since it is tangent to $x^2 = 4y$

$\Rightarrow x^2 = 4\left(mx + \frac{1}{m}\right)$

$\Rightarrow mx^2 - 4m^2x - 4 = 0$

Now, $D = 0$

$\Rightarrow 16m(m^3 + 1) = 0$

$\Rightarrow m = 0$(Not possible) $\quad m = -1$

$\Rightarrow y = -x - 1$

(ii) $y^2 = 4x$ and $x^2 + y^2 = \frac{1}{2}$

Let equation of common tangent be $y = mx + \frac{1}{m}$

Since, p = r

$$\left|\frac{\frac{1}{m}}{\sqrt{1+m^2}}\right| = \frac{1}{\sqrt{2}}$$

$\Rightarrow m = 1$ $\qquad$ $m = -1$

$x - y + 1 = 0$, $\qquad$ $x + y + 1 = 0$

(iii) $y^2 = 4x$, $x^2 = -32y$

$$c = \frac{1}{m}$$

$$y = mx + \frac{1}{m}$$

C.O.T. $\qquad x^2 = -32y$

$c = am^2$

$$\frac{1}{m} = 8m^2$$

$$m^3 = \frac{1}{8}$$

$$m = \frac{1}{2}$$

$$y = \frac{x}{2} + 2$$

$x - 2y + 4 = 0$

2. Find equation of normal to $y^2 = 4x$ which is tangent to $x^2 + y^2 = \frac{9}{2}$.

Ans. $y = x - 3$, $\qquad y + x = 3$

Sol. $y = mx - 2am - am^3$

$y = mx - 2m - m^3$

C.O.T.

$$\left|\frac{2m + m^3}{\sqrt{1+m^2}}\right| = \frac{3}{\sqrt{2}}$$

$\Rightarrow m^2 = 1$

$\Rightarrow m = \pm 1$

$\Rightarrow m = 1$ $\qquad$ $\Rightarrow m = -1$

$y = x - 3$ $\qquad$ $y = -x + 3$

3. Prove that the two parabolas $y^2 = 4ax$ and $y^2 = 4c(x - b)$ cannot have a common normal, other than the axis, unless $\frac{b}{a-c} > 2$.

Sol. Equation of normal at $y^2 = 4ax$

$y = mx - 2am - am^3$ ....(1)

and normal to $y^2 = 4c(x - b)$ with the same slope

$y = m(x - b) - 2cm - cm^3$ ....(2)

If (1) and (2) denotes the same line, we have

$-2am - am^3 = -bm - 2cm - cm^3$

$2a + am^2 = b + 2c + cm^2$ $(m \neq 0)$

$(a - c)m^2 + 2(a - c) = b$

$m^2 = \frac{b}{a-c} - 2 > 0 \Rightarrow \frac{b}{a-c} > 2$

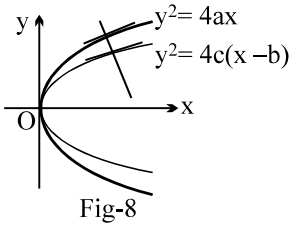


Fig-8

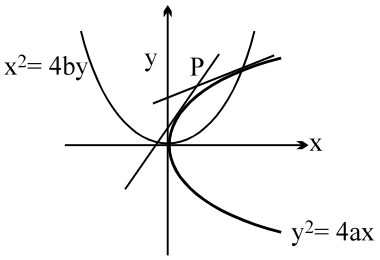
4. If $a^2 > 8b^2$, prove that a point can be found such that the two tangents from it to the parabola $y^2 = 4ax$ are normals to the parabola $x^2 = 4by$.

Sol. Equation of tangent to $y^2 = 4ax$ is

$y = mx + \frac{a}{m}$ ....(1)

Normal to $x^2 = 4by$ having slope m is

$y = mx + 2b + \frac{b}{m^2}$ ....(2)



comparing (1) and (2)

$2b + \frac{b}{m^2} = \frac{a}{m}$

$2bm^2 - am + b = 0$

for two distinct m, $D > 0$

$\therefore \quad a^2 - 8b^2 > 0 \quad \Rightarrow \quad a^2 > 8b^2$

10. COMMON CONCEPTS

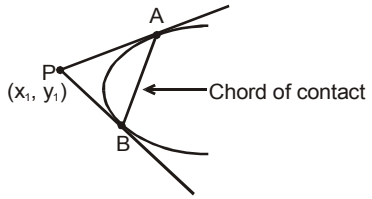
1. Chord with a given middle point

Equation of the chord of the parabola $y^2 = 4ax$ whose middle point is (x_1, y_1) is $y - y_1 = \frac{2a}{y_1}(x - x_1)$.

This reduced to $T = S_1$, where $T \equiv yy_1 - 2a(x + x_1)$ & $S_1 \equiv y_1^2 - 4ax_1$.

2. Chord of contact:

Equation to the chord of contact of tangents drawn from a point $P(x_1, y_1)$ is $T = 0 \Rightarrow yy_1 = 2a(x + x_1)$.



Note :

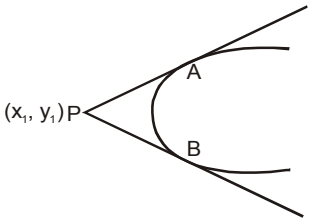
(i) Lengths of the chord of contact is $\frac{1}{a}\sqrt{(y_1^2-4ax_1)(y_1^2+4a^2)}$

(ii) Area of triangle formed by tangents drawn from (x_1, y_1) and their chord of contact is $\frac{1}{2a}(y_1^2-4ax_1)^{3/2}$

3. Pair of tangents:

The equation to the pair of tangents which can be drawn from any point (x_1, y_1) to the parabola $y^2 = 4ax$ is given by: $SS_1 = T^2$ where :

$S \equiv y^2 - 4ax$; $S_1 = y_1^2 - 4ax_1$; $T \equiv y\,y_1 - 2a(x + x_1)$.



4. Polar

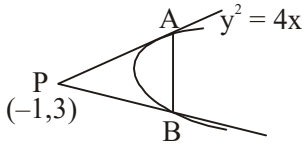
Equation of polar is $T = 0$

Questions

1. From point P(–1, 3) tangents are drawn to the parabola $y^2 = 4x$ to touch it at A and B. Find equation of AB.

Ans. $2x - 3y - 2 = 0$

Sol. equation of C.O.C. $T = 0$



$$3y = 4\left(\frac{x-1}{2}\right)$$

$3y = 2(x - 1)$

$3y = 2x - 2$

$2x - 3y - 2 = 0$

2. Line $3x + y = 6$ intersects the parabola $y^2 = 4x$ at A and B. Find the coordinates of the point of intersection of the tangents drawn at A and B.

Sol $\left(-2, -\frac{2}{3}\right)$

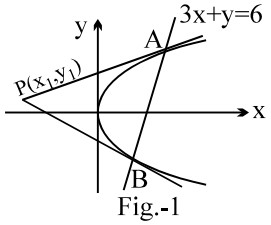


Fig.-1

Equation of C.O.C.

$$yy_1 = 4\left(\frac{x + x_1}{2}\right)$$

$yy_1 = 2(x + x_1)$

$2x - yy_1 + 2x_1 = 0$ ...(1)

$3x + y - 6 = 0$

$$\frac{2}{3} = \frac{-y_1}{1} = \frac{2x_1}{-6}$$

$$\frac{2}{3} = \frac{2x_1}{-6} \qquad \frac{-y_1}{1} = \frac{+2(-2)}{-6}$$

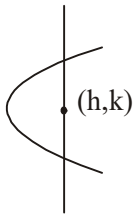
$$x_1 = -2 \qquad y_1 = -\frac{2}{3}$$

$$P\left(-2, -\frac{2}{3}\right)$$

3. Find mid-point of chord $4x + 3y + 1 = 0$ of parabola $y^2 = 8x$.

Ans. $(2, -3)$

Sol.



equation of chord whose mid-point (h, k)

$T = S_1$

$$ky - 8\left(\frac{x + h}{2}\right) = k^2 - 8h$$

$ky - 4(x + h) = k^2 - 8h$

$ky - 4x - 4h = k^2 - 8h$

$4x - ky + k^2 - 4h = 0$ ...(1)

$4x + 3y + 1 = 0$ ...(2)

$$\frac{4}{4} = \frac{-k}{3} = \frac{k^2 - 4h}{1}$$

$k = -3$

$k^2 - 4h = 1$

$9 - 4h = 1$

$h = 2$

$(2, -3)$

4. Find the locus of the middle point of the chords of the parabola $y^2 = 4ax$ which
   (i) passes through the focus
   (ii) are normal to the parabola
   (iii) subtend a constant angle α at the vertex
   (iv) are of given length (say $2l$)
   (v) are such that the normals at their extremities meet on the parabola

Ans (i) $y^2 = 2a(x - a)$ (ii) $y^2(y^2 - 2ax + 4a^2) + 8a^4 = 0$

(iii) $(8a^2 + y^2 - 2ax)^2\tan^2\alpha = 16a^2(4ax - y^2)$ (iv) $(y^2 + 4a^2)(4ax - y^2) = 4l^2a^2$

(v) $y^2 = 2a\,(x + 2a)$

Ans. $(y^2 + 4a^2)(4ax - y^2) = 4\ell^2a^2$

Sol. **(i)** $2h = a(t_1^2 + t_2^2)$ ....(1)

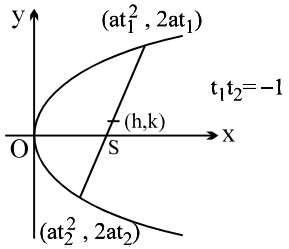
$2k = 2a(t_1 + t_2)$.....(2)

$k^2 = a^2[t_1^2 + t_2^2 + 2t_1t_2]$

$$k^2 = a^2\left[\frac{2h}{a} - 2\right]$$

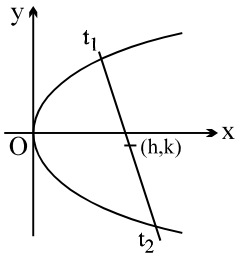
$= 2ah - 2a^2$

$\therefore$ $y^2 = 2a(x - a)$



This could also be spelled as locus of the middle point of all focal chords of all the particles $y^2 = 4ax$. **or** Locus of the middle point of all the chord of contact of the pair of tangents drawn from any point on this directrix.

(ii) $2h = a(t_1^2 + t_2^2)$ ....(1) ; $k = a(t_1 + t_2)$.....(2)



also $t_2 = -t_1 - \dfrac{2}{t_1} \Rightarrow t_1 + t_2 = -\dfrac{2}{t_1}$

using (2), $\dfrac{k}{a} = -\dfrac{2}{t_1} \Rightarrow t_1 = -\dfrac{2a}{k}$

$$\therefore \quad t_2 = +\frac{2a}{k} + \frac{2k}{2a} \Rightarrow t_2 = \frac{2a}{k} + \frac{k}{a}$$

$$\therefore \quad t_1t_2 = -\frac{2a}{k}\left[\frac{2a}{k} + \frac{k}{a}\right] = -2 - \frac{4a^2}{k^2} \quad ....(3)$$

from (1) $2h = a[(t_1 + t_2)^2 - 2t_1t_2] = a\left[\dfrac{k^2}{a^2} + 2\left(2 + \dfrac{4a^2}{k^2}\right)\right]$

$$= a\left[\frac{k^2}{a^2} + 4 + \frac{8a^2}{k^2}\right] = a\left[\frac{k^4 + 4a^2k^2 + 8a^4}{a^2k^2}\right]$$

$2ahk^2 = k^4 + 4a^2k^2 + 8a^4$

$k^2(k^2 - 2ah + 4a^2) + 8a^4 = 0$

(iii) Mid point $p(n_1, y_1)$ $y^2 = 4ax$

$T = S_1$

$$4y_1 - 4a\left(\frac{x + x_1}{2}\right) = y_1^2 - 4ax_1$$

$4y_1 - 2a(x + x_1) = y_1^2 - 4ax_1$
$4y_1 - 2ax = y_1^2 - 2ax_1$

$$\Rightarrow \frac{4y_1 - 2ax}{y_1^2 - 2ax_1}$$

Homogensing.

$$y^2 = 4ax\left(\frac{4y_1 - 2ax}{y_1^2 - 2ax_1}\right)$$

$(y_1^2 - 2ax_1)y^2 = 4ax(4y_1 - 2ax) = 0$
$\Rightarrow 8a^2x^2 - (4ay_1)xy + (y_1^2 - 2ax_1)y^2 = 0$
α is angle between lines.

$$\tan\alpha = \frac{2\sqrt{(2ay_1)^2 - 8a^2\left(4_1^2 - 2ax_1\right)}}{8a^2 + y_1^2 - 2ax_1}$$

$(y_1^2 - 2ax_1 + 8a^2)\tan^2\alpha = 16a^2(4ax_1 - y_1^2)$
lous (x_1, y_1) : $(y^2 - 2ax + 8a^2)\tan^2\alpha = 16a^2(4ax - y^2)$

(iv) are of given length ($2l$.)
$y^2 = 4ax$
(h, k) $T = S_1$
$ky - 2ax = k^2 - 2ah$

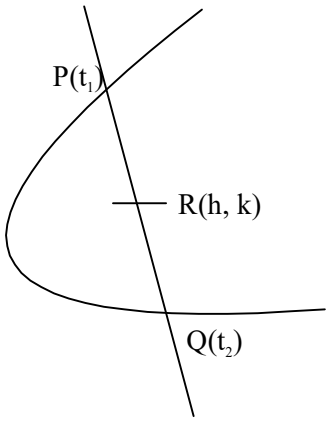
$$\left(m = \frac{2}{a}k \quad c = \frac{k^2 - 2ah}{k}\right)$$

Length of intercept of line $y = mx + c$

at $y^2 = 4ax$ is $L = \frac{4}{m^2}\sqrt{a(1+m^2)(a - m2)}$

Put m and c
$4a_1^3 l^2 = (y^2 + 4a^2)(4ax - y^2)$

(v) $y^2 = 4ax$
$P(t_1), Q(t_1)$



$$k = \frac{2at_1 + 2at_2}{2}$$

$$\frac{k}{a} = t_1 + \left(-t_1 - \frac{2}{t_1}\right)$$

$$\frac{k}{a} = -\frac{2}{t_1}$$

$$t_1 = -\frac{2a}{k}$$

Similarly. $h = \frac{at_1^2 + at_2^2}{2}$

$$\frac{2h}{a} = t_1^2 + \left(-t_1 - \frac{2}{t_1}\right)^2$$

$$\frac{2h}{a} = \left(-\frac{2a}{k}\right)^2 + \left(-\left(\frac{-2a}{k}\right) - \frac{2}{-\frac{2a}{k}}\right)^2$$

Solue $\frac{2x}{a} = \frac{4a^2}{y^2} + \left(\frac{2a}{y} + \frac{y}{a}\right)^2$

5. Find equation of polar of focus of parabola $y^2 = 4ax$.

Ans. $x + a = 0$

Sol. S (a, 0)

Polar : T = 0

$$y(0) = 4a\left(\frac{x+a}{2}\right)$$

$0 = 2a(x + a)$

$x + a = 0$

6. Find pole of the line $lx + my + n = 0$ w.r.t. the parabola $y^2 = 4ax$.

Ans. $\left(\frac{n}{\ell}, -\frac{2am}{\ell}\right)$

Sol. P(h, k)

$T = 0 \qquad ky = 2a(x + h)$

$2ax - ky + 2ah = 0$

$\ell x + my + h = 0$

$$\frac{2a}{l} = \frac{-k}{m} = \frac{2ah}{n}$$

$$h = \frac{n}{\ell}$$

$$k = -\frac{2am}{\ell}$$

7. A variable line L intersects the parabola $y^2 = 4ax$ at A and B.P is POI of tangents drawn of A and B. Find locus of P if :

(i) If line *l* is focal chord.

Ans. $x = -a$

(ii) Line *l* subtends right angle at vertex.

Ans. $x = -4a$

(iii) Slope of line *l* is 3.

Ans. $y = \frac{2a}{3}$

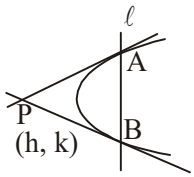
(iv) Line l is tangent to the parabola $x^2 = 4by$

Ans. $xy = -2ab$

(v) Line l is normal to the circle $x^2 + y^2 - 2ay = 0$

Ans. $y = 2x$

Sol.



Equation of AB (C.O.C.) :

P(h, k)

T = 0

$ky = 2a(x + h)$

(i) if L is focal of chord

S(a, 0) $k(0) = 2a(a + h)$

$2a^2 + 2ah = 0$

$2a(a + h) = 0$

$h = -a$

$x = -a$

(ii) line L subtends right angle at vertex

(4a, 0) $k(0) = 2a(4a + h)$

$4a + h = 0$

$x = -4a$

(iii) slope of line L is 3

$2ax / 0\ ky + 2ah = 0$

$m = 3$

$$\frac{2a}{k} = 3$$

$$k = \frac{2a}{3}$$

(iv) line L is tangent to parabola $x^2 = 4by$

$c = -bm^2$

$$\left(\frac{2ah}{k}\right) = -b\left(\frac{2a}{k}\right)^2$$

$$\frac{2ah}{k} = -\frac{4a^2b}{k^2}$$

$xy = -2ab$

(v) line ℓ is normal to circle $x^2 + y^2 - 2ay = 0$

c(0, a)

$ka = 2a(0 + h)$

$ka = 2ah$

$y = 2x$

8. Find the locus of the poles w.r.t. the parabola $y^2 = 4ax$ of the tangents to the circle $x^2 + y^2 = 4a^2$.

Ans. $x^2 - y^2 = 4a^2$

Sol. Let the pole be (h, k)

It's polar w.r.t. $y^2 = 4ax$ is

$$ky = 2a(x + h) \quad ....(1)$$

now (1) must be tangent to the circle $x^2 + y^2 = 4a^2$

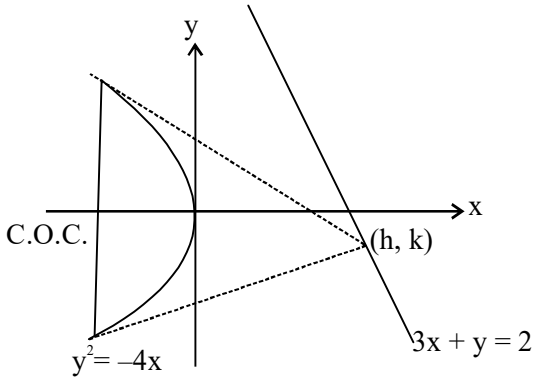
$$\therefore \quad \left| \frac{-2ah}{\sqrt{4a^2 + k^2}} \right| = 2a$$

$$h^2 = (4a^2 + k^2)$$

$\therefore$ Locus is $x^2 - y^2 = 4a^2$

9. Pair of tangents are drawn to the parabola $y^2 = -4x$ from every point on the line $3x + y = 2$. Show that their chord of contact passes through a fixed point.

Sol.



chord of contact

$$ky = -4\left(\frac{x + h}{2}\right)$$

$ky = -2x - 2h$

$2x + ky + 2h = 0 \quad ...(1)$

$3x + y = 2$

$\because (h, k)$

$3h + k = 2$

$\Rightarrow k = 2 - 3h \quad .....(2)$

Using (2) in (1)

$2x + (2 - 3h)y + 2h = 0$

$(2x + 2y) + h(2 - 3y) = 0$

$$x = \frac{-2}{3} \qquad y = \frac{2}{3}$$

$$\therefore \text{ fixed point} \equiv \left(\frac{-2}{3}, \frac{2}{3}\right)$$

10. Two equal parabolas with axes in opposite directions touch at a point O. From a point P on one of them are drawn tangents PQ and PQ' to the other. Prove that QQ' will touch the first parabola in P' where PP' is parallel to the common tangent at O.

Sol. Let two parabolas be

$y^2 = 4ax$ ...(1)

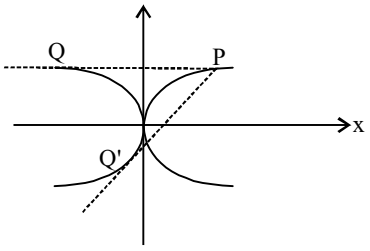
$y^2 = -4ax$ ...(2)

$P(at^2, 2at)$ tangents on (1) to (2)

$YY_1 = -2a(x + x_1)$

$2aty = -2a(x + at^2)$

$ty = -x - at^2$ ...(3)



from (1)

$y^2 = 4a(-at^2 - ty)$

$y^2 + 4aty + 4a^2t^2 = 0$

$(y + 2at)^2 = 0$

$y = -2at$

from (1)

$(-2at)^2 = 4ax$

$x = at^2$

point of contact to (1) is $P'(at^2, -2at)$

equation of common tangent at 0 is $x = 0$

slope of $PP' = \dfrac{-2at - 2at}{at^2 - at^2} = \infty$

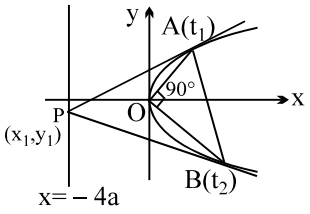
PP' is $\perp$ to x-axis. Hence it is parallel to the y-axis or $x = 0$.

11. From a point on the line $x + 4a = 0$ a pair of tangents are drawn to the parabola $y^2 = 4ax$. Show that their chord of contact subtends a right angle at the vertex.

Sol. Point of intersection of tangents at t_1 and t_2 is $\left(at_1t_2, a(t_1 + t_2)\right)$.

Hence $at_1t_2 = -4a \Rightarrow t_1t_2 = -4$

now, also $m_{OA} \cdot m_{OB} = \dfrac{2}{t_1} \cdot \dfrac{2}{t_2} = \dfrac{4}{t_1t_2} = \dfrac{4}{-4} = -1$]

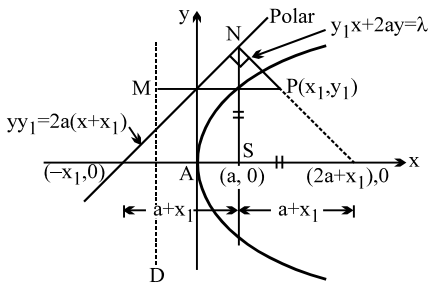


12. If a perpendicular be let fall from any point P upon its polar w.r.t. $y^2 = 4ax$, prove that the distance of the foot of this perpendicular from the focus is equal to the distance of the point P from the directrix. **[O]**

Sol. To prove that

$NS = PM$

self explanatory]



13. Polar of any point P w.r.t. the circle $x^2 + y^2 = a^2$ touches the parabola $y^2 = 4ax$ show that P lies on the curve $y^2 = ax$.

Sol. Let $P(x_1, y_1)$

its polar w.r.t. the circle $x^2 + y^2 = a^2$ is

$xx_1 + yy_1 = a^2$ ....(1) $\left[y = -\frac{x_1}{y_1}x + \frac{a^2}{y_1} \right]$

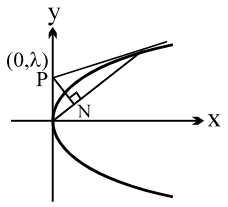
since the tangent to $y^2 = 4ax$ $\therefore$ $\frac{a^2}{y_1} = -\frac{4y_1}{x_1}$

$y_1^2 + ax_1 = 0$ $\therefore$ locus is $y^2 = ax$.

14. From a variable point on the tangent at the vertex of a parabola $y^2 = 4ax$, a perpendicular is drawn to its C.O.C. Show that these variable perpendicular lines through a fixed point on the axis of the parabola. **[O]**

Sol. chord of contact of P $(0, \lambda)$ is

$\lambda y = 2ax \quad \Rightarrow \quad 2ax - \lambda y = 0$



line perpendicular to it

$\lambda x + 2ay = c$ passes through $(0, \lambda) \Rightarrow c = 2a\lambda$

hence equation of PN is

$\lambda x + 2ay = 2a\lambda$

$2ay = \lambda (x - 2a) = 0$

This passes through $(2a, 0)$

11. DIAMETER

The locus of the mid points of a system of parallel chords of a parabola is called a diameter of the parabola.

The equation of a system of parallel chord $y = mx + c$ with respect to the parabola $y^2 = 4ax$ is given by

$y = \frac{2a}{m}$

slope = m

mid-point (h, k)

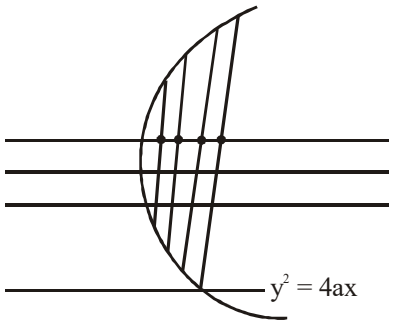
$T = S_1$

$ky - 2a(x + h) = k^2 - 4ah$

slope = m

$\frac{2a}{k} = m$

$y = \frac{2a}{m}$



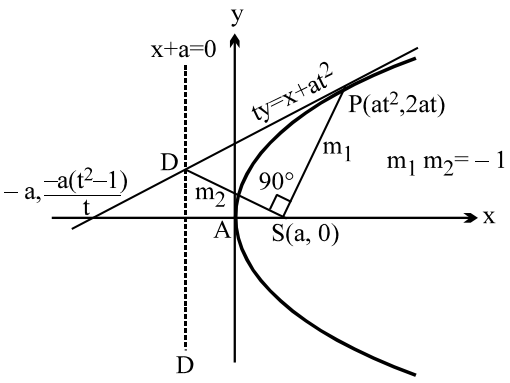
12.1 IMPORTANT HIGHLIGHTS :

1. The portion of a tangent to a parabola cut off between the directrix & the curve subtends a right angle at the focus.

$$m_1 = \frac{2at}{at^2 - a} = \frac{2t}{t^2 - 1} \quad ;$$

$$m_2 = \frac{-a(t^2-1)}{t \cdot 2a} = -\frac{(t^2-1)}{2t}$$

$$\therefore \quad m_1 m_2 = -1$$



2. Any tangent to a parabola & the perpendicular on it from the focus meet on the tangent at the vertex.
i.e. locus of the feet of the perpendicular drawn from focus upon a variable tangent is the tangent drawn to the parabola at its vertex.

Explanation : Tangent 't' is

$$ty = x + at^2 \qquad .....(1)$$

Passing through (h, k) hence

$$tk = h + at^2 \qquad .....(A)$$

A line through (a, 0) with slope – t

$$y = -t(x - a) \quad ....(2)$$

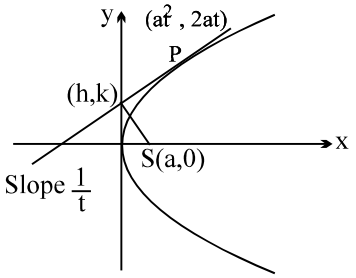
(2) also passes through (h, k)

$$k = -th + at$$

$$tk = -t^2h + at^2 \quad ....(B)$$

(A) – (B) gives $0 = (1 + t^2)h$

$\Rightarrow$ $x = 0$ which is the tangent at the vertex.



3. Semi latus rectum of the parabola $y^2 = 4ax$, is the harmonic mean between segments of any focal chord of the parabola is : $2a = \frac{2bc}{b+c}$ **i.e.** $\frac{1}{b} + \frac{1}{c} = \frac{1}{a}$.

[Sol. $2a = \frac{2bc}{b+c}$ i.e. $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$

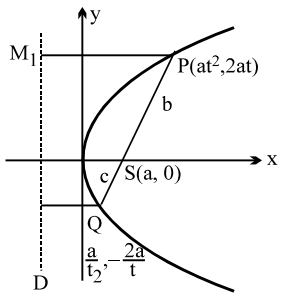
$$b = a + at^2$$

$\Rightarrow$ $b = a(1 + t^2)$

$\Rightarrow$ $\frac{a}{b} = \frac{1}{1+t^2}$ ....(1)

$$c = a + \frac{a}{t^2}$$

$\Rightarrow$ $c = a\left(1 + \frac{1}{t^2}\right) \Rightarrow \frac{a}{c} = \frac{t^2}{t^2+1}$ ....(2)



from (1) and (2)

$$\frac{a}{b}+\frac{a}{c}=1 \qquad \Rightarrow \qquad \frac{1}{a}=\frac{1}{b}+\frac{1}{c} \qquad]$$

4. The tangents at the extremities of a focal chord intersect at right angles on the directrix, and hence a circle on any focal chord as diameter touches the directrix.

5. Also a circle on any focal radii of a point P $(at^2, 2at)$ as diameter touches the tangent at the vertex

6. **(a)** If the tangent & normal at any point 'P' of the parabola intersect the axis at T & G then

(i) S is mid point of T and G where 'S' is the focus.

(ii) S is circumcenter of Triangle TPG

(iii) ST = SG = SP

(b) all rays emanating from S will become parallel to the axis of the parabola after reflection through surface of parabola

Explanation:

In other words the tangent and the normal at a point P on the parabola are the bisectors of the angle between the focal radius SP & the perpendicular from P on the directrix. From this we conclude that all rays emanating from S will become parallel to the axis of the parabola after reflection.

Deduce that, if Q is any point on the tangent and QN is the perpendicular from Q on focal radius and QL is the perpendicular on the directrix then QL = SN.

7. Tangents and Normals at the extremities of the latus rectum of a parabola $y^2 = 4ax$ constitute a square, their points of intersection being (– a, 0) & (3a, 0).

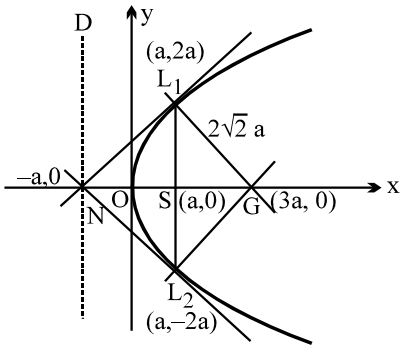
[Hint: 

figure `is Self explanatory]

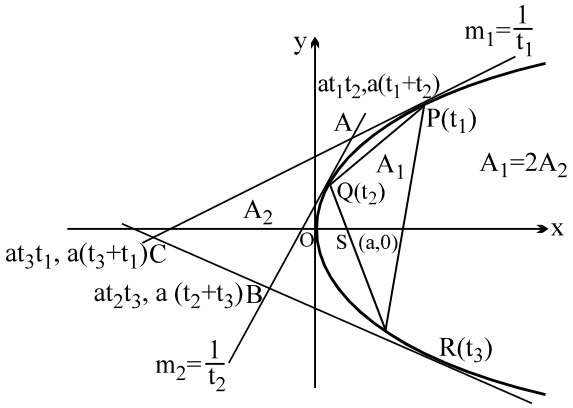
Note :

(1) The two tangents at the extremities of focal chord meet on the foot of the directrix.

(2) Figure L_1NL_2G is square of side $2\sqrt{2}\,a$

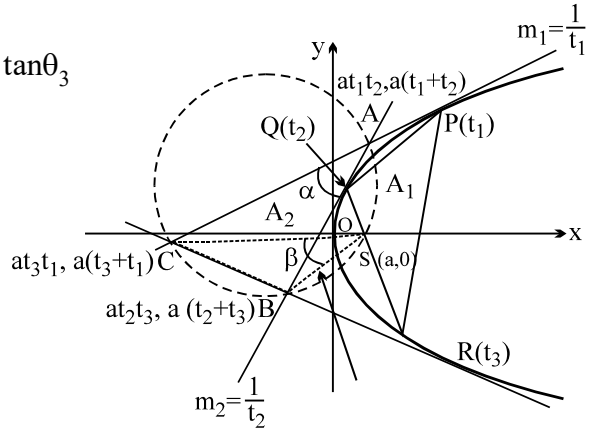
8. (i) The area of the triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents at these points.

$$\frac{A_1}{A_2} = \frac{\begin{vmatrix} at_1^2 & 2at_1 & 1 \\ at_2^2 & 2at_2 & 1 \\ at_3^2 & 2at_3 & 1 \end{vmatrix}}{\begin{vmatrix} at_1t_2 & a(t_1+t_2) & 1 \\ at_2t_3 & a(t_2+t_3) & 1 \\ at_3t_1 & a(t_3+t_1) & 1 \end{vmatrix}} = 2$$



(ii) The circle circumscribing the triangle formed by any three tangents to a parabola passes through the focus.
TPT $\alpha = \beta$

$\frac{1}{t_1} = \tan\theta_1$; $\frac{1}{t_2} = \tan\theta_2$; $\frac{1}{t_3} = \tan\theta_3$



$\therefore \quad \tan\alpha = \left| \frac{\tan\theta_1 - \tan\theta_2}{1+\tan\theta_1 \cdot \tan\theta_2} \right|$

$\tan\alpha = |\tan(\theta_1 - \theta_2)|$ ....(1)

$m_1 = \frac{a(t_3+t_1)}{at_3t_1 - a} = \frac{t_3+t_1}{t_3t_1 - 1}$

$= \frac{\frac{1}{t_1} + \frac{1}{t_3}}{1 - \frac{1}{t_1t_3}} = \frac{\tan\theta_1 + \tan\theta_3}{1 - \tan\theta_1 \cdot \tan\theta_3}$

$m_1 = \tan(\theta_1 + \theta_3)$

|||ly $m_2 = \tan(\theta_2 + \theta_3)$

$\tan\beta = \left| \frac{\tan(\theta_1+\theta_3) - \tan(\theta_2+\theta_3)}{1+\tan(\theta_1+\theta_3)\cdot\tan(\theta_2+\theta_3)} \right|$

$\tan\beta = |\tan(\theta_1 - \theta_2)|$ ....(2)

from (1) and (2), we get

$\alpha = \beta$ hence proved]

(iii) The orthocentre of any triangle formed by three tangents to a parabola $y^2 = 4ax$ lies on the directrix & has the co–ordinates $-a$, $a(t_1 + t_2 + t_3 + t_1t_2t_3)$.

12.2 EXTRA QUESTION [O]

If the tangents at P and Q meet in T, then :

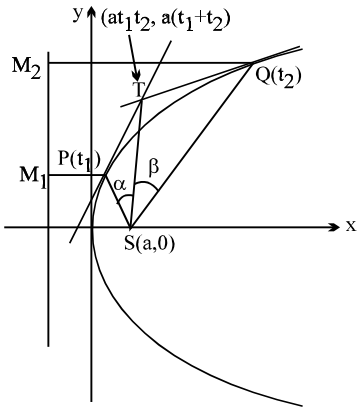
(i) TP and TQ subtend equal angles at the focus S.

(ii) $ST^2 = SP \cdot SQ$ &

(iii) The triangles SPT and STQ are similar.

(i) To prove that $\alpha = \beta$, it will be sufficient to prove that 'T' lies on the angle bisector of the angle $\angle PSQ$ i.e. perpendicular distance of 'T' from the line SP is equal to the perpendicular of T from SQ.

equation of SP



$$y = \frac{2at_1}{at_1^2 - a}(x - a)$$

$$2t_1x - (t_1^2 - 1)y - 2ab_1 = 0$$

$$p_1 = \left| \frac{2at_1^2t_2 - (t_1^2 - 1)a(t_1 + t_2) - 2at_1}{\sqrt{(t_1^2 - 1)^2 + 4t_1^2}} \right|$$

$$= a\,|\,t_1 - t_2\,|$$

|||ly $p_2 = a\,|\,t_2 - t_1\,| \Rightarrow \alpha = \beta$, Hence proved.

Now (ii) $SP \cdot SQ = (a + at_1^2)(a + at_2^2) = a^2(1 + t_1^2)(1 + t_2^2)$

also $(ST)^2 = a^2(t_1t_2 - 1)^2 + a^2(t_1 + t_2)^2 = a^2[t_1^2\,t_2^2 + 1 + t_2^2 + t_2^2]$

$$= a^2(1 + t_1^2)(1 + t_2^2)$$

Hence $(ST)^2 = SP \cdot SQ$

This is conclusive that product of the focal radii of two points P and Q is equal to the square of the distance of focus from the point of intersection of the tangents drawn at P and Q.

(iii) again, $\frac{ST}{SP} = \frac{SQ}{ST}$ and $\alpha = \beta$

hence the two triangles SPT and SQT are similar.

MATHEMATICS

ELLIPSE

THEORY SHEET

ENGLISH

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1.1 ELLIPSE : INTRODUCTION

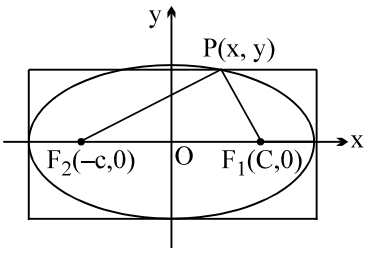
$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will represent an ellipse if $\Delta \neq 0$; $h^2 < ab$ and $0 < e < 1$

It is locus of a point moving in such a way that sum of its distances from two fixed points is always a constant.

Standard Equation :

$|PS_1| + |PS_2| = 2a$

$$\sqrt{(x-c)^2 + y^2} + \sqrt{(x+c)^2 + y^2} = 2a \quad (2a > 2c)$$



simplifying,

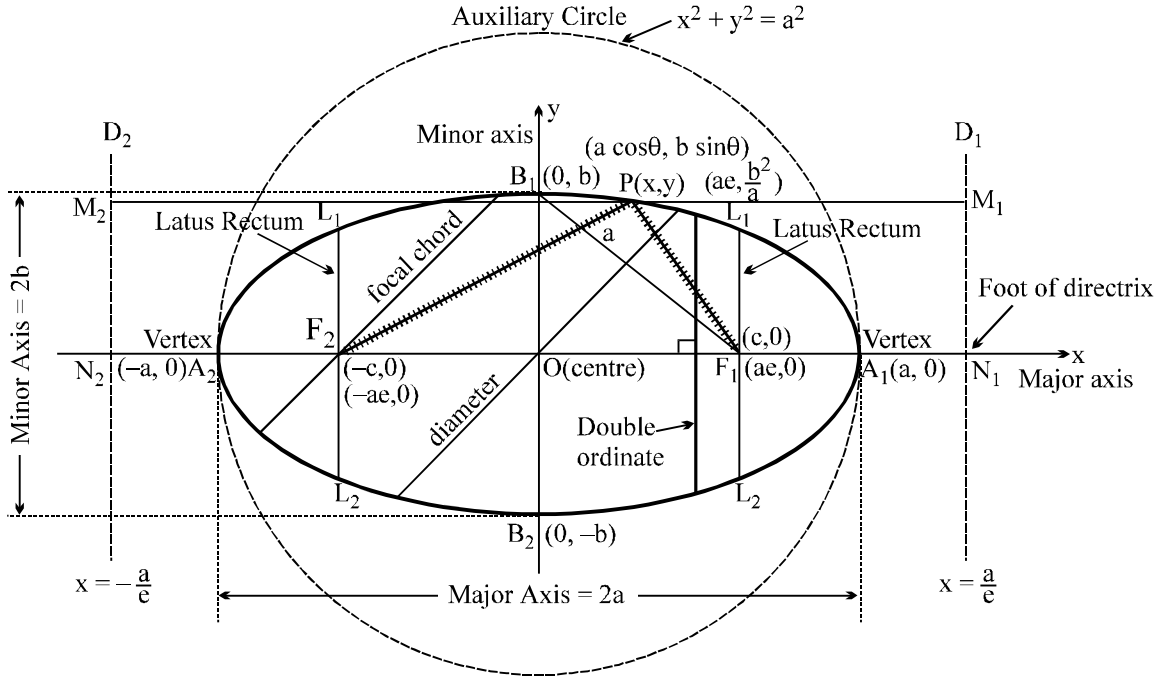
$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1$$

denote the positive quantity $a^2 - c^2$ by b^2

we get the standard equation of an ellipse as

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad \text{....(1)} \quad \text{(standard equation a > b)}$$

1.2 IMPORTANT TERMS



1. This curve is symmetric about both the x and y axis.
2. S_1 and S_2 are called foci.
3. Line joining S_1 and S_2 is called major axis/focal axis
4. A_1 & A_2 are called ends of major axis & also called vertices.
5. Perpendicular bisector of major axis is called minor axis
6. B_1 and B_2 are ends of minor axis.
7. Length of major axis = 2a

8. Length of minor axis = 2b
9. Point of intersection of the major and minor axis is called the **centre** of the ellipse.
10. Major and minor axes together are known as **principal axes** of the ellipse.
11. The distance between foci is called the **Focal Length**.
12. Any chord of the ellipse passing through center gets bisected by it and is called the **diameter**.
13. Any chord through focus is called a **focal chord**
14. Any chord perpendicular to the focal axis is called **double ordinate**.
15. A particular double ordinate through focus or a particular focal chord perpendicular to focal axis is called its **Latus Rectum**.

1.3 ECCENTRICITY

$$e = \frac{c}{a} = \frac{OF_1}{OA_1} = \frac{\text{distance from centre to focus}}{\text{distance from centre to vertex}}$$

$$\therefore \quad e^2 = \frac{c^2}{a^2} = \frac{a^2 - b^2}{a^2} = 1 - \frac{b^2}{a^2} < 1 \quad ; \text{ Hence } e^2 = 1 - \left(\frac{m}{M}\right)^2$$

$a^2e^2 = a^2 - b^2 \Rightarrow b^2 = a^2(1 - e^2)$

The two foci are $(\pm ae, 0)$

$$\text{put } x = ae \text{ in } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{y^2}{b^2} = 1 - e^2 = 1 - \left(1 - \frac{b^2}{a^2}\right) = \frac{b^2}{a^2} \quad ; \quad y = \pm \frac{b^2}{a}$$

$$\therefore \quad \text{coordinate of the extremities of LR} = \pm ae, \pm \frac{b^2}{a}$$

$$\text{Length of LR} = \frac{2b^2}{a} = 2a(1 - e^2) = \frac{4b^2}{2a} = \frac{(\text{minor axis})^2}{\text{major axis}}$$

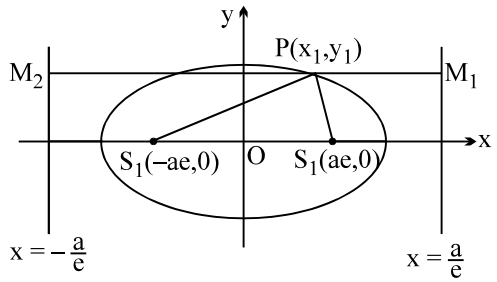
$= 2e$ (distance between focus and corresponding foot of the directrix)

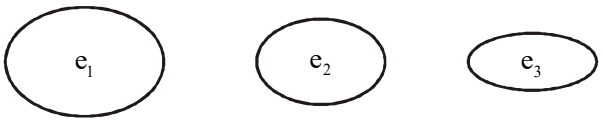
The sum of the focal radii of any point on the ellipse is equal to the length of the major axis. We have

$$PS_1 = PM_1 = e\left(\frac{a}{e} - x_1\right) = a - ex_1 \quad .....(1)$$

$$PS_2 = e\left(\frac{a}{e} + x_1\right) = a + ex_1 \quad ....(2)$$

(1) + (2) gives $PS_1 + PS_2 = 2a$





$e_1 < e_2 < e_3$

Note that two ellipses are said to be similiar if they have the same value of eccentricity.

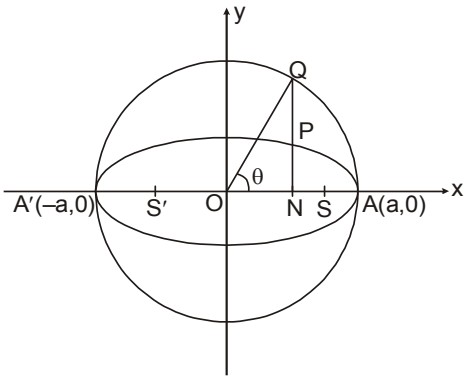
1.4 AUXILIARY CIRCLE AND ECCENTRIC ANGLE

A circle described on major axis of ellipse as diameter is called the **auxiliary circle**.

Let Q be a point on the auxiliary circle $x^2 + y^2 = a^2$ such that line through Q perpendicular to the x – axis on the way intersects the ellipse at P, then P & Q are called as the **Corresponding Points** on the ellipse & the auxiliary circle respectively. 'θ' is called the **Eccentric Angle** of the point P on the ellipse

$Q \equiv (a\cos\theta , a\sin\theta)$

$P(\theta) \equiv (a\cos\theta , b\sin\theta)$

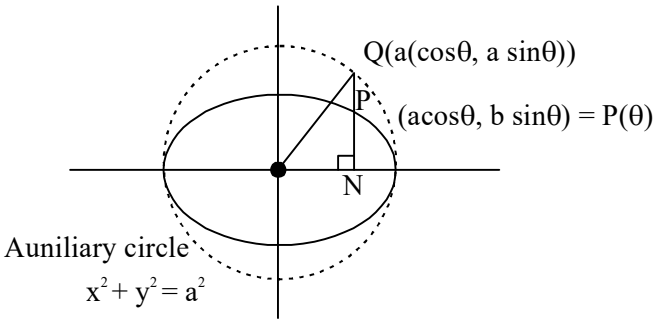


Note that :

$$\frac{\ell(PN)}{\ell(QN)} = \frac{b}{a} = \frac{\text{Semi minor axis}}{\text{Semi major axis}}$$

Q.1 If from each point on a circle, perpendicular are drawn on a fixed diametre and point P divides these perpendiculars in a constant ratio then Prove that locus of P is an ellipse whose auxilliary circle is a given fixed circle.

Sol.



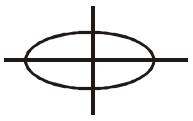
Let P pt. on ellipse

$\therefore$ Co-ordinate of P(acosθ bsinθ)

$$\frac{PN}{PQ} = \frac{b\sin\theta}{(a-b)\sin\theta} = \frac{b}{a-b} = \text{cons tan t}$$

$\therefore$ the locusot P is ellipse with auxiliray circle is a given fired circle.

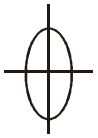
1.5 STANDARD ELLIPSES



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$a > b$

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

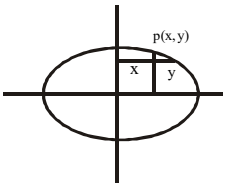


$a > b$

Center	$(0,0)$	$(0,0)$
Vertices	$(\pm a, 0)$	$(0, \pm a)$
B_1, B_2	$(0, \pm b)$	$(\pm b, 0)$
ℓ_{MA}	2a	2a
ℓ_{ma}	2b	2b
foci	$(\pm ae, 0)$	$(0, \pm ae)$

directrices	$x = \pm a/e$	$y = \pm a/e$
feet of dir.	$\left(\pm\frac{a}{e}, 0\right)$	$\left(0, \pm\frac{a}{e}\right)$
ℓ_{LR}	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$
ends of LR	$\left(\pm ae, \pm\frac{b^2}{a}\right)$	$\left(\pm\frac{b^2}{a}, \pm ae\right)$
aux. circle	$x^2 + y^2 = a^2$	$x^2 + y^2 = a^2$
eccentricity	$e^2 = 1 - \frac{b^2}{a^2}$	$e^2 = 1 - \frac{b^2}{a^2}$

1.6 ALTERNATE DEFINITION OF ELLIPSE



$a > b$ $\qquad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$p_1 \equiv$ distance of P from major axis.; $\qquad p_2 \equiv$ distance of P from minor axis.

$$\frac{p_2^2}{a^2} + \frac{p_1^2}{b^2} = 1$$

Ex.1 $\frac{(x-y+1)^2}{32} + \frac{(x+y-3)^2}{50} = 1$ $\qquad$ find e, ℓ_{LR}

Sol. $\frac{\left(\frac{x-y+1}{\sqrt{2}}\right)^2}{16} + \frac{\left(\frac{x+y-3}{\sqrt{2}}\right)^2}{25} = 1$

$a^2 = 25 \qquad b^2 = 9$

$e^2 = 1 - \frac{9}{25} \qquad e = \frac{4}{5}$

$e^2 = \frac{16}{25}$

$\ell_{L.R.} = 2\frac{b^2}{a} \Rightarrow \frac{2\times 9}{5} \Rightarrow \frac{18}{5}$

1. $\frac{x^2}{16} + \frac{y^2}{25} = 1$ Find everything

Ans. Center (0, 0), Vertices $(0, \pm 5)$, $B_1, B_2(\pm 4, 0)$, $l_{MA} = 10$, $l_{ma} = 8$, $e = \frac{3}{5}$, foci $(0, \pm 3)$

Directrices $y = \pm \frac{25}{3}$, $l_{LR} = \frac{32}{5}$,

Ends of LR $\left(\pm \frac{16}{5}, \pm 3 \right)$,

Aux. Circle $x^2 + y^2 = 25$

2. $\frac{x^2}{169} + \frac{y^2}{144} = 1$ Find everything

Ans. Center (0, 0), Vertices $(\pm 13, 0)$, $B_1, B_2(0, \pm 12)$, $l_{MA} = 26$, $l_{ma} = 24$, $e = \frac{5}{13}$, foci $(\pm 5, 0)$

Directrices $x = \pm \frac{169}{5}$, $l_{LR} = \frac{288}{13}$,

Ends of LR $\left(\pm 5, \pm \frac{144}{13} \right)$,

Aux. Circle $x^2 + y^2 = 169$

3. $9x^2 + 25y^2 - 18x + 100y - 116 = 0$ Find everything

Ans. Center (1, –2), Vertices (6, –2), (–4, –2), B_1B_2 (1, 1), (1, –5) $l_{MA} = 10$, $l_{ma} = 6$, $e = \frac{4}{5}$,

foci (5, –2), (–3, –2), Directrices $x = \frac{29}{4}$, $x = -\frac{21}{4}$, $l_{LR} = \frac{18}{5}$,

Ends of LR $\left(-3, -\frac{1}{5} \right), \left(-3, -\frac{19}{5} \right), \left(5, -\frac{1}{5} \right), \left(5, -\frac{19}{5} \right)$

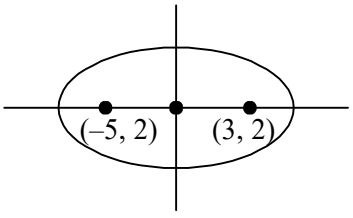
Aux. Circle $(x - 1)^2 + (y + 2)^2 = 25$

4. Find equation of ellipse

(i) Foci are (–5, 2) & (3, 2) & $l_{MA} = 10$

Ans. $\frac{(x+1)^2}{25} + \frac{(y-2)^2}{9} = 1$

Sol. $2a = 10$ $a = 5$



Center (–1, 2) $2ae = 8$ $e = \frac{8}{10}$

$\frac{(x-1)^2}{25}+\frac{(y-2)^2}{9}=1$ $\qquad e=\frac{4}{5}$

$\sqrt{1-\frac{b^2}{a^2}}=\frac{4}{5}$

$1-\frac{16}{25}=\frac{b^2}{a^2}$

$\frac{9}{25}=\frac{b^2}{a^2}$

(ii) Vertices are (4, 3), (–2, 3) & $l_{LR}=\frac{8}{3}$

Ans. Center (1, 3) $\quad \ell_{L.R.}=\frac{8}{3}=\frac{2b^2}{a}$

$\frac{4}{3}=\frac{b^2}{a}$

$\frac{(x-1)^2}{9}+\frac{(y-3)^2}{4}=1$

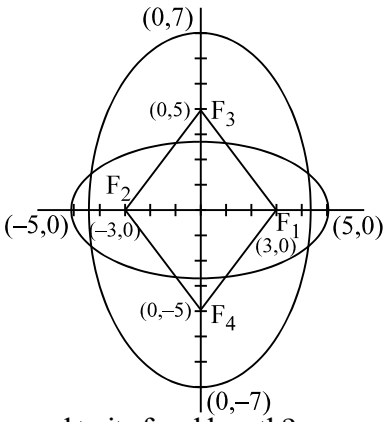
5. Find the area of the figure formed by the foci of these two ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and $\frac{x^2}{24}+\frac{y^2}{49}=1$.

Ans 30

Sol. $\frac{x^2}{25}+\frac{y^2}{16}=1$; $e_1^2=1-\frac{16}{25}=\frac{9}{25}$; $e_1=\frac{3}{5}$; $ae_1=(3,0)$

$\frac{x^2}{24}+\frac{y^2}{49}=1$; $e_2^2=1-\frac{24}{49}=\frac{25}{49}$; $e_2=\frac{5}{7}$; $ae_2=(5,0)$

Area $F_1F_3F_2F_4$ (rhombus) $=\frac{6\times10}{2}=30$ sq. units



6. Find the eccentricity of the ellipse if the length of its minor axis is equal to its focal length?

Ans $e=\frac{1}{\sqrt{2}}$

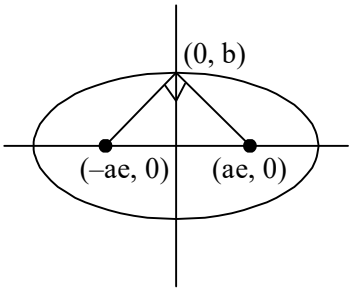
Sol. Given $2b=2ae$; $b^2=a^2e^2=a^2-b^2$; $2b^2=a^2$

$\therefore \frac{b^2}{a^2}=\frac{1}{2}$; $e^2=1-\frac{b^2}{a^2}=1-\frac{1}{2}=\frac{1}{2}$ $\Rightarrow$ $e=\frac{1}{\sqrt{2}}$

7. If triangle formed by focii of an ellipse and one end of minor axis is right angle triangle, then find eccentricity.

Ans. $e=\frac{1}{\sqrt{2}}$

Sol.



it will be rt. isosceles triangle

$(2ae)^2 = (ae)^2 + b^2 + (ae)^2 + b^2$

$4a^2e^2 = 2a^2e^2 + 2b^2$

$2a^2e^2 = a^2e^2 + b^2$

$a^2e^2 = b^2$

$$e^2 = \frac{b^2}{a^2}$$

$$e^2 = 1 - \frac{b^2}{a^2}$$

$e^2 = 1 - e^2$

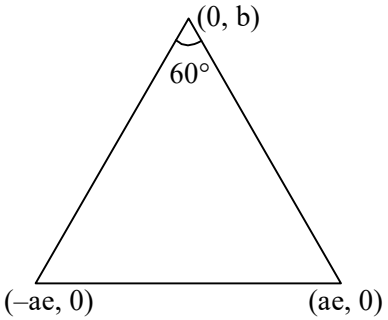
$2e^2 = 1$

$$e = \frac{1}{\sqrt{2}}$$

8. If triangle formed by focii of an ellipse and one end of minor axis is equilateral Δ, then find eccentricity.

Ans. $e = \frac{1}{2}$

Sol.



$$2ae = \sqrt{(ae)^2 + b^2}$$

$4a^2e^2 = a^2e^2 + b^2$

$3a^2e^2 = b^2$

$$3e^2 = \frac{b^2}{a^2} \qquad e^2 = 1 - \frac{b^2}{a^2}$$

$e^2 = 1 - 3e^2$ $\qquad 4e^2 = 1$ $\qquad e = \frac{1}{2}$

9. Distance of a point on the ellipse $\frac{x^2}{6}+\frac{y^2}{2}=1$ from its centre is 2. Find the eccentric angle of the point P.

Ans. $\frac{\pi}{4},\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4}$

Sol. $P\left(\sqrt{6}\cos\theta,\sqrt{2}\sin\theta\right)$ Center (0, 0)

$\left(\sqrt{6}\cos\theta\right)^2+\left(\sqrt{2}\sin\theta\right)^2=4$

$6\cos^2\theta + 2\sin^2\theta = 4$

$2 + 4\cos^2\theta = 4$

$4\cos^2\theta = 2$

$\cos^2\theta=\frac{1}{2}$

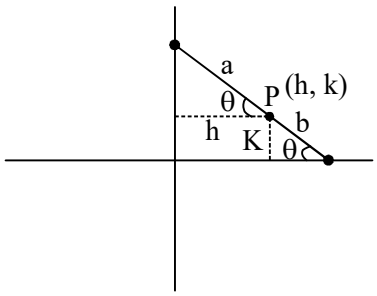
$\cos\theta=\frac{1}{\sqrt{2}}$

$\theta=\frac{\pi}{4},\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4}$

10. A rod of length a + b moves in such a way that both extremities remains on coordinates. A point P divides the length in ratio b : a measuring from x-axis. Find locus of P?

Ans. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

Sol:



$h = a\cos\theta$ and $k = b\sin\theta$

$\frac{h}{a}=\cos\theta$ $\frac{K}{b}=\sin\theta$

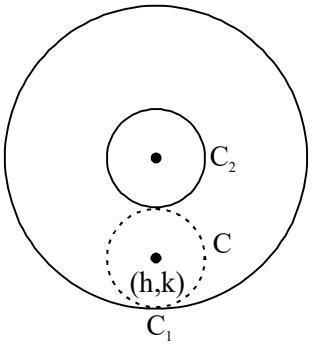
$\frac{h^2}{a^2}+\frac{K^2}{b^2}=1$

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

11. C_1 and C_2 are two fixed circles with radii r_1 and r_2 such that C_2 is contained in C_1. A third circle C moves in such a way that it touches C_1 internally and C_2 externally, then find the find locus of its centre.

Ans. locus of c is an ellipse

Sol.

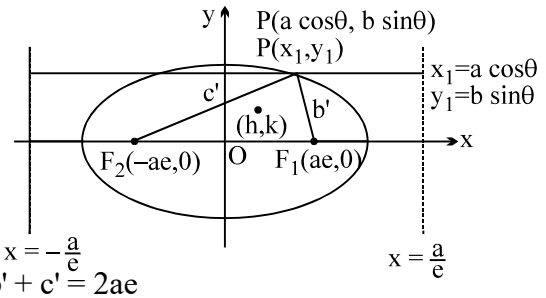


$$\left.\begin{array}{l} CC_1 = r_1 - r \\ CC_2 = r + r_2 \end{array}\right] \Rightarrow CC_1 + CC_2 = r_1 + r_2$$

Locus of center of C is ellipse.
C_1C_2 are foci.

12. Show that the locus of the incentre of the variable triangle PF_2F_1 is an ellipse whose eccentricity is $\sqrt{\frac{2e}{1+e}}$ where e is the eccentricity of standard ellipse when P moves on a standard ellipse. **[O]**

Sol. $h = \frac{a'x_1 + b'x_2 + c'x_3}{a'+b'+c'}$ & $k = \frac{a'y_1 + b'y_2 + c'y_3}{a'+b'+c'}$



$b' = e\left(\frac{a}{e} - a\cos\theta\right) = a - ae\cos\theta$

and $c' = e\left(\frac{a}{e} + a\cos\theta\right)$; hence $c' = a + ae\cos\theta$; $b' + c' = 2ae$

$$h = \frac{(2ae)(a\cos\theta) + a(1-e\cos\theta)(-ae) + a(1+e\cos\theta)(ae)}{2ae+2a}$$

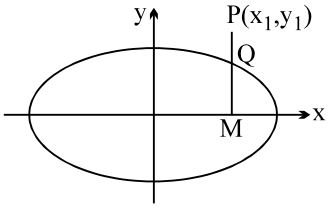
$$= \frac{2ae\cos\theta - ae(1-e\cos\theta) + ae(1+e\cos\theta)}{2a(e+1)} = \frac{2ae\cos\theta(e+1)}{2a(e+1)} = ae\cos\theta$$

$$k = \frac{(2ae)(b\sin\theta)}{2a(1+e)} = \frac{eb\sin\theta}{1+e} \quad \Rightarrow \quad \sin\theta = \frac{(1+e)k}{eb} \; ; \; \cos\theta = \frac{h}{ae}$$

$$\left(\frac{(1+e)k}{eb}\right)^2 + \left(\frac{h}{ae}\right)^2 = 1 \quad \Rightarrow \quad \frac{x^2}{(ae)^2} + \frac{y^2}{\left(\frac{be}{1+e}\right)^2} = 1$$

$$e'^2 = 1-\frac{b^2e^2}{(1+e)^2(ae)^2} = 1-\frac{b^2}{a^2(1+e)^2} = 1-\frac{1-e^2}{(1+e)^2} = \frac{2e}{1+e} \qquad \Rightarrow \quad e' = \sqrt{\frac{2e}{1+e}} \quad]$$

2.1 POSITION OF A POINT WRT AN ELLIPSE



$$S \equiv \frac{x^2}{a^2}+\frac{y^2}{b^2}-1 \qquad S_1 \equiv \frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1 \qquad T \equiv \frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1$$

1. P lies inside the ellipse $\Rightarrow S_1 < 0$
2. P lies on the ellipse $\Rightarrow S_1 = 0$
3. P lies outside the ellipse $\Rightarrow S_1 > 0$

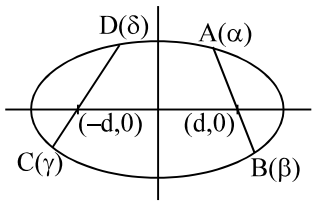
2.2 CHORD JOINING TWO POINTS

Equation of a chord of an ellipse joining two points P(α) and Q(β) on it is equal to

$$\frac{x}{a}\cos\left(\frac{\alpha+\beta}{2}\right)+\frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right)$$

If this particular chord passes through (d, 0) then we have

$$\frac{d}{a}\cos\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right) \; ; \; \frac{\cos\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}=\frac{a}{d}$$



$$\frac{\cos\left(\frac{\alpha+\beta}{2}\right)-\cos\left(\frac{\alpha-\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)+\cos\left(\frac{\alpha-\beta}{2}\right)}=\frac{a-d}{a+d}$$

or $$-\frac{2\sin\alpha/2\,\sin\beta/2}{2\cos\alpha/2\,\cos\beta/2}=\frac{a-d}{a+d} \qquad \text{i.e.} \qquad \tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{d-a}{d+a}$$

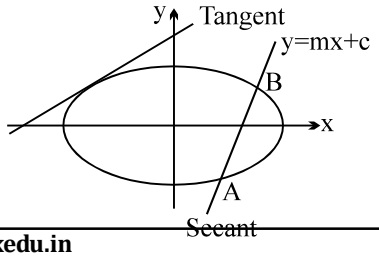
if d = ae i.e. PQ is a focal chord then $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{e-1}{e+1}$

3.1 LINE AND ELLIPSE

Line y = mx + c ....(1)

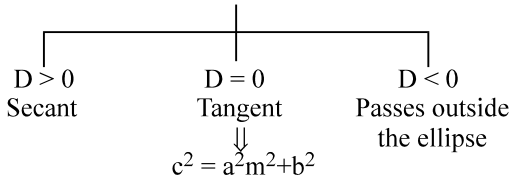
$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad ....(2)$$

solving (1) and (2)



$b^2x^2 + a^2(mx + c)^2 = a^2b^2$

i.e. $(a^2m^2 + b^2)x^2 + 2a^2cmx + a^2(c^2 - b^2) = 0$



Hence $y = mx \pm \sqrt{a^2m^2 + b^2}$ is always a tangent to the ellipse for all $m \in R$.

Examples (i) $\frac{x^2}{25} + \frac{y^2}{9} = 1$ $c^2 = 25m^2 + 9$ (ii) $\frac{x^2}{9} + \frac{y^2}{16} = 1$ $c^2 = 9m^2 + 16$

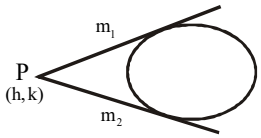
3.2 DIRECTOR CIRCLE

If it passes through (h, k) then

$k = mh \pm \sqrt{a^2m^2 + b^2}$

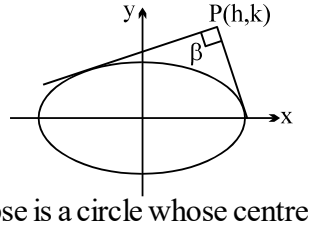
$(k - mh)^2 = a^2m^2 + b^2$

$(h^2 - a^2)m^2 - 2khm + k^2 - b^2 = 0$ ....(3)



Hence passing through a given point there can be a maximum of two tangents.

Equation (3) can be used to determine the locus of the point of intersection of two tangents enclosing an angle β.



if $\beta = 90°$ then $m_1m_2 = \frac{k^2 - b^2}{h^2 - a^2} = -1$

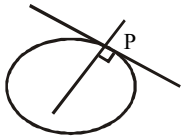
i.e. $k^2 - b^2 = a^2 - h^2$; i.e. $x^2 + y^2 = a^2 + b^2$

which is the director circle of the ellipse. Hence director circle of an ellipse is a circle whose centre is the centre of ellipse and whose radius is the length of the line joining the ends of the major and minor axis.

4.1 TANGENT

1. $P(x_1, y_1)$

$T = 0 \Rightarrow \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$



2. $P(\theta) \equiv P(a\cos\theta, b\sin\theta)$

$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$

$COT \equiv c^2 = a^2m^2 + b^2$

3. slope is m : $y = mx + c$ apply COT $c^2 = a^2m^2 + b^2 \Rightarrow$ $y = mx \pm \sqrt{a^2m^2 + b^2}$

4.2 NORMAL

1. $P(x_1, y_1)$

slope of tangent $= \frac{-b^2x_1}{a^2y_1} \Rightarrow$ slope of normal $= \frac{a^2y_1}{b^2x_1}$

equation of normal

$y - y_1 = \frac{a^2y_1}{b^2x_1}(x - x_1) \Rightarrow \quad \frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$

2. $P(\theta) \equiv P(a\cos\theta, b\sin\theta) \Rightarrow \quad \frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$

1. Find equation of tangent to the ellipse $4x^2 + 3y^2 - 12 = 0$ perpendicular to the line $y = 2x + 7$

Ans. $y = 2x \pm 4$

Sol. $\frac{x^2}{3} + \frac{y^2}{4} = 0 \qquad y = 2x + 7$

$m_1 = -\frac{1}{2}$

$C = \pm\sqrt{\frac{3}{4} + 4}$

$C = \pm\sqrt{\frac{19}{2}}$

$y = \frac{-x}{2} \pm \sqrt{\frac{19}{2}}$

2. Find equation of the tangent to an ellipse $9x^2 + 16y^2 = 144$ passing from (2, 3). Also find the equation of tangents to the ellipse $2x^2 + 7y^2 = 14$ from (5, 2)

Ans. $y = 3$, $x + y = 5$; $x - y = 3$ and $x - 9y + 13 = 0$

Sol. $\frac{x^2}{16} + \frac{y^2}{9} = 1$

$y = mx \pm \sqrt{16m^2 + 9}$

$3 = 2m \pm \sqrt{16m^2 + 9}$

$(3 - 2m)^2 = 16m^2 + 9$

$9 + 4m^2 - 12m = 16m^2 + 9$

$-12m^2 - 12m = 0$

$-12m(m + 1) = 0$

$m = 0$	$m = -1$
$y = mx + C$	$y = mx + C$
$y = C$	$y = -x + C$
(2, 3)	
$3 = C$	$2 = -3 + C$
$y = 3$	$C = 5$
	$y = -x + 5$

$2x^2 + 7y^2 = 14$

$\frac{x^2}{7}+\frac{y^2}{2}=1$

$(5, 2)$

$y = mx \pm \sqrt{7m^2+2}$

$(2 - 5m)^2 = (7m^2 + 2)$

$4 + 25m^2 - 20m = 7m^2 + 2$

$18m^2 - 20m + 2 = 0$

$9m^2 - 10m + 1 = 0$

$9m^2 - 9m - m + 1 = 0$

$9m(m - 1) - 1(m - 1) = 0$

$(m - 1)(9m - 1) = 0$

$m = 1$	$m = \frac{1}{9}$
$y = mx + C$	$y = mx + C$
$y = x + C$	$y = \frac{x}{9} + C$
$(5, 2)$	$(5, 2)$
$2 = 5 + C$	$2 = \frac{5}{9} + C$
$C = -3$	$\frac{18-5}{9} = C$
$y = x - 3$	$y = \frac{x}{9} + \frac{13}{9}$
	$9y = x + 13$

3. Find equation of tangent to an ellipse $3x^2 + 4y^2 = 12$, parallel to the line $y + 2x = 4$.

Ans $y + 2x = \pm\sqrt{19}$

Sol. $\frac{x^2}{4}+\frac{y^2}{3}=1 \quad m = -2$

Eq of tangent

$y = mx \pm \sqrt{4m^2+3}$

$y = -zx \pm \sqrt{4(4)+3}$

$y = -2x \pm \sqrt{19}$

$y + 2x = \pm\sqrt{19}$

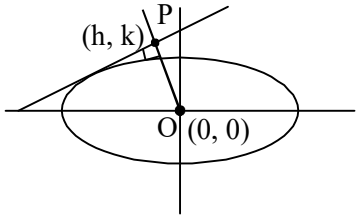
4. Prove that Locus of the feet of the perpendicular from centre upon a variable tangent to the standard ellipse is $(x^2 + y^2)^2 = a^2x^2 + b^2y^2$.

Sol. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

Eq^n of tangent

$$y = mx \pm \sqrt{a^2m^2 + b^2}$$

$$h = mh \pm \sqrt{a^2m^2 + b^2} \qquad \text{.....(1)}$$



$$\text{Slope of OP} = -\frac{1}{m}$$

$$y = \frac{-x}{m}$$

$$k = \frac{-h}{m}$$

$$m = \frac{-h}{k} \qquad \text{.....(2)}$$

put is eq^n (1)

$$K = \left(\frac{-h}{K}\right)h \pm \sqrt{a^2\left(\frac{h^2}{K^2}\right) + b^2}$$

$$K = \frac{-h^2}{K} \pm \frac{\sqrt{a^2h^2 + b^2k^2}}{K}$$

$$K^2 + h^2 = \pm\sqrt{a^2h^2 + b^2K^2}$$

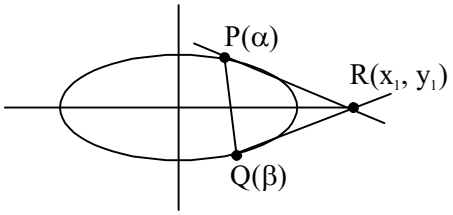
Sq. both side

$(x^2 + y^2)^2 = (a^2 x^2 + b^2 y^2)$

5. Find point of intersection of tangents drawn at P(α) and Q(β).

Ans. $$\left(\frac{a\cos\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}, \frac{b\sin\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}\right)$$

Sol.



$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

$a > b$

$P(a\cos\alpha, b\sin\alpha)$ $Q(a\cos\beta, b\sin\beta)$

Eq^n of PQ : $\frac{x}{a}\cos\left(\frac{\alpha+\beta}{2}\right)+\frac{y}{b}\left(\sin\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right)$

C.O.C. for $R(x_1\ y_1)$

$T = 0$

$$\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$$

$$\frac{\frac{x_1}{a}}{\cos\left(\frac{\alpha+\beta}{2}\right)}=\frac{\frac{y_1}{b}}{\sin\left(\frac{\alpha+\beta}{2}\right)}=\frac{1}{\cos\left(\frac{\alpha-B}{2}\right)}$$

$$x_1=\frac{a\cos\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)} \qquad y_1=\frac{b\sin\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}$$

6. Tangent to an ellipse makes angles θ_1, θ_2 with major axis. Find the locus of their intersection when $\cot\theta_1 + \cot\theta_2 = k^2$.

Ans. $k^2y^2 - 2xy = b^2k^2$

Sol. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $\qquad \cot\theta_1 + \cot\theta_2 = K^2$

$a > b$ $\qquad \frac{1}{m_1}+\frac{1}{m_2}=K^2$

$y = mx \pm \sqrt{a^2m^2+b^2}$ $\qquad \frac{m_1+m_2}{m_1m_2}=K^2$

$(h_1\ k_1)$ pt fo intersection

$(k_1 - hm)^2 = a^2m^2 + b^2$

$(h^2 - a^2)m^2 - 2hK_1m + K_1^2 - b^2 = 0 \lessdot \begin{matrix} m_1 \\ m_2 \end{matrix}$

$$m_1+m_2=\frac{2hK_1}{h^2-a^2} \qquad m_1m_2=\frac{K_1^2-b^2}{h^2-a^2}$$

$$\frac{2hK_1}{K_1^2-b^2}=K^2 \qquad \frac{2xy}{y^2-b^2}=K^2$$

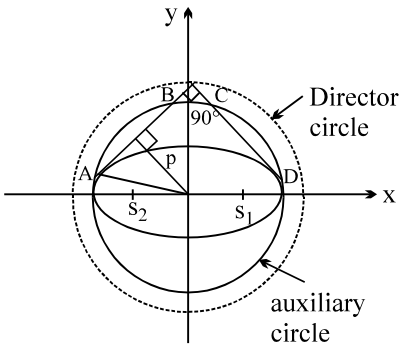
$2xy = k^2y^2 - h^2\ b^2$

$h^2b^2 = k^2y^2 - 2xy$

7. Two tangents to an ellipse intersect at right angles. Prove that the sum of the square of the chords which the auxiliary circle intersects on them is constant and equal to the square on the line joining the foci.

Ans. $4a^2e^2$ or $4(a^2 - b^2)$

Sol. TPT $(AB)^2 + (CD)^2 = (S_1S_2)^2$



$$y = mx + \sqrt{a^2m^2 + b^2} \quad ; \quad mx - y + \sqrt{a^2m^2 + b^2} = 0$$

$$p = \frac{\sqrt{a^2m^2 + b^2}}{\sqrt{1 + m^2}}$$

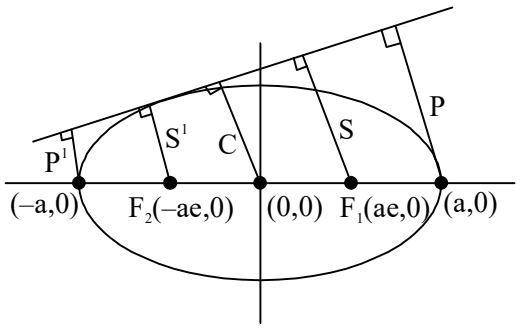
$$l_1^2 = a^2 - \frac{a^2m^2 + b^2}{1 + m^2} \quad ; \quad l_1^2 = \frac{a^2 - b^2}{1 + m^2}$$

now $$l_2^2 = \frac{a^2 - b^2}{1 + \frac{1}{m^2}} = \frac{(a^2 - b^2)m^2}{1 + m^2}$$

$$l_1^2 + l_2^2 = a^2 - b^2 \quad ; \quad \therefore \quad 4l_1^2 + 4l^2 = 4(a^2 - b^2) = (2ae)^2$$

8. If s, s' are the lengths of perpendicular on a tangent to the ellipse from the foci ; p, p' from the vertex and c that from the centre then show that $c^2 - ss' = e^2(c^2 - pp')$.

Sol.



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Eq^n of tangent

$$y = mx \pm \sqrt{a^2m^2 + b^2}$$

$$C = \left|\frac{\sqrt{a^2m^2 + b^2}}{\sqrt{1 + m^2}}\right| \qquad S = \left|\frac{mae \pm \sqrt{a^2m^2 + b^2}}{\sqrt{1 + m^2}}\right|$$

$$C^2 = \frac{a^2m^2 + b^2}{\left(1 + m^2\right)} \qquad S^1 = \left|\frac{-mae \pm \sqrt{a^2m^2 + b^2}}{\sqrt{1 + m^2}}\right|$$

$$SS^1 = \frac{\left(a^2m^2 + b^2\right) - \left(m^2a^2e^2\right)}{1 + m^2}$$

$$P = \left|\frac{am \pm \sqrt{a^2m^2 + b^2}}{\sqrt{1+m^2}}\right| \qquad P^1 = \left|\frac{-am \pm \sqrt{a^2m^2 + b^2}}{\sqrt{1+m^2}}\right|$$

$$PP^1 = \frac{\left(a^2m^2 + b^2\right) - \left(a^2m^2\right)}{1+m^2} \Rightarrow \frac{h^2}{1+m^2}$$

$$C^2 - SS^1 = \frac{a^2m^2 + b^2}{1+m^2} - \left(\frac{a^2m^2 + b^2 - a^2m^2e^2}{1+m^2}\right)$$

$$= \frac{a^2m^2 + e^2}{1+m^2}$$

$$e^2\left(c^2 - pp^1\right) = e^2\left(\frac{a^2m^2 + b^2}{1+m^2} - \frac{b^2}{1+m^2}\right)$$

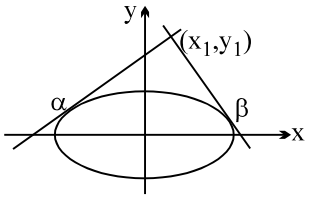
$$= \frac{a^2m^2 + e^2}{1+m^2}$$

$C^2 - SS^1 = e^2(c^2 - pp_1)$

9. Find locus of the point of intersection of the pair of tangents on an ellipse if the difference of the eccentric angle of their points of contact is $2\pi/3$.

Ans. $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 4$

Sol. $|\alpha - \beta| = 2\pi/3$



$$h = a\,\frac{\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}} = 2a\cos\frac{\alpha+\beta}{2} \quad \text{and} \quad k = 2b\sin\frac{\alpha+\beta}{2}$$

$$\therefore \qquad \frac{x^2}{4a^2} + \frac{y^2}{4b^2} = 1 \qquad \Rightarrow \qquad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 4$$

10. Find the equations to the normals at the ends of the latera recta, and prove that each passes through an end of the minor axis if $e^4 + e^2 = 1$.

Ans. $\dfrac{a^2x}{ae} - \dfrac{b^2ya}{b^2} = a^2e^2$

Sol. $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

$$\frac{a^2x}{ae} - \frac{b^2ya}{b^2} = a^2e^2$$

$\Rightarrow \dfrac{ax}{e} - ay = a^2e^2$. This passes through $(0, -b)$

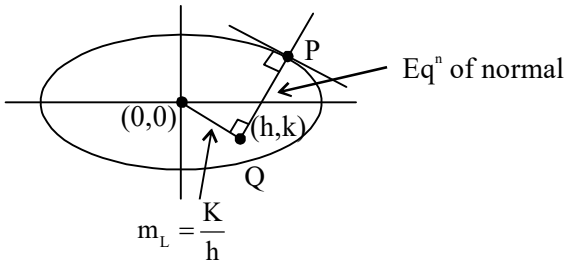
$ab = a^2e^2 \Rightarrow a^2 \cdot a^2(1 - e^2) = a^4\,e^4$

$\Rightarrow e^4 + e^2 - 1 = 0$

11. Find the locus of the feet of the perpendiculars from centre on a normal to a standard ellipse.

Ans $\left(\frac{x^2+y^2}{a^2-b^2}\right)^2\left(\frac{a^2}{x^2}+\frac{b^2}{y^2}\right)=1$

Sol.



$\frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2-b^2$ .....(1)

$m_1 m_2=-1$ Eqn of PQ line

$m_1=-\frac{h}{k}$ $y-k=-\frac{h}{k}(x-h)$

$Ky-K^2=-hx+h^2$

$hx+Ky=h^2+K^2$ .....(2)

compare (1) and (2)

$\frac{h\cos\theta}{a}=\frac{K\sin\theta}{-b}=\frac{h^2+K^2}{a^2-b^2}$

$\cos\theta=\frac{a}{h}\left(\frac{h^2+K^2}{a^2-b^2}\right)$ $\sin\theta=-\frac{b}{K}\left(\frac{h^2+K^2}{a^2-b^2}\right)$

$\left(\frac{x^2+y^2}{a^2-b^2}\right)^2\left(\frac{a^2}{x^2}+\frac{b^2}{y^2}\right)=1$

12. Find Locus of the point of intersection of the pair of tangents to an ellipse if the sum of the ordinates of their point of contact is b. **[O]**

Ans. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{2y}{b}$

Sol. Given $\sin\alpha+\sin\beta=1$ ....(1)

$x_1=a\frac{\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}$; $y_1=b\frac{\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}$

hence $\left(\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}\right)\cos^2\frac{\alpha-\beta}{2}=1$ ....(2)

now (1) gives $2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} = 1$ ....(3)

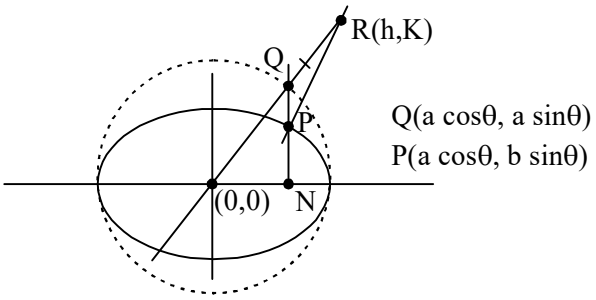
or $\frac{2y_1}{b}\cos^2\frac{\alpha-\beta}{2} = 1$ [Using $\sin\frac{\alpha+\beta}{2} = \frac{y_1}{b}\cos\frac{\alpha-\beta}{2}$]

$\cos^2\frac{\alpha-\beta}{2} = \frac{b}{2y_1}$; $\therefore$ $\left(\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}\right)\frac{b}{2y_1} = 1$

locus is $\frac{x^2}{a^2}+\frac{y^2}{b^2} = \frac{2y}{b}$

13. Any ordinate NP of an ellipse meets the auxiliary circle in Q ; prove that the locus of the intersection of the normals at P and Q is the circle $x^2 + y^2 = (a + b)^2$.

Sol.



N at circle at the point Q

$y = x\tan\theta$. It passes through (h, k)

$k = h\tan\theta$ ....(1)

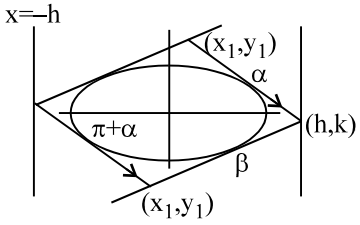
normal at P : $\frac{ah}{\cos\theta}-\frac{bk}{\sin\theta} = a^2 - b^2$ ....(2)

using $\tan\theta = \frac{k}{h}$, $\frac{ah\sqrt{h^2+k^2}}{h}-\frac{bk\sqrt{h^2+k^2}}{k} = a^2 - b^2$

$\sqrt{h^2+k^2} = a + b \Rightarrow x^2 + y^2 = (a + b)^2$

14. A parallelogram circumscribes the ellipse and two of its opposite angular points lie on the straight lines $x^2 = h^2$; prove that the locus of the other two is the conic $\frac{x^2}{a^2}+\frac{y^2}{b^2}\left(1-\frac{a^2}{h^2}\right) = 1$ **[O]**

Sol. $h = \frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}$ ....(1) ; $k = \frac{b\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}$



$\alpha \to \pi + \alpha$, we will get the point of intersection of tangent

at β and $\pi + \alpha$

$$x_1 = \frac{a\cos\frac{\pi+\alpha+\beta}{2}}{\cos\frac{\pi+\alpha-\beta}{2}} \quad ; \quad y_1 = \frac{b\sin\frac{\pi+\alpha+\beta}{2}}{\cos\frac{\pi+\alpha-\beta}{2}}$$

$$x_1 = \frac{a\sin\frac{\alpha+\beta}{2}}{\sin\frac{\alpha-\beta}{2}} \quad ....(2) \quad ; \quad y_1 = -\frac{b\cos\frac{\alpha+\beta}{2}}{\sin\frac{\alpha-\beta}{2}} \quad ....(3)$$

from (2) and (3), $\left(\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}\right)\sin^2\left(\frac{\alpha-\beta}{2}\right) = 1$ ....(4)

(1) and (3) gives, $\frac{h}{y_1} = -\frac{a}{b}\tan\left(\frac{\alpha-\beta}{2}\right)$

hence $\cot\left(\frac{\alpha-\beta}{2}\right) = -\frac{ay}{bh}$; $\therefore \left(\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}\right) = 1+\cot^2\left(\frac{\alpha-\beta}{2}\right) = 1+\frac{a^2y^2}{b^2h^2}$

5 COMMON CONCEPTS

1. Chord with a given middle point :

Equation of the chord of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$ whose middle point is (x_1, y_1) is $T = S_1$,

where $S_1 = \frac{x_1^2}{a^2}+\frac{y_1^2}{b^2} - 1$; $T \equiv \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$.

2. Chord of Contact :

Equation to the chord of contact of tangents drawn from a point $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $T = 0$, where $T = \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$

3. Pair of Tangents :

The equation to the pair of tangents which can be drawn from any point (x_1, y_1) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is given by: $SS_1 = T^2$ where :

$S \equiv \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$; $S_1 = \frac{x_1^2}{a^2}+\frac{y_1^2}{b^2} - 1$; $T \equiv \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$.

4. Polar :

Equation to the polar for a point $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $T = 0$,

where $T = \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$

5. Diameter :

mid–pt. (h, k) slope is m $\Rightarrow$ $\frac{hx}{a^2}+\frac{ky}{b^2}=\frac{h^2}{a^2}+\frac{k^2}{b^2}$ $[T = S_1]$

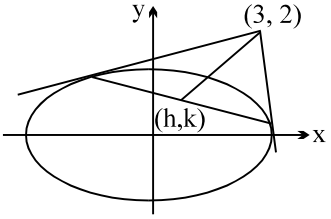
$\frac{-hb^2}{ka^2} = m \Rightarrow k = \frac{-b^2h}{a^2m} \Rightarrow y = \frac{-b^2}{a^2m}x$

Questions on Common concepts

1. Tangents are drawn from the point (3, 2) to the ellipse $x^2 + 4y^2 = 9$. Find the equation to their chord of contact and the equation of the straight line joining (3, 2) to the middle point of this chord of contact.

Ans. $3x + 8y = 9$; $2x = 3y$]

Sol. $x^2 + 4y^2 = 9$



equation of the chord of contact of the pair of tangents from (3, 2)

$3x + 8y = 9$ ....(1)

This must be the same as chord whose middle point is (h, k).

$T = S_1$

$$\frac{hx}{9}+\frac{ky}{9/4}=\frac{h^2}{9}+\frac{k^2}{9/4} \Rightarrow \frac{hx}{9}+\frac{4ky}{9}=\frac{h^2}{9}+\frac{4k^2}{9} \Rightarrow hx + 4ky = h^2 + 4k^2 \quad ....(2)$$

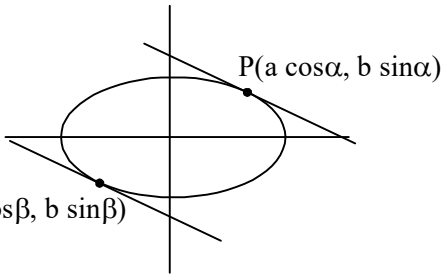
comparing coefficient of (1) and (2),

$$\frac{h}{3}=\frac{4k}{8}=\frac{h^2+4k^2}{9} \quad ; \quad 2h = 3k \quad ; \quad \text{also} \quad 3h = h^2 + 4k^2$$

$$3h = h^2 + 4\cdot\frac{4h^2}{9} \Rightarrow \frac{25h^2}{9} = 3h \Rightarrow h = \frac{27}{25} \quad \& \quad k = \frac{18}{25}$$

2. PT : eccentric angle of point of contact of two parallel tangents differ by 180°.

Sol.



$\frac{x}{a}\cos\alpha + \frac{y}{b}\sin\alpha = 1$ $\qquad$ $\frac{x}{a}\cos\beta + \frac{y}{b}\sin\beta = 1$

$\text{Slope} = -\frac{b\cos\alpha}{a\sin\alpha}$ $\qquad$ $\text{Slope} = -\frac{b\cos\beta}{a\sin\beta}$

$\frac{\cos\alpha}{\sin\alpha} = \frac{\cos\beta}{\sin\beta}$

$\cos\alpha \sin\beta = \sin\alpha \cos\beta$

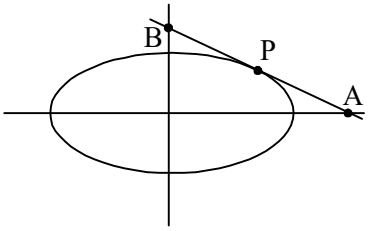
$\sin(\alpha - \beta) = 0$

$\alpha - \beta = \pi \qquad \alpha - \beta = 0$

H.P. $\alpha = \beta$.

3. Find locus of mid-point of intercept mades by a **tangent**/normal to the standard ellipse on the coordinate axes.

Ans



Eq^r of tangent $\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$

$A\left(\frac{a}{\cos\theta}, 0\right) \quad B\left(0, \frac{b}{\sin\theta}\right)$

Mid pt (h, K) = $\left(\frac{a}{2\cos\theta}, \frac{b}{2\sin\theta}\right)$

$\cos\theta = \frac{a}{2h} \qquad \sin\theta = \frac{b}{2K}$

$\frac{a^2}{4h^2} + \frac{b^2}{4K^2} = 1$

$\frac{a^2}{x^2} + \frac{b^2}{y^2} = 4$

Eq^n of normal $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$

$C\left(\frac{\left(a^2 - b^2\right)\cos\theta}{a}, 0\right) D\left(0, \frac{\left(a^2 - b^2\right)\sin\theta}{-b}\right)$

Mid pt. (h, K) = $\left(\frac{\left(a^2 - b^2\right)\cos\theta}{2a}, \frac{\left(a^2 - b^2\right)\sin\theta}{-2b}\right)$

$\frac{2ha}{a^2 - b^2} = \cos\theta \qquad \frac{-2Kb}{a^2 - b^2} = \sin\theta$

$\frac{4x^2a^2}{\left(a^2 - b^2\right)^2} + \frac{4y^2b^2}{\left(a^2 - b^2\right)} = 1$

$4(a^2x^2 + b^2y^2) = (a^2 - b^2)^2$

4 Chords of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ touch the parabola $ay^2 = -2b^2x$; Prove that the locus of their poles is the parabola $ay^2 = 2b^2x$.

Sol. Let the pole be (h, k). It's polar w.r.t. the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

$$\frac{hx}{a^2}+\frac{ky}{b^2}=1 \qquad .....(1)$$

Now (1) must be the tangent to $ay^2 = -2b^2x$ ....(2)

from (1) $\frac{hx}{a^2}=1-\frac{ky}{b^2} \Rightarrow x=\frac{a^2}{h}\left(1-\frac{ky}{b^2}\right)$

Hence (2) becomes

$ay^2 = -2a^2b^2 + 2a^2ky$; $ahy^2 - 2a^2ky + 2a^2b^2 = 0$

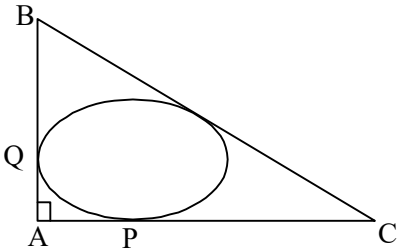
$D = 0$

$4a^4k^2 = 4ah \cdot 2a^2b^2$; $ak^2 = b^2h$; $\therefore$ $ay^2 = 2b^2x$

5. ΔABC right angled at A circumscribes the standard ellipse, find locus of point A.

Ans $x^2 + y^2 = a^2 + b^2$

sol.



(h, K)

Tangent at AP and AQ

$y = mx \pm \sqrt{a^2m^2 + b^2}$

(h, K)

$(K - mh)^2 = a^2m^2 + b^2$

$(h^2 - a^2)m^2 - 2Khm + K^2 - b^2 = 0 \begin{cases} m_1 \\ m_2 \end{cases}$

$m_1m_2 = -1$

$\frac{K^2-b^2}{h^2-a^2}=-1$

$K^2 - b^2 = -h^2 + a^2$

$x^2 + y^2 = a^2 + b^2$

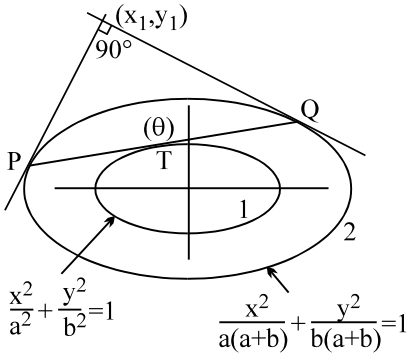
6. A tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meets the ellipse $\frac{x^2}{a}+\frac{y^2}{b}=a+b$ in the points P and Q ; prove that the tangents at P and Q are at right angles.

Sol. Chord of contact of (h, k) w.r.t ellipse–2

is $\frac{xx_1}{a(a+b)}+\frac{yy_1}{b(a+b)}=1$ ....(1)

(1) must be the same on the tangent at 'θ' on standard ellipse–1

T at (1) is, $\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$ ....(2)



comparing (1) and (2), we get

$$\frac{x_1}{a(a+b)}\cdot\frac{a}{\cos\theta}=\frac{y_1}{b(a+b)}\cdot\frac{b}{\sin\theta}=1$$

$x_1^2=\cos^2\theta\,(a+b)^2 \quad y_1^2=\sin^2\theta\,(a+b)^2$

$x_1^2+y_1^2=(a+b)^2$

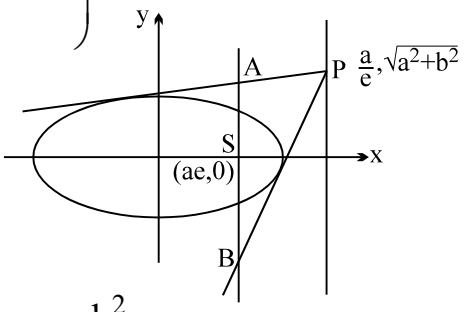
which is the director circle of ellipse–2

7. Tangents are drawn to the ellipse from the point $\left(\frac{a^2}{\sqrt{a^2-b^2}},\sqrt{a^2+b^2}\right)$.

Prove that they intercept on the ordinate through the nearer focus a distance equal to the major axis. **[O]**

Sol. TPT = AB = 2a ; using $SS_1=T^2$

$$\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}-1\right)\left(\frac{1}{e^2}+\frac{a^2+b^2}{b^2}-1\right)=\left(\frac{x\cdot a}{e\cdot a^2}+\frac{y\cdot\sqrt{a^2+b^2}}{b^2}-1\right)^2$$



solving this with x = ae, we get

$$\left(e^2+\frac{y^2}{b^2}-1\right)\left(\frac{1}{e^2}+\frac{a^2}{b^2}\right)=\frac{y^2(a^2+b^2)}{b^4}$$

$$\left(\frac{y^2}{b^2}-\frac{b^2}{a^2}\right)\left(\frac{b^2+a^2-b^2}{b^2e^2}\right)=\frac{y^2(a^2+b^2)}{b^2\cdot b^2} \qquad \left(\text{Using } e^2=1-\frac{b^2}{a^2}\right)$$

$$\left(\frac{a^2y^2-b^4}{a^2b^2}\right)\left(\frac{a^2}{b^2e^2}\right)=\frac{y^2(a^2+b^2)}{b^4}$$

$a^2(a^2y^2-b^4)=y^2(a^4-b^4) \qquad (a^2e^2=a^2-b^2)$

$a^2b^4=y^2b^4 \quad \Rightarrow \quad y^2=a^2 \Rightarrow \quad y=\pm a \Rightarrow \quad AB=2a$

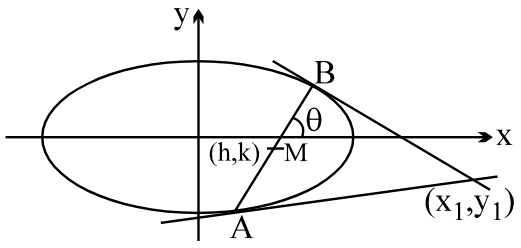
8. Find the locus of the middle points of chords of an ellipse

(a) which subtend a right angle at their centre.

(b) the tangent at the ends of which intersect at right angles. **[O]**

Ans. (a) $(a^2 + b^2)(b^2x^2 + a^2y^2)^2 = a^2b^2(b^4x^2 + a^4y^2)$;
(b) $a^4b^4(x^2 + y^2) = (a^2 + b^2)(b^2x^2 + a^2y^2)^2$

Sol. **(b)** $\dfrac{x-h}{\cos\theta} = \dfrac{y-k}{\sin\theta} = \pm c$



$$\left.\begin{aligned} x &= h + c\cdot\cos\theta \\ y &= k + c\cdot\sin\theta \end{aligned}\right] \;\&\; \left[\begin{aligned} x &= h - c\cdot\cos\theta \\ y &= k - c\cdot\sin\theta \end{aligned}\right]$$

substituting in the equation $b^2x^2 + a^2y^2 = a^2b^2$
we get, $b^2(h + c\cdot\cos\theta)^2 + a^2(k + c\cdot\cos\theta)^2 = a^2b^2$ ....(1)
$b^2(h - c\cdot\cos\theta)^2 + a^2(k - c\cdot\cos\theta)^2 = a^2b^2$ ....(2)
(1) add ; (2) subtract and eliminate θ.

(c) If (x_1, y_1) is the point of intersection then its C.O.C. and midpoint must be same. Get x_1, y_1 in term of h, k. Now

$x_1^2 + y_1^2 = a^2 + b^2$

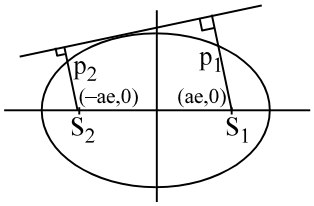
6.1 IMPORTANT HIGHLIGHTS :

Referring to an ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

1. If P be any point on the ellipse with S & S′ as its foci then $\ell(PS) + \ell(PS') = 2a$.

2. Product of the length's of the perpendiculars from either focus on a variable tangent to an Ellipse= (semi minor axis)$^2 = b^2$

Explanation: $mx - y + \sqrt{a^2m^2 + b^2} = 0$



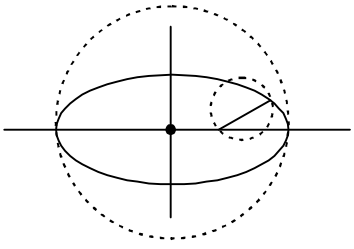
$$p_1p_2 = \left| \frac{mae + \sqrt{a^2m^2 + b^2}}{\sqrt{1+m^2}} \cdot \frac{-mae + \sqrt{a^2m^2 + b^2}}{\sqrt{1+m^2}} \right|$$

$$= \left| \frac{(a^2m^2 + b^2) - m^2a^2e^2}{1+m^2} \right|$$

$$= \left| \frac{a^2m^2 + b^2 - m^2(a^2 - b^2)}{1+m^2} \right| = \left| \frac{b^2(1+m^2)}{1+m^2} \right| = b^2$$

3. The portion of the tangent to an ellipse between the point of contact & the directrix subtends a right angle at the corresponding focus.

4. The circle drawn on any focal distance as diameter touches the auxiliary circle.



5. (i) Feet of the perpendiculars from either foci on a variable tangent to an ellipse lies on its auxiliary circle.

(ii) If Y_1 and Y_2 be the feet of the perpendiculars on the auxiliay circle from the foci upon any tangent, at P on the ellipse, then the point of intersection 'Q' of the tangents at Y_1 and Y_2 lies on the ordinate through P. If P varies then the locus of Q is an ellipse having the same eccentricity as the original ellipse.

Explanation:

Chord of Contact (C.O.C) w.r.t the circle $x^2 + y^2 = a^2$ is

$$hx + ky = a^2 \qquad \text{....(1)}$$

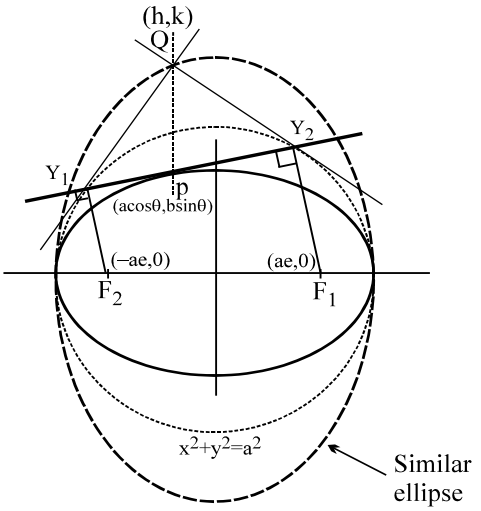
This must be the same tangent at P(θ)

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1 \qquad \text{....(2)}$$

Comparing (1) and (2) $\frac{ha}{\cos\theta} + \frac{kb}{\sin\theta} = a^2$

$$\left.\begin{aligned} h &= a\cos\theta \\ k &= \frac{a^2\sin\theta}{b} \end{aligned}\right\}$$

$$\frac{x^2}{a^2} + \frac{y^2}{\left(\frac{a^2}{b}\right)^2} = 1$$



Which is an ellipse whose eccentricity e' is given by

$$e'^2 = 1 - \frac{a^2b^2}{a^4} \Rightarrow \qquad e'^2 = 1 - \frac{b^2}{a^2} = e^2 \qquad]$$

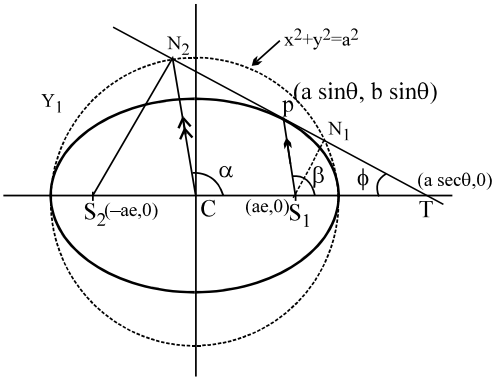
(iii) Lines joining centre to the feet of perpendicular from a focus on any tangent at P and the line joining other focus to the point of contant 'P' are parallel.

Explanation:

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$$

$$\frac{CT}{S_1T} = \frac{a\sec\theta}{a\sec\theta - ae} = \frac{a\sec\theta}{a(\sec\theta - e)} \Rightarrow \qquad \frac{CT}{S_1T} = \frac{1}{1 - e\cos\theta}$$

Again $\frac{CN_2}{S_1P} = \frac{a}{e\left(\frac{a}{e} - a\cos\theta\right)} = \frac{a}{a - ae\cos\theta} = \frac{1}{1 - e\cos\theta} \Rightarrow \frac{CT}{ST} = \frac{CN_2}{S_1P}$



Now use sine law in triangles CN_2T and S_1PT we can prove $\alpha + \phi = \beta + \phi$

i.e. $\dfrac{CT}{\sin(\alpha+\phi)} = \dfrac{a}{\sin\phi}$ and $\dfrac{S_1P}{\sin\phi} = \dfrac{S_1T}{\sin(\beta+\phi)}$

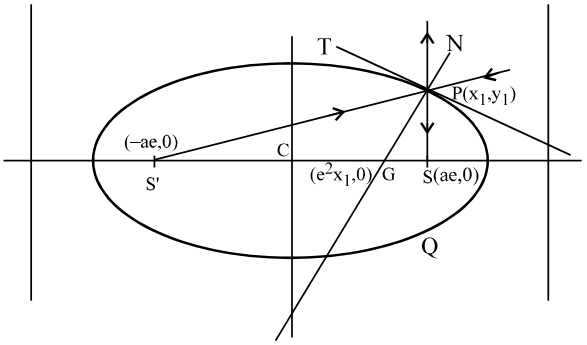
$\therefore$ $\dfrac{CT}{a} = \dfrac{\sin(\alpha+\phi)}{\sin\phi}$ and $\dfrac{S_1T}{S_1P} = \dfrac{\sin(\beta+\phi)}{\sin\phi}$

hence $\sin(\alpha + \phi) = \sin(\beta + \phi)$]

6. The tangent & normal at a point P on the ellipse bisect the external & internal angles between the focal distances of P. This refers to the well known reflection property of the ellipse which states that rays from one focus are reflected through other focus & vice–versa. Hence we can deduce that the straight lines joining each focus to the foot of the perpendicular from the other focus upon the tangent at any point P meet on the normal PG and bisects it where G is the point where normal at P meets the major axis.

Reflection property :

$$\frac{S_2G}{GS_1} = \frac{e^2x_1 + ae}{ae - e^2x_1} = \frac{a + ex_1}{a - ex_1} \quad \text{or} \quad \frac{PS_2}{PS_1} = \frac{e\left(\frac{a}{e} + x_1\right)}{e\left(\frac{a}{e} - x_1\right)} = \frac{a + ex_1}{a - ex_1}$$



Also In ΔS_1PN_1 and ΔN_1PQ $\angle N_1PQ = \angle N_2PS_1 = \theta$ (neglect property)

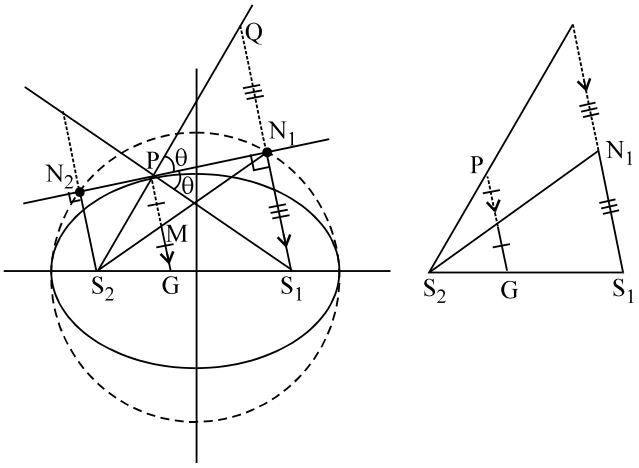
PN_1 is common

$\angle QN_1P = PN_1S_1 = 90°$

$\therefore \quad S_1PN_1 = N_1PQ_1$

$\therefore \quad N_1$ is the mid point of S_1Q

Now proceed



6.2 EXTRA QUESTION [O]

If the normal at any point P on the ellipse with centre C meet the major & minor axes in G & g respectively, Tangent at P meet the major and minor axiz at T and t respectively if CF and CY are perpendicular upon this normal and tangent respectively, then

(i) $PF \cdot PG = b^2$ **(ii)** $PF \cdot Pg = a^2$

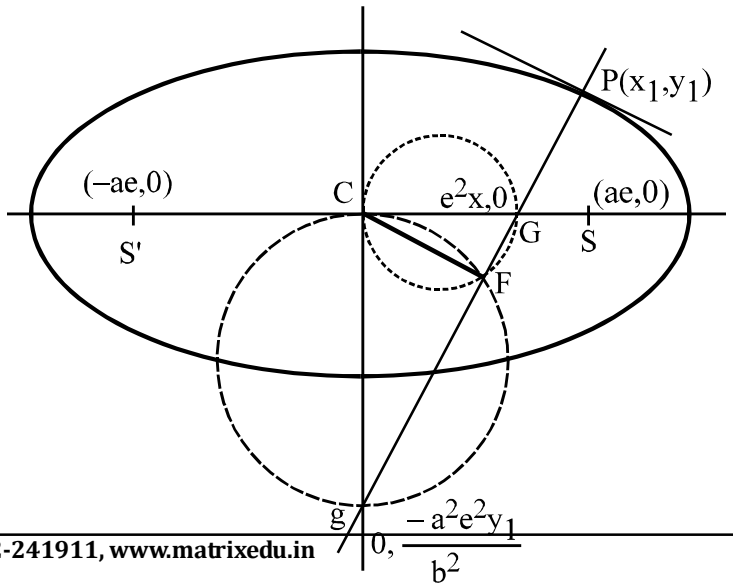
(iii) $PG \cdot Pg = SP \cdot S'P$ **(iv)** $CG \cdot CT = (CS)^2$

(v) $Tt \cdot PY = a^2 - b^2$ **(vi)** least value of Tt is $a + b$

(vii) locus of the mid point of Gg is another ellipse having the same eccentricity as that of the original ellipse.

[where S and S′ are the focii of the ellipse]

Sol (i) $PF \cdot PG = b^2 \dfrac{a^2x}{x_1} - \dfrac{b^2y}{y_1} = a^2e^2$



LHS=Power of the point P w.r.t. the circle on CG as diameter

$= x_1(x_1 - e^2x_1) + y_1^2$

$= x_1^2(1 - e^2) + y_1^2$

$= a^2\cos^2\theta \left(1 - 1 + \dfrac{b^2}{a^2}\right) + b^2\sin^2\theta$

$= b^2\cos^2\theta + b^2\sin^2\theta$
$= b^2$]

(ii) $PF \cdot Pg = a^2$

LHS = Power of the point P w.r.t.
the circle on Cg as diameter

$$= x_1^2 + y_1\left(y_1 + \frac{a^2e^2y_1}{b^2}\right) = a^2\cos^2\theta + b^2\sin^2\theta\left(1 + \frac{(a^2 - b^2)}{b^2}\right) + b^2\sin^2\theta$$

$$= a^2\cos^2\theta + b^2\sin^2\theta \cdot \frac{a^2}{h^2} = a^2 \ ]$$

(iii) $PG \cdot Pg = SP \cdot S'P$

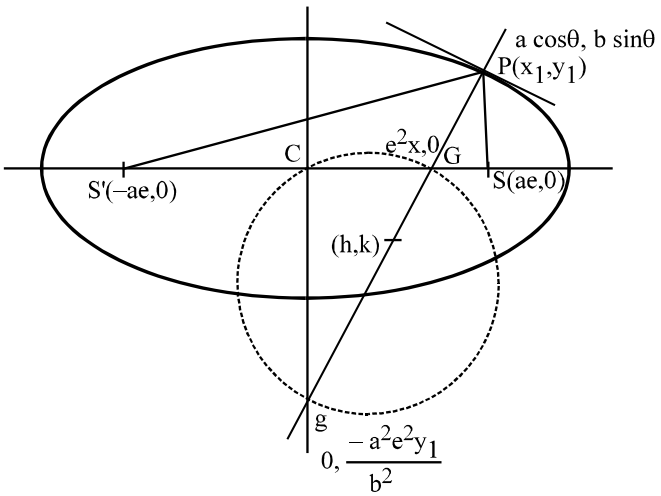
RHS $= (a - ae\cos\theta)(a + ae\cos\theta)$

$a^2 - a^2e^2\cos^2\theta$

$a^2 - (a^2 - b^2)\cos^2\theta$

$a^2\sin^2\theta + b^2\cos^2\theta$

LHS = Power of P w.r.t
the circle on Gg as diameter



$$= x_1(x_1 - e^2x_1) + y_1\left(y_1 + \frac{a^2e^2y_1}{b^2}\right)$$

$$= x_1^2(1 - e^2) + y_1^2\left(1 + \frac{a^2e^2}{b^2}\right) = x_1^2\left(\frac{b^2}{a^2}\right) + y_1^2\left(1 + \frac{a^2 - b^2}{b^2}\right)$$

$= b^2\cos^2\theta + a^2\sin^2\theta = RHS$]

(iv) $CG \cdot CT = (CS)^2$

MATHEMATICS

HYPERBOLA

THEORY SHEET

ENGLISH

MATRIX

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1.1 HYPERBOLA : INTRODUCTION

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will represent an ellipse

if $\Delta \neq 0$; $h^2 > ab$ and $e > 1$

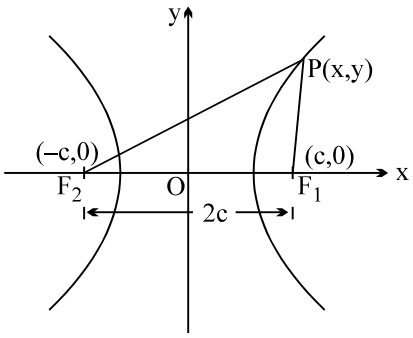
It is locus of a point moving in such a way that difference of its distances from two fixed points is always a constant.

Standard Equation :

i.e. $|PS_1 - PS_2| = 2a$ $\quad (2a < 2c \quad \text{i.e. } c > a)$

i.e. $PS_1 - PS_2 = \pm 2a$

i.e. $\sqrt{(x-c)^2 + y^2} - \sqrt{(x+c)^2 + y^2} = \pm 2a$

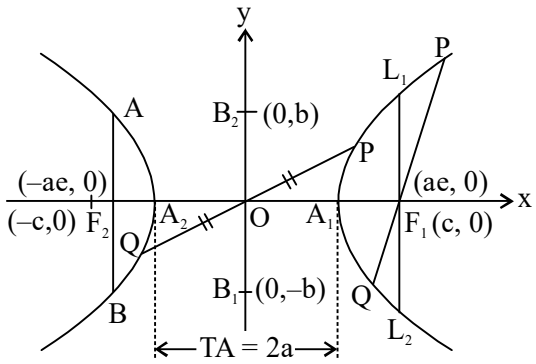


simplifying we get $\dfrac{x^2}{a^2} - \dfrac{y^2}{c^2 - a^2} = 1$

or $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ $\quad (\text{where } c^2 - a^2 = b^2) \quad ....(1)$

which is the same as the standard equation of ellipse except b^2 has been replaced by $-b^2$.

1. 2 IMPORTANT TERMS



1. This curve is symmetric about both the x and y axis.
2. S_1 and S_2 are called foci.
3. Line joining S_1 and S_2 is called transverse axis/focal axis
4. A_1 & A_2 are called ends of transverse axis & also called vertices.
5. Perpendicular bisector of transverse axis is called conjugate axis
6. B_1 and B_2 are ends of conjugate axis.
7. Length of transverse axis = 2a
8. Length of conjugate axis = 2b
9. Point of intersection of the transverse and conjugate axis is called the **centre** of the Hyperbola.
10. transverse and conjugate axes together are known as **principal axes** of the Hyperbola.
11. The distance between foci is called the **Focal Length**.
12. Any chord of the ellipse passing through center gets bisected by it and is called the **diameter**.
13. Any chord through focus is called a **focal chord**
14. Any chord perpendicular to the focal axis is called **double ordinate**.
15. A particular double ordinate through focus or a particular focal chord perpendicular to focal axis is called its **Latus Rectum**.

1.3 ECCENTRICITY

$$e = \frac{c}{a} \Rightarrow e^2 = \frac{c^2}{a^2} = \frac{a^2+b^2}{a^2} \Rightarrow e^2 = 1 + \frac{b^2}{a^2}$$

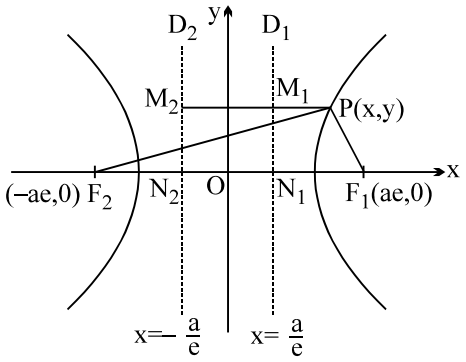
$$c^2 = a^2 + b^2 \Rightarrow b^2 = c^2 - a^2 \Rightarrow b^2 = a^2(e^2-1)$$

for ends of LR, put $x = ae$ in $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \Rightarrow \quad y = \pm\frac{b^2}{a}$

$$\Rightarrow \text{ends of LR} \left(\pm ae, \pm \frac{b^2}{a} \right)$$

$$\ell_{LR} = \frac{2b^2}{a} = \frac{(2b)^2}{2a} = \frac{(CA)^2}{TA} = \frac{2a^2(e^2-1)}{a} = 2a(e^2-1) = 2e\left(ae - \frac{a}{e} \right)$$

$\ell_{LR} = 2e$ (distance between focus & corresponding directrix)



focal distance of $P(x_1, y_1)$

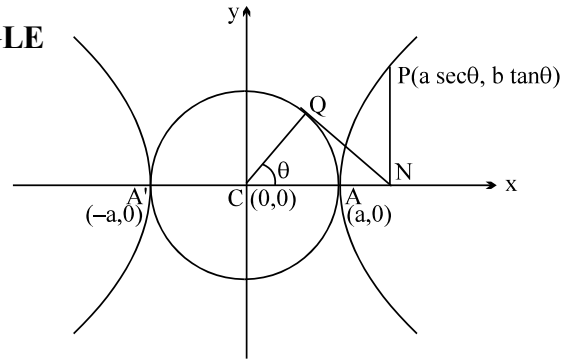
$$PS_1 = ePM_1 = e\left(x_1 - \frac{a}{e} \right) = ex_1 - a$$

$$PS_2 = ePM_2 = e\left(x_1 - \frac{a}{e} \right) = ex_1 + a$$

Two hyperbolas are said to be similar if they have the same value of eccentricity.

1.4 AUXILLIARY CIRCLE AND ECCENTRIC ANGLE

A circle drawn with centre C & T.A. as a diameter is called the Auxiliary Circle of the hyperbola. Equation of the auxiliary circle is $x^2 + y^2 = a^2$.



Note from the figure that P & Q are called the "Corresponding Points" on the hyperbola & the auxiliary circle. 'θ' is called the eccentric angle of the point 'P' on the hyperbola. ($0 \le \theta < 2\pi$).

$$H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1; \qquad AC: x^2 + y^2 = a^2$$

$Q(a\cos\theta, a\sin\theta)$ tangent at $Q \Rightarrow$ $x\cos\theta + y\sin\theta = a$

$N(a\sec\theta, 0) \Rightarrow$ $P(a\sec\theta, b\tan\theta) \equiv P(\theta)$

$\theta \in 1^{st}$ quad. $\rightarrow P \in 1^{st}$ quad.; $\theta \in 2^{nd}$ quad. $\rightarrow P \in 3^{rd}$ quad.;

$\theta \in 3^{rd}$ quad $\rightarrow P \in 2^{nd}$ quad. $\theta \in 4^{th}$ quad $\rightarrow P \in 4^{th}$ quad.

Note : The equations $x = a\sec\theta$ & $y = b\tan\theta$ together represents the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where θ is a parameter.

1.5 STANDARD HYPERBOLAS

	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
Center	$(0,0)$
Vertices	$(\pm a, 0)$
B_1, B_2	$(0, \pm b)$
ℓ_{TA}	$2a$
ℓ_{CA}	$2b$
foci	$(\pm ae, 0)$
directrices	$x = \pm a/e$
feet of dir.	$\left(\pm\frac{a}{e}, 0\right)$
ℓ_{LR}	$\frac{2b^2}{a}$
ends of LR	$\left(\pm ae, \pm\frac{b^2}{a}\right)$
aux. circle	$x^2 + y^2 = a^2$
eccentricity	$e^2 = 1 + \frac{b^2}{a^2}$

1.6 ALTERNATE DEFINITION OF HYPERBOLA

$p_1 \equiv$ distance of P from transverse axis.

$p_2 \equiv$ distance of P from conjugate axis.

$$\frac{p_2^2}{a^2} - \frac{p_1^2}{b^2} = 1$$

Questions on equation of Hyperbola

1. Find everything

(i) $\frac{x^2}{144} - \frac{y^2}{25} = 1$

Ans. Center (0, 0), Vertices $(\pm 12, 0)$, $B_1, B_2 (0, \pm 5)$, $l_{TA} = 24$, $l_{CA} = 10$, $e = \frac{13}{12}$, foci $(\pm 13, 0)$

Directrices $x = \pm \frac{144}{13}$, $l_{LR} = \frac{50}{12}$, Ends of LR $\left(\pm 13, \pm \frac{25}{12}\right)$, Aux. Circle $x^2 + y^2 = 144$

Sol. $\frac{x^2}{144} - \frac{y^2}{25} = 1$

$\frac{x^2}{(12)^2} - \frac{y^2}{(5)^2} = 1$ a = 12, b = 5

$e = \sqrt{1 + \frac{25}{144}} = \frac{13}{12}$

$ae = 13, \frac{a}{e} = \frac{144}{13}$ $\frac{b^2}{a} = \frac{25}{12}$

c(0, 0) $= Z_2 \left(\pm \frac{144}{13}, 0\right)$

Foci (±13, 0)

A_1A_2 (±12, 0) and of L.R. $= \left(\pm 13m \pm \frac{25}{12}\right)$

$B_1 B_2$ (0, ±5) $L_1R : 2 \times \frac{25}{12}$ $L_{TA} : 2 \times 12$

$L_{CA} : 2 \times 5$

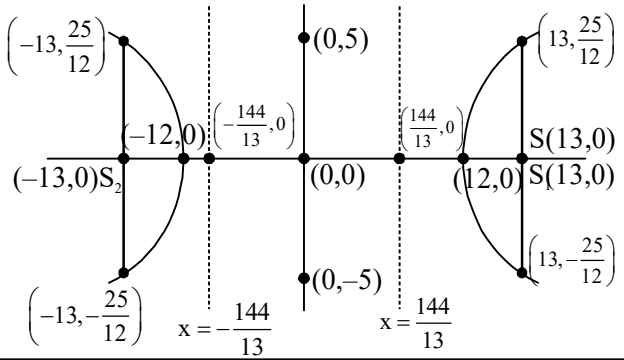
and of direction $\Rightarrow x = \pm \frac{144}{13}$ aux. wire $x^2 + y^2 = 144$

OR

a = 12, b = 5 $L_{TA} = 24$, $T_{CA} = 10$, $L_{L.R.} = \frac{50}{12} = \frac{25}{6}$

$\frac{x^2}{12^2} - \frac{y^2}{5^2} = 1$ $e = \frac{13}{12}\left\{\sqrt{1 - \frac{25}{144}}\right.$

Aux. circle $\rightarrow x^2 + y^2 = 144$

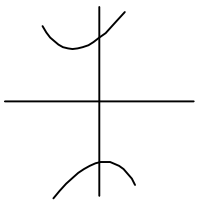


(ii) $\frac{x^2}{16}-\frac{y^2}{9}=-1$

Ans. Center (0, 0), Vertices $(0,\pm3)$, $B_1,B_2(\pm4,0)$, $l_{TA}=6$, $l_{CA}=8$, $e=\frac{5}{3}$, foci $(0,\pm5)$

Directrices $y=\pm\frac{9}{5}$, $l_{LR}=\frac{32}{3}$, Ends of LR $\left(\pm\frac{16}{3},\pm5\right)$, Aux. Circle $x^2+y^2=9$

Sol. $\frac{x^2}{16}-\frac{y^2}{9}=-1$



$\frac{y^2}{9}-\frac{x^2}{16}=1$

$a=3,\quad b=4$

$\frac{y^2}{32}-\frac{x^2}{4^2}=1$ $\qquad e=\sqrt{1+\frac{16}{9}}\qquad e=\frac{5}{3}$

$ae=5,\ \frac{a}{e}=\frac{9}{5},\ \frac{b^2}{a}=\frac{16}{3}$

c(0, 0)

foci (0, ±5) $\qquad z_1z_2$: (0, ±9/5)

A_1A_2 (0, ±3) $\qquad$ ean of diren : y ± 9/5

B_1B_2 (±4, 0) $\qquad$ end of L.R : (±16/3, ±5)

L_{TR} : 2 × 16/3

L_{TA} : 2 × 3

L_{CA} : 2 × 4

z_1z_2 : (0, ±915) $\qquad x=0\quad y=\pm9/5$

$x-1=0\qquad y+2=\pm9/5$

$x=1\qquad y=\frac{-1}{5},-\frac{19}{5}$

$\left(1,-\frac{1}{5}\right),\left(1,-\frac{19}{5}\right)$

ean of direction $\qquad y=\pm9/5$

$y+2=\pm9/15$

$y=-1/5\qquad y=-19/5$

$L_{TA}=2\times3\qquad L_{CA}=2\times4$

$L_{LR}=\frac{2\times16}{3}$

aux. circle : $(x-1)^2+(y+2)^2=9$

and of L.R. : $\left(\pm\frac{15}{3},\pm5\right)$

$= \pm\frac{16}{3}$ $\qquad y = \pm 5$

$x - 1 = \pm\frac{16}{3}$ $\qquad y + 2 = \pm 5$

$x \frac{19}{3}, -\frac{13}{3}$ $\qquad y = 3, -7$

$$\left(\frac{19}{3}, -7\right), \left(-\frac{13}{3}, -7\right)$$

$$\left(\frac{19}{3}, 3\right), \left(-\frac{13}{3}, 3\right)$$

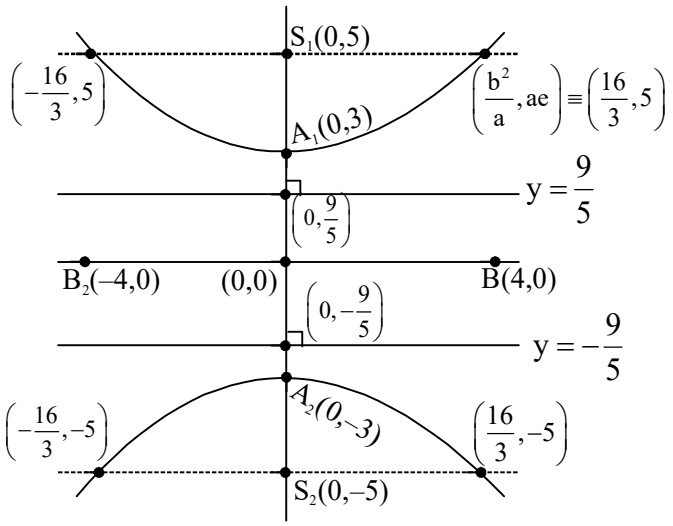
OR

$\Rightarrow \frac{y^2}{9} - \frac{x^2}{16} = 1$ $\qquad a = 3,\ b = 4$

$$e = \sqrt{1 + \frac{16}{9}} = \frac{5}{3}$$

$\ell_{TA} = 6,\ \ell_{CA} = 8,\ \ell_{L.R.} = \frac{2b^2}{a} = \frac{32}{3}$

Aux. circle $\rightarrow x^2 + y^2 = 9$



(iii) $9x^2 - 18x - 16y^2 - 64y + 89 = 0.$

Ans. Center (1, –2), Vertices (1, 1), (1, –5), B_1, B_2 (5, –2), (–3, –2), $l_{TA} = 6$, $l_{CA} = 8$, $e = \frac{5}{3}$, foci (1, 3), (1, –7)

Directrices $y = -\frac{1}{5}$, $y = -\frac{19}{5}$, $l_{LR} = \frac{32}{3}$,

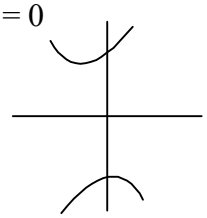
Ends of LR $\left(\frac{19}{3}, 3\right), \left(\frac{19}{3}, -7\right), \left(-\frac{13}{3}, 3\right), \left(-\frac{13}{3}, -7\right)$, Aux. Circle $(x-1)^2 + (y+2)^2 = 9$

Sol. $9x^2 - 18x - 16y^2 - 64y + 89 = 0$

$9(x^2 - 2x) - 16(y^2 + 4y) + 89 = 0$

$9(x^2 - 2x + 1 - 1) - 16(y^2 + 4y + 4 - 4) + 89 = 0$

$9(x-1)^2 - 16(y+2)^2 = -144$



$$-\frac{(x-1)^2}{16} + \frac{(y+2)^2}{9} = 1$$

$$\frac{(y+1)^2}{9} - \frac{(x-1)^2}{16} = 1$$

$\frac{y^2}{3^2} - \frac{x^2}{4^2} = 1$ $\quad a = 3,\ b = 4 \quad e = \sqrt{1 + \frac{16}{9}} = \frac{5}{3}$

$ae = 5,\ a/e = 9/5 \quad b^2/a = 16/3$

untre (0, 0)

$x = 0$	$y = 0$	
$x - 1 = 0$	$y + 2 = 0$	
$x = 1$	$y = -2$	c(1/–2)

foci $(0, \pm 5)$

$x = 0$	$y = \pm 5$
$x - 1 = 0$	$y + 2 = \pm 5$
$x = 1$	$y = 3, -7$

(1, 3), (1, –7)

A_1A_2 : $(0, \pm 3)$

$x = 0$	$y = \pm 3$
$x - 1 = 0$	$y + 2 = \pm 3$
$x = 1$	$y = 1, -5$

(1, 1), (1, –5)

B_1B_2 : $(\pm 4, 0)$

$x = \pm 4$	$y = 0$
$x - 1 = \pm 4$	$y + 2 = 0$
$x = 5, -3$	$y = -2$
	(5, –2), (–3, –2)

OR

$9(x^2 - 2x) - 16(y^2 + 4y) + 89 = 0$

$\Rightarrow 9(x-1)^2 - 9 - 16(y+2)^2 + 64 + 89 = 0$

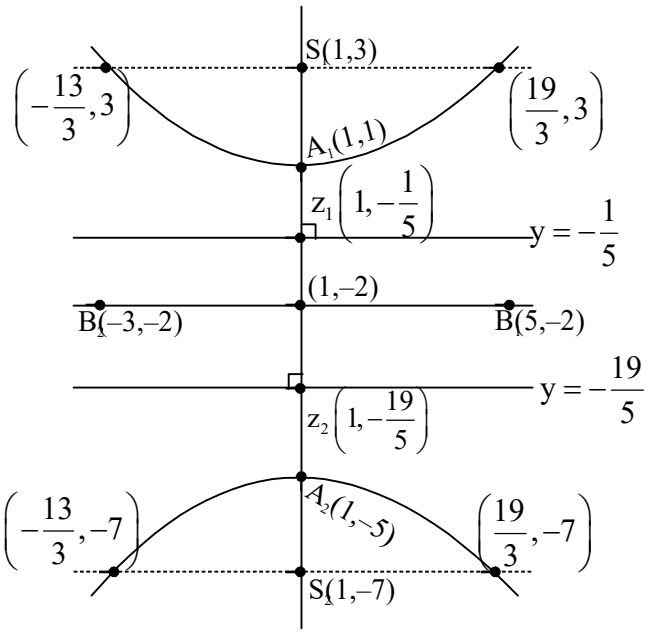
$$\Rightarrow \frac{9(x-1)^2}{-144} + \frac{16(y+2)^2}{144} = 1$$

$$\Rightarrow \frac{-(x+1)^2}{16} + \frac{(y+2)^2}{9} = 1, \qquad a = 3,\ b = 4$$

$$e = \sqrt{1+\frac{16}{9}} = \frac{5}{3}$$

$\ell_{TA} = 6,\ \ell_{CA} = 8,\ \ell_{L.R.} = \frac{32}{3}$

aux. circle $\rightarrow (x-1)^2 + (y+2)^2 = 9$



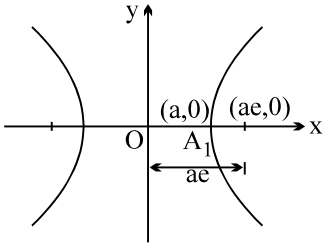
2. Find the equation of hyperbola referred to its principal axes as the coordinate axes

(a) if the distances of one of its vertices from the foci are 3 and 1.

Ans $x^2 - \frac{y^2}{3} = 1$ or $\frac{x^2}{3} - \frac{y^2}{1} = -1$

Sol. $ae - a = 1$ and $ae + a = 3$

$\therefore \quad \frac{e+1}{e-1} = 3 \quad \Rightarrow \quad e = 2$



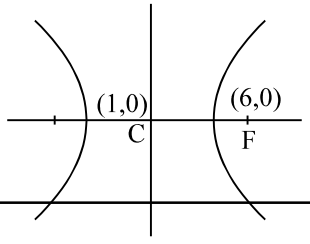
equation is $x^2 - \frac{y^2}{3} = 1$ or $\frac{x^2}{3} - \frac{y^2}{1} = -1$

(b) whose centre is (1, 0) ; focus is (6, 0) and length of transverse axis 6.

Ans $\frac{(x-1)^2}{9} - \frac{y^2}{16} = 1$

Sol. Equation of hyperbola with centre (1, 0)

$$\frac{(x-1)^2}{a^2} - \frac{y^2}{a^2(e^2-1)} = 1 \qquad ....(1)$$



given $a = 3$ and $ae = 5$ hence $e = \frac{5}{3}$

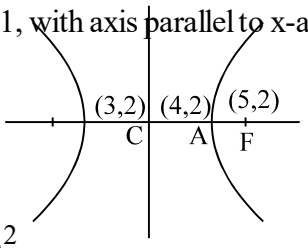
$\therefore$ equation (1) becomes

$$\frac{(x-1)^2}{9} - \frac{y^2}{9\left(\frac{25}{9}-1\right)} = 1 \Rightarrow \frac{(x-1)^2}{9} - \frac{y^2}{16} = 1$$

(c) whose centre is (3, 2), one focus is (5, 2) and one vertex is (4, 2).

Ans $(x-3)^2 - \frac{(y-2)^2}{3} = 1$

Sol. Equation of hyperbola with centre (3, 2) is $\frac{(x-3)^2}{a^2} - \frac{(y-2)^2}{a^2(e^2-1)} = 1$, with axis parallel to x-axis.



a = distance between centre are vertex = 1
ae = distance between centre and focus
hence $ae = 2 \Rightarrow e = 2$

$\therefore$ equation is, $\frac{(x-3)^2}{1} - \frac{(y-2)^2}{(4-1)} = 1 \Rightarrow (x-3)^2 - \frac{(y-2)^2}{3} = 1$

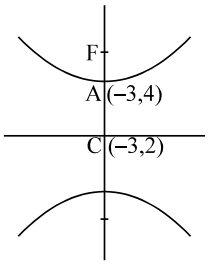
(d) whose centre is (–3, 2), one vertex is (–3, 4) and eccentricity is 5/2.

Ans $\frac{(x+3)^2}{21} - \frac{(y-2)^2}{4} = -1$

Sol. equation of hyperbola is $\frac{(x+3)^2}{a^2} - \frac{(y-2)^2}{b^2} = -1$

(as the line joining centre to the vertex is parallel to y-axis)
now $b = 2$ (distance between centre and vertex)

$$\text{focus} = be = 2 \cdot \frac{5}{2} = 5$$



also $a^2 = b^2(e^2 - 1) = 4\left(\frac{25}{4} - 1\right) = 21$

$\therefore$ $\frac{(x+3)^2}{21} - \frac{(y-2)^2}{4} = -1$

$4(x+3)^2 - 21(y-2)^2 = -84$

$4x^2 - 21y^2 + 24x + 84y + 36 = 0$

(e) whose foci are (4, 2) and (8, 2) and eccentricity is 2.

Ans $\frac{(x-6)^2}{1} - \frac{(y-2)^2}{3} = 1$

Sol. line joining foci is parallel to x-axis.
distance between the two foci = 4 = 2ae
hence a = 1 as e = 2

now centre is (5, 2) hence equation of hyperbola is $\frac{(x-6)^2}{1}-\frac{(y-2)^2}{3}=1$

3. Find the coordinates of the foci and the centre of the hyperbola $\frac{(3x-4y-12)^2}{100}-\frac{(4x+3y-12)^2}{225}=1$

Ans Center $\left(\frac{84}{25},-\frac{12}{25}\right)$

Sol. $\frac{(3x-4y-12)^2}{100}-\frac{(4x+3y-12)^2}{225}=1$

$$\frac{\left(\frac{3x-4y-12}{5}\right)^2}{4}-\frac{\left(\frac{nx+3y-12}{5}\right)^2}{9}=1$$

$a = 2 \quad b = 3$

$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1 \qquad e=\sqrt{1+\frac{9}{4}}=\frac{\sqrt{13}}{2}$

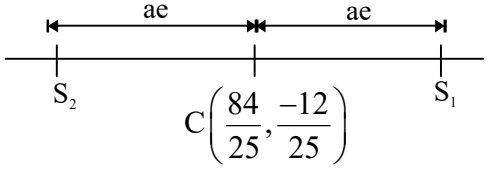
$3x - 4y - 12 = 0 \qquad 4x + 3y - 12 = 0$

$3x - 4y - 12 = 0) \times 4$
$4x + 3y - 12 = 0) \times 3$
$12x - 16y - 48 = 0$
$12x + 9y - 36 = 0$
$- \quad - \quad +$

$-25y - 12 = 0 \qquad y=\frac{-12}{25}$

$3x=4\times\frac{-12}{25}+12$

$3x=\frac{252}{25} \quad \Rightarrow x=\frac{8u}{25} \quad \cdot C\left(\frac{84}{25},\frac{-12}{25}\right)$



$ae = \sqrt{13}$

T.A. $4x + 3y - 12 = 0$

slope $=\frac{-4}{3} \quad \tan\theta=\frac{-4}{3} \quad \sin\theta=\frac{4}{5} \quad \cos\theta=\frac{-3}{5}$

$$S_1\left[\frac{84}{25}+\sqrt{13}\times\frac{-3}{5},\frac{-12}{25}+\sqrt{13}\times\frac{4}{5}\right]$$

$$S_1\left[\frac{84-15\sqrt{13}}{25},\frac{-12+20\sqrt{13}}{25}\right]$$

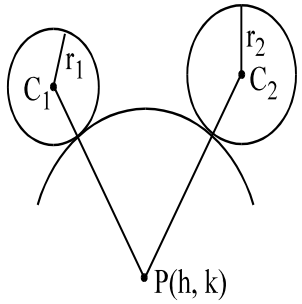
$$S_2\left[84-\sqrt{13}\times\frac{-3}{5}\ ,\frac{-12}{25}-\sqrt{13}\times\frac{4}{5}\right]$$

$$S_2\left[\frac{84+15\sqrt{13}}{25},\frac{-12-20\sqrt{13}}{25}\right]$$

4. Show that the locus of the centre of a circle which touches externally two given circles is a hyperbola.

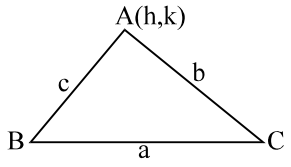
Sol. $(PC_1) - (PC_2) = (r + r_1) - (r + r_2) = r_1 - r_2$

$\Rightarrow$ a hyperbola (with centres of the given circles on its foci)]



5. Given the base of a triangle and the ratio of the tangents of half the base angles, prove that the vertex moves on a hyperbola whose foci are the extremities of the base. **[O]**

Sol. Given $\dfrac{\tan\frac{B}{2}}{\tan\frac{C}{2}}=\lambda\ ;\ \dfrac{\Delta}{s(s-b)}\dfrac{s(s-c)}{\Delta}=\lambda$



$$\frac{a+x}{a-x}=\lambda \quad \text{(where } b-c=x\text{)}$$

$\Rightarrow$ $x=\dfrac{a(\lambda-1)}{\lambda+1}$ i.e. $b-c=\dfrac{a(\lambda-1)}{\lambda+1}$ $\Rightarrow$ hyperbola

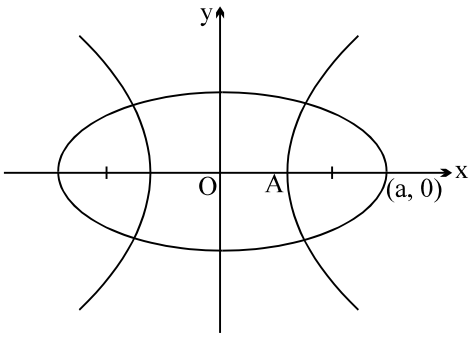
6. An ellipse and a hyperbola are confocal (have the same focus) and the conjugate axis of hyperbola is equal to the minor axis of the ellipse. If e_1 and e_2 are the eccentricities of ellipse and hyperbola then prove that

$$\frac{1}{e_1^2}+\frac{1}{e_2^2}=2.$$

Sol. We have, $ae_1 = Ae_2$ ....(1)

and $a^2(1 - e_1^2) = A^2(e_2^2 - 1)$ .....(2)

from (1) and (2) $\Rightarrow$ result]



$ae_1 = Ae_2$

$\Rightarrow a^2e_1^2 = A^2e_2^2$

$b^2 = a^2(1 - e_1^2)$ $\qquad$ $b^2 = A^2(e_2^2 - 1)$

$$\Rightarrow \frac{b^2e_1^2}{1-e_1^2} = \frac{b^2e_2^2}{e_2^2-1} \Rightarrow \frac{1-e_1^2}{e_1^2} = \frac{-1+e_2^2}{e_2^2}$$

$$\Rightarrow \frac{1}{e_1^2} + \frac{1}{e_2^2} = 2$$

1.7 CONJUGATE HYPERBOLA

Corresponding to every hyperbola there exist a hyperbola such that, the conjugate axis and transverse axis of one is equal to the transverse axis and conjugate axis of other, such hyperbolas are known as conjugate to each other.

Hence for the hyperbola, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ .....(1)

the conjugate hyperbola is, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ ....(2)

Example –

1. $\frac{x^2}{16} - \frac{y^2}{25} = 1$ CH: $\frac{x^2}{16} - \frac{y^2}{25} = -1$ $\qquad$ 2. $\frac{x^2}{9} - \frac{y^2}{4} = -1$ CH: $\frac{x^2}{9} - \frac{y^2}{4} = 1$

1. If e_1 and e_2 are the eccentricities of a hyperbola and its conjugate respectively, then $e_1^{-2} + e_2^{-2} = 1$
2. The foci of a hyperbola and its conjugate are concylic and form the vertices of a square.

1.8 RECTANGULAR HYPERBOLA

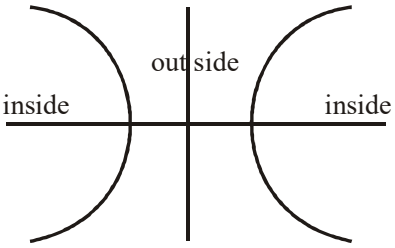
If a = b, hyperbola is said to be equilateral or rectangular and has the equation,

$x^2 - y^2 = a^2$. Eccentricity for such a hyperbola is $\sqrt{2}$.

$l_{LR} = 2a(e^2 - 1) = 2a = l_{TA} = l_{CA}$

2.1 POSITION OF A POINT WRT A HYPERBOLA

To determine position of a pt., we will take center as reference centre always lies outside the hyperbola.

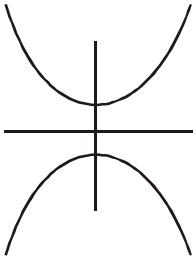


$P(x_2, y_1)$ $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

$S \equiv \frac{x^2}{a^2}-\frac{y^2}{b^2}-1;$ $\quad S_1 \equiv \frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}-1;$ $\quad T \equiv \frac{xx_1}{a^2}-\frac{yy_1}{b^2}-1$

$S_1 < 0 \rightarrow P$ lies outside hyperbola; $\quad S_1 = 0 \Rightarrow P$ lies on the hyperola

$S_1 > 0 \Rightarrow P$ lies inside the hyperbola



$\frac{x^2}{a^2}-\frac{y^2}{b^2}=-1$ $\qquad$ $\frac{y^2}{b^2}-\frac{x^2}{a^2}=1$

$S \equiv \frac{x^2}{a^2}-\frac{y^2}{b^2}+1$ $\qquad$ $S \equiv \frac{y^2}{b^2}-\frac{x^2}{a^2}-1$

$S_1 > 0 \Rightarrow$ outside $\qquad$ $S_1 < 0 \Rightarrow$ outside

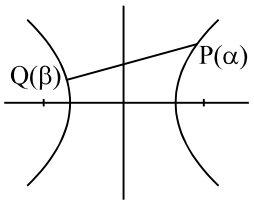
$S_1 < 0 \Rightarrow$ inside $\qquad$ $S_1 > 0 \Rightarrow$ inside

2.2 CHORD JOINING TWO POINTS

Chord joining two points with eccentric angles α and β is given by

$$\frac{x}{a}\cos\frac{\alpha-\beta}{2}-\frac{y}{b}\sin\frac{\alpha+\beta}{2}=\cos\frac{\alpha+\beta}{2} \quad ....(1)$$

if (1) passes through (d, 0) then



$$\frac{d}{a}\cos\frac{\alpha-\beta}{2} = \cos\frac{\alpha+\beta}{2}$$

$$\frac{d}{a} = \frac{\cos\frac{\alpha}{2}\cos\frac{\beta}{2} - \sin\frac{\alpha}{2}\sin\frac{\beta}{2}}{\cos\frac{\alpha}{2}\cos\frac{\beta}{2} + \sin\frac{\alpha}{2}\sin\frac{\beta}{2}}$$

$$\frac{x}{a}\cos\frac{\alpha+\beta}{2} + \frac{y}{b}\sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha-\beta}{2}$$

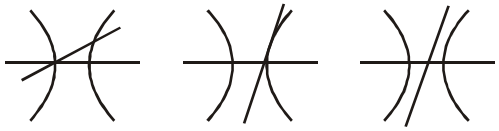
$$\frac{d+a}{d-a} = -\frac{\cos\frac{\alpha}{2}\cos\frac{\beta}{2}}{\sin\frac{\alpha}{2}\sin\frac{\beta}{2}} \quad \Rightarrow \quad \tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{a-d}{a+d}$$

3.1 LINE AND HYPERBOLA

Line $y = mx + c$; Hyperbola $b^2x^2 - a^2y^2 = a^2b^2$

$b^2x^2 - a^2(mx + c)^2 = a^2b^2$

$(b^2 - a^2m^2)x^2 - 2a^2mc\,x - a^2(b^2 + c^2) = 0$ ....(1)



$D > O$ $D = O$ $D < O$

$$C^2 = a^2m^2 - b^2$$

$D > 0$	$D = 0$	$D < 0$
Secant	Tangent	$\overline{T} \cap \overline{S}$

condition of tangency gives, $c^2 = a^2m^2 - b^2$

hence, $y = mx \pm \sqrt{a^2m^2 - b^2}$ is a tangent to the standard hyperbola. ....(1)

Hence for given m, there can be two parallel tangents to the hyperbola.

Example –

1. $\frac{x^2}{16} - \frac{y^2}{25} = 1 \Rightarrow$ COT $c^2 = 16m^2 - 25$

2. $\frac{x^2}{16} - \frac{y^2}{9} = -1 \Rightarrow \frac{x^2}{(-16)} - \frac{y^2}{(-9)} = 1 \Rightarrow$ COT $c^2 = -16m^2 - (-9)$

3.2 DIRECTOR CIRCLE

If it passes through (h, k) then

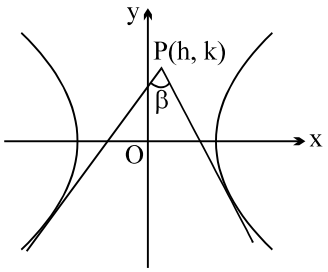
$(k - mh)^2 = a^2m^2 - b^2$

$(h^2 - a^2)m^2 - 2kmh + k^2 + b^2 = 0$ ....(2)

Hence passing through a given point (h, k) there can be a maximum of two tangents

now, $m_1 + m_2 = \dfrac{2kh}{h^2 - a^2}$ ....(3)

$m_1 m_2 = \dfrac{k^2 + b^2}{h^2 - a^2}$ ....(4)



If tangentss are perpendicular to each other then $m_1 m_2 = -1$ hence from (4)

$k^2 + b^2 = a^2 - h^2$

$x^2 + y^2 = a^2 - b^2$ which is the equation to the director circle, of the hyperbola.

If $l(TA) > l(CA)$; director circle is real with finite radius.

if $l(TA) = l(CA)$; director circle is a point circle

if $l(TA) < l(CA)$; no real circle

4.1 TANGENT

1. $P(x_1, y_1)$

$T = 0 \Rightarrow \dfrac{xx_1}{a^2} - \dfrac{yy_1}{b^2} = 1$

2. $P(\theta) \equiv P(a\sec\theta, b\tan\theta)$

$\dfrac{x\sec\theta}{a} - \dfrac{y\tan\theta}{b} = 1$

$COT \equiv c^2 = a^2m^2 - b^2$

3. slope is m : $y = mx + c$

apply COT $c^2 = a^2m^2 - b^2 \Rightarrow y = mx \pm \sqrt{a^2m^2 - b^2}$

4.2 NORMAL

1. $P(x_1, y_1)$

slope of tangent $= \dfrac{b^2x_1}{a^2y_1} \Rightarrow$ slope of normal $= -\dfrac{a^2y_1}{b^2x_1}$

equation of normal

$y - y_1 = -\dfrac{a^2y_1}{b^2x_1}(x - x_1) \Rightarrow \dfrac{a^2x}{x_1} + \dfrac{b^2y}{y_1} = a^2 + b^2$

2. $P(\theta) \equiv P(a\sec\theta, b\tan\theta)$

$\dfrac{ax}{\sec\theta} + \dfrac{by}{\tan\theta} = a^2 + b^2 = a^2e^2$

Q.1 If line $y = mx + c$ is normal to standard hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then find condition.

Ans. $\frac{\pm m\left(a^2 + b^2\right)}{\sqrt{a^2 - b^2 m^2}}$

5.1 COMMON CONCEPTS

1. Chord with a given middle point :

Equation of the chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ whose middle point is (x_1, y_1) is $\mathbf{T = S_1}$,

2. Chord of Contact :

Equation to the chord of contact of tangents drawn from a point $P(x_1, y_1)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is , $T = 0$

3. Pair of Tangents :

The equation to the pair of tangents which can be drawn from any point (x_1, y_1) to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is given by: $SS_1 = T^2$

4. Polar :

Equation to the polar for a point $P(x_1, y_1)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $T = 0$,

5. Diameter :

mid–pt. (h, k) slope is m

$$\frac{hx}{a^2} - \frac{ky}{b^2} = \frac{h^2}{a^2} - \frac{k^2}{b^2} \qquad [T = S_1]$$

$$\frac{hb^2}{ka^2} = m \Rightarrow \quad k = \frac{b^2 h}{a^2 m} \Rightarrow \quad y = \frac{b^2}{a^2 m} x$$

Questions

1. Find point of intersection of tangents drawn to the standard hyperbola at $P(\alpha)$ and $Q(\beta)$

Ans $\left(\frac{a\cos\left(\frac{\alpha - \beta}{2}\right)}{\cos\left(\frac{\alpha + \beta}{2}\right)}, \frac{b\sin\left(\frac{\alpha + \beta}{2}\right)}{\cos\left(\frac{\alpha + \beta}{2}\right)} \right)$

Sol. Chord of Contact

$$\frac{xh}{a^2} - \frac{yk}{b^2} = 1 \qquad \text{...(i)}$$

Chord joining 2 points P(α) & Q(β)

$$\frac{x}{a}\cos\left(\frac{\alpha-\beta}{2}\right)-\frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha+\beta}{2}\right) \quad ...(ii)$$

(i) & (ii) $\rightarrow$ compare

$$\frac{h\times a}{a^2\cos\left(\frac{\alpha-\beta}{2}\right)}=\frac{k\cdot b}{b^2\times\sin\left(\frac{\alpha+\beta}{2}\right)}=\frac{1}{\cos\left(\frac{\alpha+\beta}{2}\right)}$$

$$(h,k)\equiv\left(\frac{a\cos\left(\frac{\alpha-\beta}{2}\right)}{\cos\left(\frac{\alpha+\beta}{2}\right)},\frac{b\sin\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha+\beta}{2}\right)}\right)$$

2. Find equation of tangent(s) to the hyperbola $\frac{x^2}{36}-\frac{y^2}{9}=1$ passing through (0, 4)

Ans $5x - 6y + 24 = 0$, $5x + 6y - 24 = 0$

Sol. $\frac{x^2}{36}-\frac{y^2}{9}=1$ (0, 4)

a tangent of slope m passing through (0, 4)

$y - 4 = mx$

COT $\rightarrow c^2 = a^2m^2 - b^2 \Rightarrow 16 = 36m^2 - 9 \Rightarrow m = \pm\frac{5}{6}$

Tangents $\rightarrow y-4=\frac{5}{6}x$ & $y-4=-\frac{5}{6}x$

$\Rightarrow 5x - 6y + 24 = 0$ & $5x + 6y - 24 = 0$

3. Find equation of the common tangent to the curves $y^2 = 8x$ and $3x^2 - y^2 = 3$

Ans $2x - y + 1 = 0$, $2x + y + 1 = 0$

Sol. $y^2 = 8x$, $3x^2 - y^2 = 3$

$y = mx + c$ $\qquad \frac{x^2}{1}-\frac{y^2}{3}=1$

$c=\frac{2}{m}$ $\qquad$ COT $\rightarrow c^2 = 1 \times m^2 - 3$

$\Rightarrow \frac{4}{m^2} = m^2 - 3$ $\qquad \Rightarrow 4 = m^4 - 3m^2$

$\Rightarrow (m^2)^2 - 3m^2 - 4 = 0$

$\Rightarrow (m^2 + 1)(m^2 - 4) = 0$

$\Rightarrow m^2 = -1, \quad m^2 = 4$

$\Rightarrow m = \pm 2$

$\Rightarrow c = \pm 1$

$y = 2x + 1$, $y = -2x - 1$

4. Find the equation of the common tangent to the hyperbolas $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and $\frac{y^2}{a^2}-\frac{x^2}{b^2}=1$.

Ans. $y=\pm x\pm\sqrt{a^2-b^2}$

Sol. Tangent to $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is

$$y=mx\pm\sqrt{a^2m^2-b^2} \quad ....(1)$$

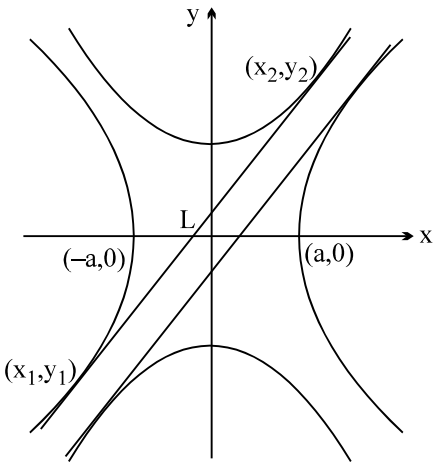
(1) should also be tangent to other hyperbola

$$\frac{x^2}{(-b^2)}-\frac{y^2}{(-a^2)}=1 \qquad ....(2)$$

$a^2m^2-b^2=(-b)^2m^2-(-a^2)$

$\Rightarrow \quad (a^2+b^2)m^2=a^2+b^2 \quad \Rightarrow \quad m=\pm 1$

hence equations are $y=\pm x\pm\sqrt{a^2-b^2}$



5. Find locus of point of intersection of tangents to the standard hyperbola at points

(i) $P(\alpha)$ and $Q(\frac{\pi}{2}+\alpha)$

Ans $\frac{2x^2}{a^2}-\frac{y^2}{b^2}=1$

Sol. $\alpha=\alpha$

$$\beta=\frac{\pi}{2}+\alpha$$

$$|\alpha-\beta|=\frac{\pi}{2}$$

$$h=\frac{a\cos\left(\frac{\alpha-\beta}{2}\right)}{\cos\left(\frac{\alpha+\beta}{2}\right)},k=\frac{b\sin\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha+\beta}{2}\right)}$$

$$\Rightarrow \sec\left(\frac{\alpha+\beta}{2}\right)=\frac{\sqrt{2}h}{a},\tan\left(\frac{\alpha+\beta}{2}\right)=\frac{k}{b}$$

$$\Rightarrow \sec^2\left(\frac{\alpha+\beta}{2}\right)-\tan^2\left(\frac{\alpha+\beta}{2}\right)=1$$

$$\Rightarrow \frac{x^2}{\left(\frac{a^2}{2}\right)} - \frac{y^2}{b^2} = 1$$

(ii) $P(\alpha)$ and $Q(\frac{\pi}{2} - \alpha)$

Ans $y = b$

Sol. $P(\alpha), Q\left(\frac{\pi}{2} - \alpha\right)$

$$\left.\begin{aligned} \alpha &= \alpha \\ \beta &= \frac{\pi}{2} - \alpha \end{aligned}\right\} \alpha + \beta = \frac{\pi}{2}$$

$$k = \frac{b\sin\frac{\pi}{4}}{\cos\frac{\pi}{4}} \Rightarrow y = b$$

6 Perpendicular from the centre upon the tangent and normal at any point of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meet them in Q and R. Find their loci.

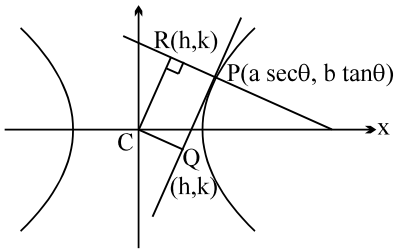
Ans. $(x^2 + y^2)^2 = a^2x^2 - b^2y^2$; $(x^2 + y^2)^2 (a^2y^2 - b^2x^2) = (a^2 + b^2)^2 x^2y^2]$

Sol. Compare the equation $hx + ky = h^2 + k^2$ with

tangent, $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$ .....(1)

and normal $\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2$ ....(2)

and eliminate θ to get the locus.



Alternate :

$xh + yk = h^2 + k^2$

Tangent to the H

COT $\rightarrow c^2 = a^2m^2 - b^2$

$$\Rightarrow \left(\frac{h^2 + k^2}{k}\right)^2 = a^2 \times \frac{h^2}{k^2} - b^2$$

$\Rightarrow xh + yk = h^2 + k^2$

is the normal to H.

$$c = \pm\frac{m\left(b^2 + a^2\right)}{\sqrt{a^2 - b^2m^2}} \Rightarrow \left(\frac{h^2 + k^2}{k}\right)^2 = \frac{\frac{h^2}{k^2}\left(a^2 + b^2\right)^2}{a^2 - b^2 \times \frac{h^2}{k^2}}$$

7. Tangents are drawn to the hyperbola $x^2 - y^2 = a^2$ enclosing at an angle of 45°. Show that the locus of their point of intersection is $(x^2 + y^2)^2 + 4a^2(x^2 - y^2) = 4a^4$

Sol. $x^2 - y^2 = a^2$

$$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{a^2} = 1$$

$$y = mx \pm \sqrt{a^2m^2 - a^2}$$

(h, k)

$$\Rightarrow m^2\left(h^2 - a^2\right) - 2kmh + \left(k^2 + a^2\right) = 0 \left\langle \begin{matrix} m_1 \\ m_2 \end{matrix} \right.$$

$$\Rightarrow \tan 45^\circ = \left(\frac{m_1 - m_2}{1 + m_1 \cdot m_2}\right)$$

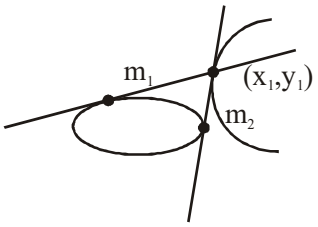
$\Rightarrow (m_1 - m_2)^2 = (1 + m_1m_2)^2$

$\Rightarrow (m_1 + m_2)^2 - 4m_1m_2 = (1 + m_1 \cdot m_2)^2$

$$\Rightarrow \left(\frac{2kh}{h^2 - a^2}\right)^2 - 4 \times \left(\frac{k^2 + a^2}{h^2 - a^2}\right) = \left(1 + \frac{h^2 + a^2}{h^2 - a^2}\right)^2$$

8. Prove that the two tangents drawn from any point on the hyperbola $x^2 - y^2 = a^2 - b^2$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} =$ 1 make complementary angles with the axes.

Sol.



$$x_1^2 - y_1^2 = a^2 - b^2$$

$$\Rightarrow \frac{y_1^2 - b^2}{x_1^2 - a^2} = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Tangent $\rightarrow y = mx \pm \sqrt{a^2m^2 + b^2}$

(x_1, y_1)

$$\Rightarrow m^2\left(x_1^2 - a^2\right) - 2x_1y_1m + \left(y_1^2 - b^2\right) = 0 \left\langle \begin{matrix} m_1 \\ m_2 \end{matrix} \right.$$

$$m_1 \cdot m_2 = \frac{y_1^2 - b^2}{x_1^2 - a^2}$$

$\Rightarrow m_1 \cdot m_2 = +1$

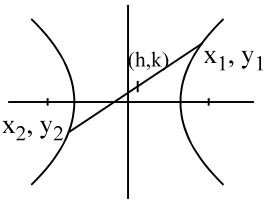
9. Find the equation to the locus of the middle points of the chords of the hyperbola $2x^2 - 3y^2 = 1$, each of which makes an angle of 45° with the x-axis.

Ans $2x = 3y$

Sol. $\frac{y_2 - y_1}{x_2 - x_1} = \tan 45° = 1$

now $2x_1^2 - 3y_1^2 = 1$ ....(1)

and $2x_2^2 - 3y_2^2 = 1$ ....(2)



(1) – (2) $2(x_1^2 - x_2^2) = 3(y_1^2 - y_2^2)$

$$\frac{y_1 - y_2}{x_1 - x_2} = \frac{2(x_1 + x_2)}{3(y_1 + y_2)}$$

$$1 = \frac{2h}{3k} \quad \Rightarrow \quad 2x = 3y$$

10. From points on the circle $x^2 + y^2 = a^2$ tangents are drawn to the hyperbola $x^2 - y^2 = a^2$; prove that the locus of the middle points of the chords of contact is the curve $(x^2 - y^2)^2 = a^2(x^2 + y^2)$

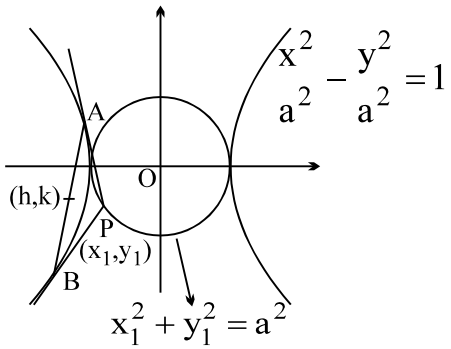
Sol. $\frac{xx_1}{a^2} - \frac{yy_1}{a^2} = 1$ ....(1)

$xx_1 - yy_1 = a^2$ .....(2)

$$\frac{hx}{a^2} - \frac{ky}{a^2} - 1 = \frac{h^2}{a^2} - \frac{k^2}{a^2} - 1$$

$hx - ky = h^2 - k^2$ ....(3)

$$\frac{x_1}{h} = \frac{y_1}{k} = \frac{a^2}{h^2 - k^2}$$



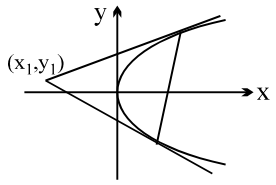
$x_1 = \frac{a^2 h}{h^2 - k^2}$; $y_1 = \frac{a^2 k}{h^2 - k^2}$ $\Rightarrow$ result

5.2 OTHER QUESTIONS

1. A point P moves such that the chord of contact of the pair of tangents from P on the parabola $y^2 = 4ax$ touches the rectangular hyperbola $x^2 - y^2 = c^2$. Show that the locus of P is the ellipse $\frac{x^2}{c^2} + \frac{y^2}{(2a)^2} = 1$.

Sol. Equation of chord of contact of P (x_1, y_1)

$yy_1 = 2a(x + x_1)$ .....(1)



(1) touches the curve $x^2 - y^2 = c^2$ ....(2)

using the condition of tangency, we get

$$y = \frac{2ax}{y_1} + \frac{2ax_1}{y_1} \quad .....(1) \quad ; \quad (1) \text{ is tangent to } \frac{x^2}{c^2} - \frac{y^2}{c^2} = 1$$

$$\frac{4a^2x_1^2}{y_1^2} = c^2 \cdot \frac{4a^2}{y_1^2} - c^2$$

$4a^2 x_1^2 = 4a^2c^2 - c^2 y_1^2 \quad \Rightarrow \quad 4a^2 x_1^2 + c^2 y_1^2 = 4a^2c^2$

$$\frac{x_1^2}{c^2} + \frac{y_1^2}{4a^2} = 1 \quad \Rightarrow \quad \frac{x^2}{c^2} + \frac{y^2}{(2a)^2} = 1$$

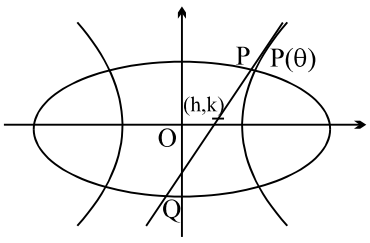
2. A tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ cuts the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at P and Q. Show that the locus of the mid point of PQ is $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2 = \frac{x^2}{a^2} - \frac{y^2}{b^2}$

Sol. Tangent to hyperbola at P(θ)

$$\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1 \quad ....(1)$$

also the chord of the ellipse with middle point (h, k)

$$\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} = \lambda \text{ (say)} \quad ....(2)$$



comparing (1) and (2)

$$\frac{\sec\theta}{a} \cdot \frac{a^2}{h} = -\frac{\tan\theta}{b} \cdot \frac{a^2}{k} = \frac{1}{\lambda} \quad ; \quad \sec\theta = \frac{h}{\lambda a} \text{ and } \tan\theta = -\frac{k}{\lambda b}$$

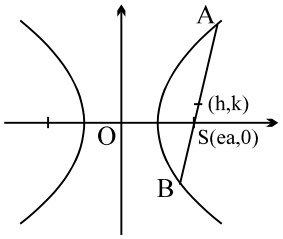
$$\therefore \quad \frac{h^2}{\lambda^2 a^2}-\frac{k^2}{\lambda^2 b^2}=1 \quad \Rightarrow \quad \frac{h^2}{a^2}-\frac{k^2}{b^2}=\lambda^2=\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)^2$$

hence locus is $\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)^2=\frac{x^2}{a^2}-\frac{y^2}{b^2}$

3. Show that the mid points of focal chords of a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ lie on another similar hyperbola.

Sol Equation of chord AB with $T = S_1$

$$\frac{hx}{a^2}-\frac{ky}{b^2}=\frac{h^2}{a^2}-\frac{k^2}{b^2}$$



It passes through (ae, 0)

$$\frac{he}{a}=\frac{h^2}{a^2}-\frac{k^2}{b^2}$$

Locus is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=\frac{ex}{a}$

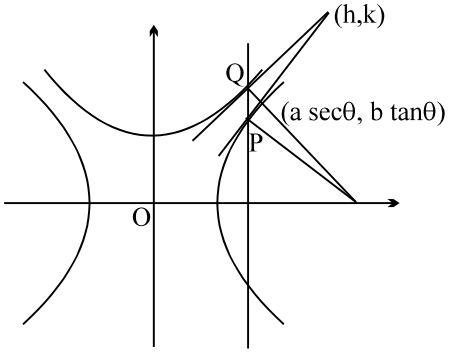
$$\frac{1}{a^2}[x^2-aex]-\frac{y^2}{b^2}=0\ ;\ \frac{\left(x-\frac{ae}{2}\right)^2-\frac{a^2e^2}{4}}{a^2}-\frac{y^2}{b^2}=0\ ;\ \frac{\left(x-\frac{ae}{2}\right)^2}{a^2}-\frac{y^2}{b^2}=\frac{e^2}{4}$$

which is also a hyperbola with eccentricity e

4. A straight line is drawn parallel to the conjugate axis of a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ to meet it and the conjugate hyperbola in the points P & Q. Show that the tangents at P & Q meet on the curve $\frac{y^4}{b^4}\left(\frac{y^2}{b^2}-\frac{x^2}{a^2}\right)=\frac{4x^2}{a^2}$ and that the normals meet on the axis of x. **[O]**

Sol. $\frac{x^2}{a^2}-\frac{y^2}{b^2}=-1$ ...(1) put $x = a\sec\theta$

$(\sec^2\theta + 1)b^2 = y^2 \quad \Rightarrow \quad y = b\sqrt{\sec^2\theta+1}$



Tangent to hyperbola : $\frac{x\sec\theta}{a}-\frac{y\tan\theta}{b}=1$

$bh - ak\sin\theta = ab\cos\theta$ ....(1)

|||ly $\quad \dfrac{x\sec\theta}{a} - \dfrac{y\sqrt{1+\sec^2\theta}}{b} = -1$

$bh\sec\theta - ak\sqrt{1+\sec^2\theta} = -ab$

$bh - ak\sqrt{1+\cos^2\theta} = -ab\cos\theta$ ....(2)

from (1) $\quad b^2(h - a\cos\theta)^2 = a^2k^2\sin^2\theta$ ....(3)

from (2) $\quad b^2(h + a\cos\theta)^2 - a^2k^2(1+\cos^2\theta)$ ....(4)

(1) + (2) $\quad 2b^2(h^2 + a^2\cos^2\theta) = 2a^2k^2$ ....(5)

(4) – (3) $\quad 4b^2ah\cos\theta = a^2k^2 \cdot 2\cos^2\theta$

$2b^2h = ak^2\cos\theta$

Normal to hyperbola: $\dfrac{ah}{\sec\theta} + \dfrac{bk}{\tan\theta} = a^2 + b^2$ ....(1)

conjugate hyperbola : $\dfrac{x^2}{(-a)^2} - \dfrac{y^2}{(-b)^2} = 1$

normal to CH : $\dfrac{-a^2x}{x_1} + \dfrac{(-b^2)y}{y_1} = -(a^2 + b^2)$

$$\frac{-ah}{\sec\theta} + \frac{bk}{\sqrt{1+\sec^2\theta}} = -(a^2 + b^2)$$

$$\frac{ah}{\sec\theta} + \frac{bk}{\tan\theta} = a^2 + b^2$$

$$- \quad - \quad -$$

$$k = 0 \quad \Rightarrow \quad y = 0$$

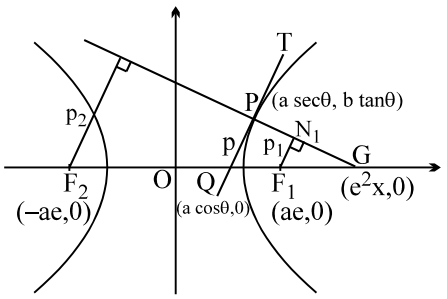
5. Prove that the part of the tangent at any point of the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ intercepted between the point of contact and the transverse axis is a harmonic mean between the lengths of the perpendiculars drawn from the foci on the normal at the same point. **[O]**

Sol. $\dfrac{x\sec\theta}{a} - \dfrac{y\tan\theta}{b} = 1$; $\dfrac{a^2x}{x_1} + \dfrac{b^2y}{y_1} = a^2e^2$

$$\frac{p}{p_1} = \frac{QG}{F_1G} = \frac{e^2x_1 - a\cos\theta}{e^2x_1 - ae} = \frac{e^2a\sec\theta - a\cos\theta}{e^2a\sec\theta - ae}$$

$$\frac{e^2-\cos^2\theta}{e^2-e\cos\theta}=\frac{e+\cos\theta}{e}$$

$$\frac{p}{p_1}=1+\frac{\cos\theta}{e}$$

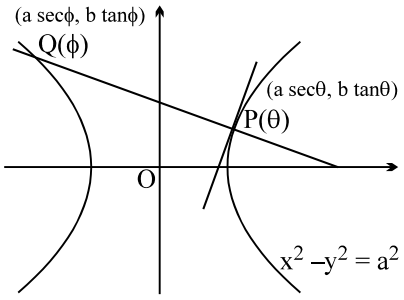


||ly $\quad \frac{p}{p_2}=1-\frac{\cos\theta}{e} \quad \Rightarrow \quad \frac{p}{p_1}+\frac{p}{p_2}=2 \quad \Rightarrow \quad \frac{2}{p}=\frac{1}{p_1}+\frac{1}{p_2}$

6. If a chord joining the points P (a sec θ, a tan θ) & Q (a sec ϕ , a tan ϕ) on the hyperbola $x^2 - y^2 = a^2$ is a normal to it at P, then show that tan ϕ = tan θ (4 sec² θ – 1). **[O]**

Sol. $x^2 - y^2 = a^2$ ....(1)

$$\frac{x}{\sec\theta}+\frac{y}{\tan\theta}=2a \quad ....(2)$$



$$\frac{x}{\sec\theta}=\left(2a-\frac{y}{\tan\theta}\right)$$

$$\sec^2\theta\left(2a-\frac{y}{\tan\theta}\right)^2-y^2=a^2$$

$$\sec^2\theta\left[4a^2+\frac{y^2}{\tan^2\theta}-\frac{4ay}{\tan\theta}\right]-y^2=x^2$$

$$\left(\frac{\sec^2\theta}{\tan^2\theta}-1\right)y^2-\left(\frac{4a\sec\theta}{\tan\theta}\right)y+a^2(4\sec^2\theta-1)=0$$

$y_1y_2 = a^2(4\sec^2\theta - 1)\cdot\tan^2\theta = \tan\theta\tan\phi\, a^2$

$\tan\phi = \tan\theta(4\sec^2\theta - 1)$ Hence proved

7. If one axis of varying central conic (hyperbola) be fixed is magnitude and position, prove that the locus of the point of contact of a tangent drawn to it from a fixed points on the other axis is a parabola. **[O]**

Sol. Equation of a tangent at P(θ) $h = a\sec\theta$; $k = b\tan\theta$

$$\frac{x\sec\theta}{a}-\frac{y\tan\theta}{b}=1 \quad \textbf{....(1)}$$

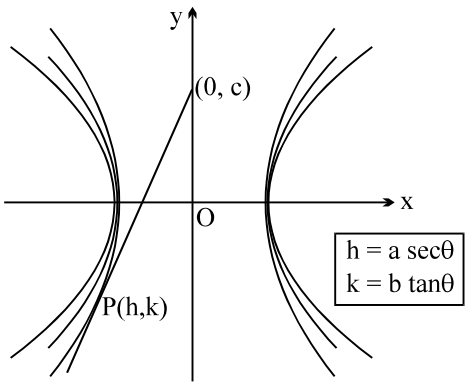
$\because$ (1) passes through the point (0, c)

$\therefore$ $-c\tan\theta = b$ **....(2)**

Now substituting the value of b in (2)

$$\frac{k}{c}=-\tan^2\theta$$

and $$\frac{h^2}{a^2}=\sec^2\theta$$

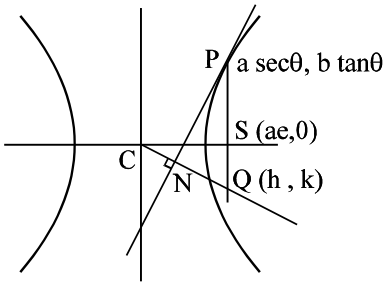


adding $\frac{k}{c}+\frac{h^2}{a^2}=1 \Rightarrow \frac{h^2}{a^2}=1-\frac{k}{c} \Rightarrow h^2=\frac{a^2}{c}(c-k) \Rightarrow x^2=-\frac{a^2}{c}(y-c)$ which is a parbola

8. In the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, prove that the focal distance of any point and the perpendicular from the centre upon the tangent at it meet on the circle whose centre is the focus and whose radius is the semi-transverse axis.

Sol. TPT (h, k) lies on $(x-ae)^2+y^2=a^2$

$$T: \frac{x\sec\theta}{a}-\frac{y\tan\theta}{b}=1$$

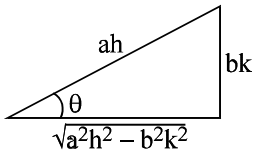


$$m_T=\frac{b}{a}\frac{1}{\sin\theta} \quad ; \quad m_{CN}=-\frac{a\sin\theta}{b}$$

$\therefore$ equation of CN is $y=-\frac{a}{b}\sin\theta\, x$

It passes through (h, k) $k=-\frac{a}{b}\sin\theta\, h$ **....(1)**

$$\sin\theta=-\frac{bk}{ah}$$



equation of PS : $y-0=\frac{b\tan\theta}{a(\sec\theta-e)}(x-ae)$

$$ay=\frac{b\sin\theta}{(1-e\cos\theta)}(x-ae) \qquad \text{or} \qquad k=\frac{b\sin\theta}{a(1-e\cos\theta)}(h-ae) \qquad \textbf{....(2)}$$

substituting the value of sinθ and cosθ in equation (2)

$$\therefore \quad ak=\frac{-b\frac{bk}{ah}(ah)}{ah-e\sqrt{a^2h^2-b^2k^2}}(h-ae)$$

$$\therefore \quad ak\left[ah-e\sqrt{a^2h^2-b^2k^2}\right]=-b^2k\,(h-ae)$$

$$a^2h+b^2(h-ae)=ae\sqrt{a^2h^2-b^2k^2}$$

$$(a^2+b^2)\,h-b^2ae=ae\sqrt{a^2h^2-b^2k^2} \qquad [\,(a^2+b^2)=a^2e^2\,]$$

$$(a\,e\,h-b^2)^2=a^2h^2-b^2k^2$$

$$a^2e^2h^2+b^4-2aeb^2h=a^2h^2-b^2k^2$$

$$b^2\,[b^2-2aeh+k^2]=a^2h^2\,(1-e^2)=\frac{-a^2h^2b^2}{a^2}$$

$$k^2+h^2-2aeh+b^2=0$$

$k^2 + h^2 - 2aeh + a^2e^2 = a^2e^2 - b^2$

$k^2 + (h - ae)^2 = a^2 + b^2 - b^2$

$\therefore$ locus is $(x - ae)^2 + y^2 = a^2$

6. IMPORTANT HIGHLIGHTS :

Referring to a Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

1. If P be any point on the hyperbola with S & S′ as its foci then $|PS - PS'| = 2a$.

2. Product of the length's of the perpendiculars from either focus on a variable tangent to a hyperbola = (semi conjugate axis)$^2 = b^2$

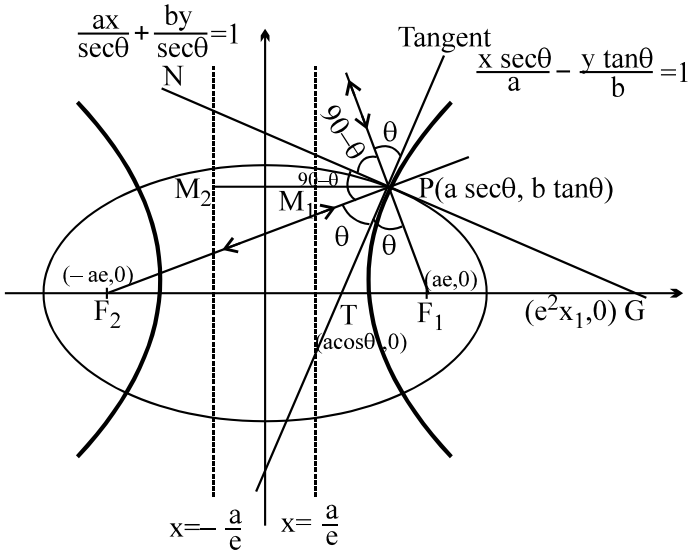
3. Feet of the perpendicular drawn from focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ upon any tangent will lie on its auxiliary circle i.e. $x^2 + y^2 = a^2$

4. The portion of the tangent to an hyperbola between the point of contact & the directrix subtends a right angle at the corresponding focus.

5. The tangent & normal at any point of a hyperbola bisect the angle between the focal radii. This spells the reflection property of the hyperbola as **"An incoming light ray "** aimed towards one focus is reflected from the outer surface of the hyperbola towards the other focus.

6. If an ellipse and a hyperbola have the same foci, they cut at right angles at any of their common point.

Note that the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{a^2 - k^2} - \frac{y^2}{k^2 - b^2} = 1 (a > k > b > 0)$ are confocal and therefore orthogonal



7. The foci of the hyperbola and the points P and Q in which any tangent meets the tangents at the vertices are concyclic with PQ as diameter of the circle.

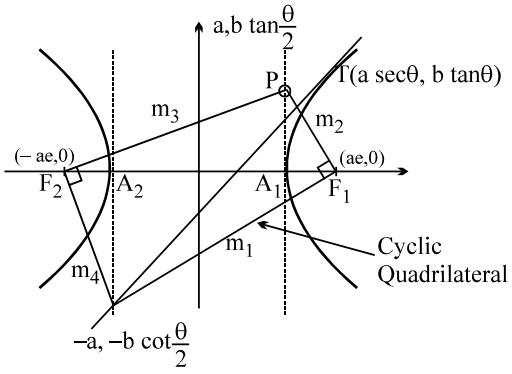
$$T: \frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$$

$$\frac{(\sec\theta - 1)b}{\tan\theta} = y$$

$$y = b\left(\frac{1-\cos\theta}{\sin\theta}\right)$$

Family of circles through F_1F_2

$(x - ae)(x + ae) + y^2 + \lambda y = 0 \Rightarrow$ PQ as diameter]



7. ASYMPTOTES

From a point P on the curve, perpendicular is drawn on the line L. If length of perpendicular approaches zero when P is moving towards ∞ along the curve then the given line L is asymptote to the curve.

To find the asymptote of the hyperbola :

Let $y = mx + c$ is the asymptote of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Solving these two we get the quadratic as

$(b^2 - a^2m^2)\,x^2 - 2a^2mcx - a^2(b^2 + c^2) = 0$ ....(1)

In order that $y = mx + c$ be an asymptote, both roots of equation (1) must approach infinity, the conditions for which are :

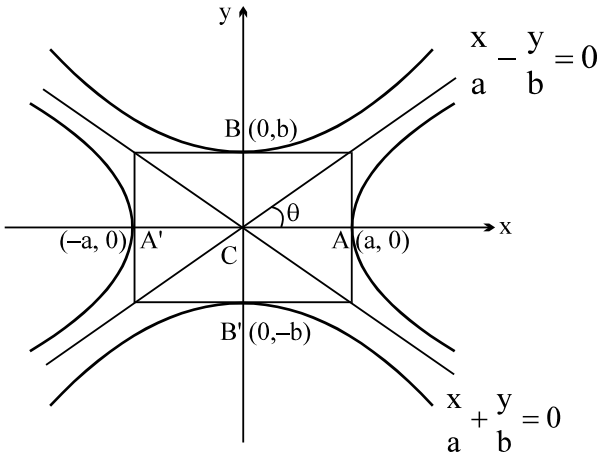
coeff of $x^2 = 0$ & coeff of $x = 0$.

$\Rightarrow$ $b^2 - a^2m^2 = 0$ or $m = \pm\frac{b}{a}$ &

$a^2mc = 0 \Rightarrow c = 0$.

$\therefore$ equations of asymptote are $\frac{x}{a} + \frac{y}{b} = 0$

and $\frac{x}{a} - \frac{y}{b} = 0$.



$\frac{x}{a} - \frac{y}{b} = 0$

$\frac{x}{a} + \frac{y}{b} = 0$

combined equation to the asymptotes $\frac{x^2}{a^2}-\frac{y^2}{b^2}=0$.

Example –

H: $\frac{x^2}{9}-\frac{y^2}{4}=1$; A: $\frac{x^2}{9}-\frac{y^2}{4}=0$

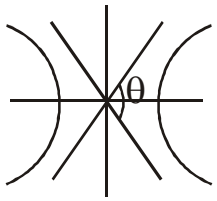
H: $\frac{x^2}{16}-\frac{y^2}{9}=-1$ A: $\frac{x^2}{16}-\frac{y^2}{9}=0$

Important Points

1. The asymptotes pass through the centre of the hyperbola
2. Asymptotes can be treated as the tangent to the hyperbola from the centre.
3. A hyperbola and its conjugate have the same asymptote.

$$H:\frac{x^2}{a^2}-\frac{y^2}{b^2}=1;C:\frac{x^2}{a^2}-\frac{y^2}{b^2}=-1;A:\frac{x^2}{a^2}-\frac{y^2}{b^2}=0$$

4. The equation of the pair of asymptotes differ the hyperbola & the conjugate hyperbola by the same constant only. $2A=C+H$
5. To find center of hyperbola, use partial derivative approach as taught in straight line.
6. The bisectors of the angles between the asymptotes are the principal axes of the hyperbola.
7. Line parallel to asymptote will intersect hyperbola at exactly one point (it will not be a tangent).
8. Angle between asympotes



$$\tan\theta=\frac{\frac{b}{a}-\left(-\frac{b}{a}\right)}{1-\frac{b^2}{a^2}}\Rightarrow\tan\theta=\frac{2ab}{a^2-b^2}$$

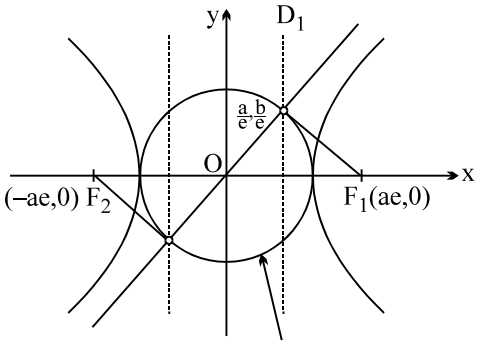
9. If angle between asymptotes is θ then $e=\sec\left(\frac{\theta}{2}\right)$
10. If hyperbola is rectangular then asymptotes will be mutually perpendicular
11. The asymptotes of a hyperbola are the diagonals of the rectangle formed by the lines drawn through the extremities of each axis parallel to the other axis.
12. Perpendicular from the foci on either asymptote meet it in the same points as the corresponding directrix & the common points of intersection lie on the auxiliary circle.

$y=\frac{b}{a}x$ ....(1)

$by+ax=a^2e$ ....(2)

$\left(\frac{b^2}{a}+a\right)x=a^2e$

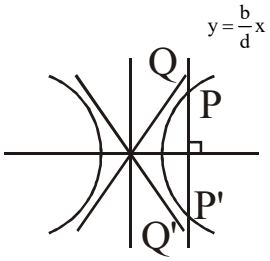
$(b^2+a^2)x=a\cdot a^2e$



$\frac{a^2}{e^2}+\frac{b^2}{e^2}=\frac{a^2+b^2}{a^2}=a^2$

$(a^2e^2)x = a^2ae \Rightarrow \quad x = \dfrac{a}{e}$; hence $y = \dfrac{b}{a}\cdot\dfrac{a}{c} = \dfrac{b}{c}$

13.



$(PQ)(P'Q) = b^2$

$P(a\sec\theta, b\tan\theta)$ $\quad$ $Q(a\sec\theta, b\sec\theta)$

$P'(a\sec\theta, -b\tan\theta)$ $\quad$ $Q'(a\sec\theta, -b\sec\theta)$

$PQ = b(\sec\theta - \tan\theta)$; $P'Q = b(\sec\theta + \tan\theta)$

$(PQ)(P'Q) = b^2$

14. The tangent at any point P on a hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ with centre C, meets the asymptotes in Q and R

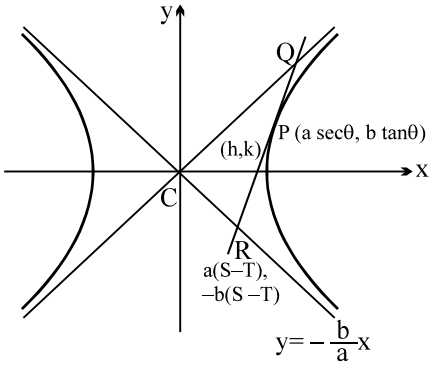
(i) area of $\Delta CQR = ab$ (always)

$$\frac{a\sec\theta}{a} + \frac{y\tan\theta}{b} = 1 \quad \text{(tangent at P)}$$

$Q(a(\sec\theta + \tan\theta), b(\sec\theta + \tan\theta))$

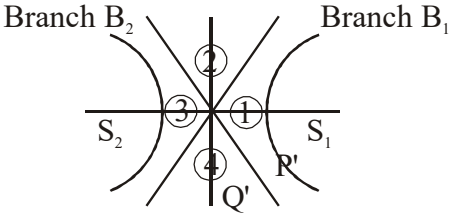
$R(a(\sec\theta - \tan\theta), -b(\sec\theta - \tan\theta))$

area of $\Delta CQR = ab$



(ii) P is mid pt. of QR

15.



(i) If $PS_1 - PS_2 = 2a$ then pt. P will lie on branch B_2

(ii) If $PS_1 - PS_2 = -2a$ then pt. P will lie on branch B_1

(iii) From a point in region (1), both tangents fot the hyperbola will touch branch B_1

(iv) From a point in region (3), both tangents to the hyperbola will touch branch B_2

(v) From a point in region (2)/(4), one tangent to the hyperbola will touch branch B_1 & other will touch branch B_2 .

Questions on Asymptotes

1 Find the asymptotes of the hyperbola, $3x^2 - 5xy - 2y^2 - 5x + 11y - 8 = 0$. Also find the equation of the conjugate hyperbola.

Ans. $3x^2 - 5xy - 2y^2 - 5x + 11y + \lambda = 0; \quad \lambda = -\frac{38}{49}$

Sol. $H \to 3x^2 - 5xy - 2y^2 - 5x + 11y - 8 = 0$

$A \to 3x^2 - 5xy - 2y^2 - 5x + 11y + x = 0$

$\Delta = 0 \Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$$\Rightarrow (3\times -2\times \lambda) + \left(2\times \frac{11}{2}\times -\frac{5}{2}\times \frac{-5}{2}\right) - \left(3\times \frac{121}{4}\right) + \left(2\times \frac{25}{4}\right) - \left(\lambda \times \frac{25}{4}\right) = 0$$

$$\lambda = -\frac{38}{49}$$

$CH \to 2A = H + C$

2. Find the equation to the hyperbola whose asymptotes are the straight lines $2x + 3y + 3 = 0$ and $3x + 4y + 5 = 0$ and which passes through the point $(1, -1)$. Also write the equation of its transverse axis

Ans $(2x + 3y + 3)(3x + 4y + 5) = 8$

$$\frac{2x+3y+3}{\sqrt{13}} = \pm\left(\frac{3x+4y+5}{5}\right) \quad + \Rightarrow TA; \quad - \Rightarrow CA$$

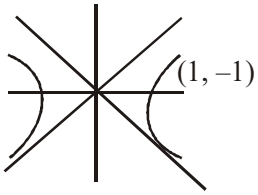
Sol. $A \to 2x + 3y + 3 = 0,\ 3x + 4y + 5 = 0$

$H \to (2x + 3y + 3) \times (3x + 4y + 5) = \lambda$

$(1, -1) \Rightarrow \lambda = 8$

$H \to (2x + 3y + 3) \times (3x + 4y + 5) - 8 = 0$

Bisectors $\to$



$$\frac{2x+3y+3}{\sqrt{2^2+3^2}} = \pm\frac{(3x+4y+5)}{\sqrt{3^2+4^2}}$$

$$+ \to x\left(10-3\sqrt{13}\right) + y\left(15-4\sqrt{13}\right) + 15 - 5\sqrt{13} = 0$$

$$- \to x\left(10+3\sqrt{13}\right) + y\left(15+4\sqrt{13}\right) + 15 + 5\sqrt{13} = 0$$

one who contain $(1, -1)$ will be Transverse axis

$(1, -1) \to 2x + 3y + 3 \to$ +ve

$\to 3x + 4y + 5 \Rightarrow$ +ve

Both have same sign

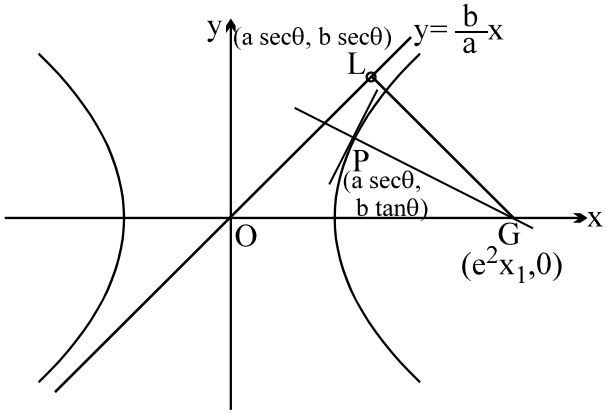
+ve sign bisector will be T.A.

3. A normal is drawn to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ at P which meets the transverse axis (TA) at G. If perpendicular from G on the asymptote meets it at L, show that LP is parallel to conjugate axis. **[O]**

Sol. equation of GL with slope – a/b and passing through $(e^2x_1, 0)$ is

$y - 0 = -a/b(x - e^2x_1)$

$ax + by = ae^2x_1$ ....(1)



put $y = \frac{b}{a}x$

$ax + b\cdot\frac{b}{a}x = ae^2x_1$

$x\left[\frac{a^2+b^2}{a}\right] = a^2e^2x_1$

$x = x_1$

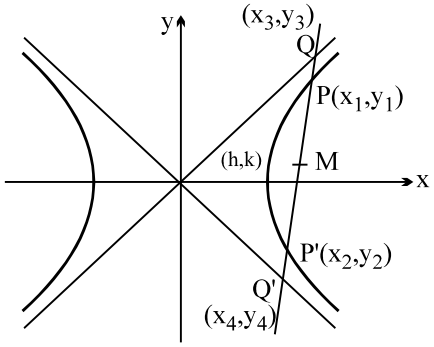
4. A transversal cuts the same branch of a hyperbola $x^2/a^2 - y^2/b^2 = 1$ in P, P′ and the asymptotes in Q, Q′. Prove that : (i) PQ = P'Q' & (ii) PQ' = P'Q **[O]**

Sol. TPT PQ = P'Q' and PQ' = P'Q

$b^2x^2 - a^2y^2 = a^2b^2$; Let the transversal be $y = mx + c$

$b^2x^2 - a^2(mx + c)^2 = a^2b^2$

$(b^2 - a^2m^2)x^2 - 2a^2mcx - a^2(c^2 + b^2) = 0$



$$\frac{x_1+x_2}{2}=\frac{2a^2mc}{b^2-a^2m^2} \quad ....(1)$$

solving $y = mx + c$ with $b^2x^2 - a^2y^2 = 0$

$(b^2 - a^2m^2)x^2 - 2a^2mcx - a^2c^2 = 0$ ....(2)

$$\frac{x_3+x_4}{2}=\frac{2a^2mc}{b^2-a^2m^2} \quad ....(3)$$

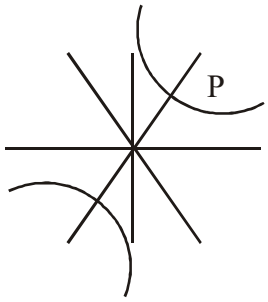
MQ = MQ'

also MP = MP'

PQ = P'Q'

5. The tangent at any point P of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ meets one of the asymptotes in Q and L, M are the feet of the perpendiculars from Q on the axes. Prove that LM passes through P. **[O]**

8. RECTANGULAR HYPERBOLA $xy = c^2$



$P(t) \equiv \left(ct, \frac{c}{t} \right)$

$A : xy = 0 ; \quad H : xy = c^2$

1. center $\equiv (0,0)$
2. $TA : x - y = 0 \quad CA : x + y = 0$
3. Vertices $(c, c) \ , \ (-c, -c)$
4. $\ell_{TA} = \ell_{CA} = \ell_{LR} = 2\sqrt{2}c$
5. $B_1(-c, c) \quad B_2(c, -c)$
6. foci $\quad S_1(\sqrt{2}\, c, \sqrt{2}\, c) \quad S_2(-\sqrt{2}\, c, -\sqrt{2}\, c)$
7. equation of directrices $\quad x + y = \lambda$

distance from center $= c \Rightarrow \left| \frac{\lambda}{\sqrt{2}} \right| = c \Rightarrow \quad \lambda = \pm\sqrt{2}c$

$x + y = \pm\sqrt{2}c$

8. ends of LR

$((\sqrt{2}+1)c, (\sqrt{2}-1)c), ((\sqrt{2}-1)c, (\sqrt{2}+1)c)$

$(-(\sqrt{2}+1)c, -(\sqrt{2}-1)c), (-(\sqrt{2}-1)c, -(\sqrt{2}+1)c)$

9. $P(t_1)$ and $Q(t_2)$ are two points on hyperbola then $m_{PQ} = \frac{\frac{c}{t_1} - \frac{c}{t_2}}{ct_1 - ct_2} = \frac{(t_2 - t_1)}{(t_1 t_2)(t_1 - t_2)} \Rightarrow m_{PQ} = \frac{-1}{t_1 t_2}$

10. equation of PQ $\quad y - \frac{c}{t_1} = -\frac{1}{t_1 t_2}(x - c\, t_1)$

11. Tanget at $P(x_1, y_1)$ on $\quad xy = c^2$

$T = 0 \Rightarrow \frac{xy_1 + yx_1}{2} = c^2$

$xy_1 + yx_1 = 2c^2 = 2x_1 y_1 \Rightarrow \quad \frac{x}{x_1} + \frac{y}{y_1} = 2$

12. Tangent at P(t)

$\frac{x}{ct} + \frac{y}{c/t} = 2 \Rightarrow \quad \frac{x}{t} + ty = 2c$

13. Normal at P(t)

$m|_{tangent} = \frac{-1}{t^2} \Rightarrow \quad m|_{normal} = t^2$

Equation of normal $y - \frac{c}{t} = t^2(x - ct)$

14. Chord whose mid–pt. is (h, k)

$T = S_1 \Rightarrow \quad \frac{x}{h} + \frac{y}{k} = 2$

15. Auxiliary circle $x^2 + y^2 = 2c^2$

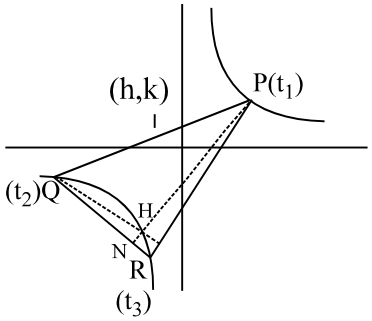
Questions **[O]**

Q.1 A rectangular hyperbola $xy = c^2$ circumscribing a triangle also passes through the orthocentre of this triangle. If $\left(ct_i, \frac{c}{t_i}\right)$ i = 1, 2, 3 be the angular points P, Q, R then orthocentre is $\left(\frac{-c}{t_1t_2t_3}, -ct_1t_2t_3\right)$.

Sol. slope of $QR = -\frac{1}{t_2t_3}$

$\therefore$ slope of $PN = t_2t_3$

$\therefore$ equation of altitude through P



$$y - \frac{c}{t_1} = t_2t_3(x - ct_1)$$

$$y + c\,t_1\,t_2\,t_3 = \frac{c}{t_1} + x\,t_2t_3$$

$$y + c\,t_1\,t_2\,t_3 = t_2t_3\left(x + \frac{c}{t_1t_2t_3}\right) \qquad .....(1)$$

(1) is suggestive that orthocentre is $\left(\frac{-c}{t_1t_2t_3}, -ct_1t_2t_3\right)$

Q.2 If a circle and the rectangular hyperbola $xy = c^2$ meet in the four points t_1, t_2, t_3 & t_4, then

(a) $t_1\,t_2\,t_3\,t_4 = 1$

(b) the centre of the mean position of the four points bisects the distance between the centres of the two curves.

(c) the centre of the circle through the points t_1, t_2 & t_3 is :

$$\left\{\frac{c}{2}\left(t_1 + t_2 + t_3 + \frac{1}{t_1t_2t_3}\right), \frac{c}{2}\left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + t_1t_2t_3\right)\right\}$$

(d) If PQRS are the four points of intersection of the circle with rectangular hyperbola then $(OP)^2 + (OQ)^2 + (OR)^2 + (OS)^2 = 4r^2$ where r is the radius of circle.